

ALGEBRAIC TOPOLOGY 1, 2022

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INTRODUCTION

The goal of the course is to equip ourselves with mathematical tools that will enable us to give concrete reasons why a football and a swim ring have fundamentally different shapes! We will do this by associating algebraic objects (like groups) to geometric objects (like a ball or a swim ring), and compare the algebraic objects instead. The two main such “algebraic tools” that we will define and study are the *fundamental group* and *homology*. The fundamental group is closely related to the study of *coverings*, while homology will require to talk about *chain complexes*. Though this course might at times be a little abstract, in the end it will allow us to prove the very concrete *Hairy ball theorem*, that formally proves the experimentally well-known fact that there is no way to comb hair without introducing a dividing line or a hair whorl (or two).



There is much material on the subject of Algebraic Topology in form of books, online lecture notes, and various videos. For supplementary reading, we recommend for example Marvin Greenberg’s 1967 book *Lectures on Algebraic Topology* or Allen Hatcher’s book *Algebraic Topology*. The writing style of Greenberg’s book is closer to these notes, but Hatcher’s book covers more material and is available freely for download on the author’s web page. A more advanced approach is taken in Peter May’s book *A Concise Course on Algebraic Topology*, which is also available on the author’s web page.

These notes are a revision by Nathalie Wahl of notes written by Søren Galatius in 2017+19, including corrections from students who took the course in 2020 (thanks for the comments!!). Further comments, corrections, and critique are very welcome, so the notes can keep improving!

SOME PREREQUISITE NOTIONS

Prerequisite notions from **algebra**: Groups, abelian groups, normal subgroups, quotient group, homomorphisms.

Prerequisite notions from **topology**: Topological spaces, continuous maps, product topology, subspace topology, quotient topology, compactness.

By “map” we shall always mean a continuous function. The letter I shall denote $[0, 1]$ in the usual topology (from the metric $d(s, t) = |t - s|$). We shall also write $\partial I = \{0, 1\} \subset I$. Whenever we write $I \times X$, $X \times Y$ etc., the product shall always be given the product topology.

Date: October 12, 2022.

Basic **set theory** terminology: If X is a set (or topological space, or group, or...) then the *identity* of X is the function $\text{id}_X : X \rightarrow X$ given by $\text{id}_X(x) = x$. More generally, if $A \subset X$ is a subset (or subspace, or subgroup, or...), then the *inclusion* $i : A \hookrightarrow X$ is the function $i(a) = a$. If $f : X \rightarrow Y$ and $A \subset X$, then the *restriction* $f|_A$ is the function $A \rightarrow Y$ given by $f|_A(a) = f(a)$.

A *diagram* of the form

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ g \downarrow & & \downarrow g' \\ Z & \xrightarrow{f'} & W \end{array}$$

is the data of functions $f : X \rightarrow Y$ etc.; we say it is a *commutative diagram* if $f' \circ g = g' \circ f$. In other words, the two maps $X \rightarrow W$ arising from the diagram are equal. Similarly for diagrams of other shapes.

The notation “ $X \subset Y$ ” shall mean that X is a (not necessarily proper) subset of Y .

KEYWORDS

- Week 1 Homotopy, homotopy equivalence, paths and loops, fundamental group of a based space. Fundamental group of the circle.
- Week 2 Covering spaces, lifting paths and homotopies, action of fundamental group on the fiber, universal covering spaces, calculation of fundamental group. Maps between covering spaces, classification of covering spaces.
- Week 3 Classification of covering spaces. Free groups, van Kampen’s theorem.
- Week 4 Categories and functors. Elementary category theory. Definition of singular chains on a space, boundary maps in singular chains, singular homology. Path components and H_0 .
- Week 5 Relationship between π_1 and H_1 . Elementary homological algebra: short and long exact sequences, the long exact sequence in homology associated to a short exact sequence of chain complexes, 5-lemma. The long exact sequence for a pair of spaces.
- Week 6 Properties of singular homology: Long exact sequence, homotopy invariance, excision. Calculation of $H_*(S^n)$ and $H_*(D^n, \partial D^n)$. Proof of excision.
- Week 7 Δ -complexes and simplicial homology. Classical applications of homology.

CONTENTS

Introduction	1
Some prerequisite notions	1
Keywords	2
1. The fundamental group (week 1)	5
1.1. Homotopy	5
1.2. Definition of the fundamental group	7
1.3. Induced maps on the fundamental groups	9
1.4. Paths and change of basepoint	9
1.5. Homotopy invariance	10
1.6. The fundamental group of the circle	11
2. Covering spaces (weeks 2-3)	15
2.1. Covering maps	15
2.2. Lifting paths and homotopies	16
2.3. Right action on the fiber	20
2.4. Covering maps from group actions	23

2.5.	The fundamental group of a quotient space	24
2.6.	Examples of covering maps from group actions	25
2.6.1.	Circle	25
2.6.2.	Torus	25
2.6.3.	Klein bottle	26
2.6.4.	Real projective space	26
2.6.5.	Thrice punctured sphere	26
2.7.	More lifting theorems	27
2.8.	Classification of covering maps	28
2.9.	Existence of covering maps	30
2.9.1.	Existence of universal covers	30
2.9.2.	More quotient constructions	31
2.9.3.	Proof of Theorem 2.8.3	31
3.	Free groups and van Kampen theorem (week 3)	33
3.1.	van Kampen lite	33
3.2.	van Kampen	34
3.3.	Appendix: proof of Theorem 2.9.1	36
3.4.	Appendix: words, free monoids, and free groups	37
3.4.1.	Appendix: coproducts of monoids	37
3.4.2.	Appendix: coproducts of groups	39
3.4.3.	Appendix: reduced words	40
3.4.4.	Appendix: free groups on a set of generators	41
4.	Categories and functors (week 4)	42
4.1.	Categories	42
4.2.	Isomorphisms	43
4.3.	Functors	44
4.4.	Some more exotic looking examples	45
4.5.	Appendix: Functor categories	46
5.	Singular homology (week 4)	48
5.1.	Free abelian group	48
5.2.	Standard simplices and their faces	49
5.3.	Singular chains and singular homology	49
5.4.	The zeroth singular homology group	51
6.	Chain complexes, homology, and first computations (week 5)	53
6.1.	Relationship between $\pi_1(X, x)$ and $H_1(X)$	53
6.2.	Chain complexes	54
6.3.	Singular homology is a functor	55
6.4.	Exact sequences	56
6.5.	Short and long exact sequences	56
6.6.	Naturality properties	58
6.7.	The 5-lemma	59
6.8.	Relative singular chains and homology	60
6.9.	Two important theorems: Homotopy invariance and Excision	62
6.10.	Homology of a disk relative to its boundary	63
6.11.	Homology of a sphere	64
6.12.	A geometric consequence	64
7.	Homotopy invariance and excision (week 6)	65
7.1.	Homotopy invariance	65
7.2.	Small chains	66
7.3.	Subdivision	68
7.4.	Excision revisited	72
7.5.	Cell attachments	73

8. More tools for computations (week 6-7)	75
8.1. The Mayer–Vietoris sequence	75
8.2. Relative homology of a simplex	75
8.3. Δ -complexes	76
8.4. Simplicial chains and simplicial homology	79
9. Applications of homology (week 7)	83
9.1. Euler characteristic	83
9.2. Degree	84
9.3. Reduced homology	85
9.4. The Jordan curve theorem	87
10. Engelsk–Dansk ordbog	90

1. THE FUNDAMENTAL GROUP (WEEK 1)

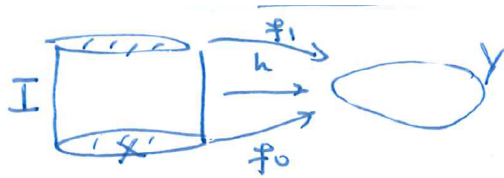
In this section, we define the fundamental group of a space: it is a group that one can associate to any space equipped with a chosen base point.

Corresponding sections in Hatcher are 1.1, 1.3. or in May 1.1–1.5.

1.1. Homotopy. We start by defining the notion of homotopy of maps and use it to define an equivalence relation on maps.

Definition 1.1.1.

- (i) Let X and Y be topological spaces and $f_0, f_1 : X \rightarrow Y$ continuous maps. A *homotopy* from f_0 to f_1 is a continuous function $h : I \times X \rightarrow Y$ such that $h(i, x) = f_i(x)$ for $i = 0, 1$ and all $x \in X$ (see Figure 1).
- (ii) Let $A \subset X$. A homotopy $h : I \times X \rightarrow Y$ is *relative to A* if $h(t, a) = h(0, a)$ for all $a \in A$.

FIGURE 1. Homotopy h from f_0 to f_1 **Definition 1.1.2.**

- (i) Given two maps $f_0, f_1 : X \rightarrow Y$ we shall say that f_0 is *homotopic* to f_1 if there exists a homotopy from f_0 to f_1 . We shall use the notation

$$f_0 \simeq f_1$$

to mean “ f_0 is homotopic to f_1 ”.

- (ii) If $A \subset X$, then two maps $f_0, f_1 : X \rightarrow Y$ are *homotopic relative to A* if there exists a homotopy from f_0 to f_1 which is relative to A . This relationship shall be denoted

$$f_0 \simeq f_1 \quad \text{rel } A.$$

Example 1.1.3. (1) For any map $f : X \rightarrow Y$ we have the *constant homotopy* $h : I \times X \rightarrow Y$ defined by $h(t, x) = f(x)$. It is a homotopy from f to f . It is relative to any subset $A \subset X$.

- (2) Let $f_0, f_1 : \mathbb{R} \rightarrow \mathbb{R}^2$ be defined by $f_0(x) = (x, 0)$ and $f_1(x) = (0, x)$. Then $h : I \times \mathbb{R} \rightarrow \mathbb{R}^2$ defined by $h(t, x) = (1 - t)(0, x) + t(x, 0)$ is a homotopy between f_0 and f_1 .

- (3) More generally, let $Y \subset \mathbb{R}^n$ have the subspace topology, let X be a space, and let $f_0, f_1 : X \rightarrow Y$ be maps such that $(1 - t)f_0(x) + tf_1(x) \in Y$ for all $t \in [0, 1]$ and all $x \in X$. Then we may define a homotopy from f_0 to f_1 by $h(t, x) = (1 - t)f_0(x) + tf_1(x)$. This construction is often called the “straight line homotopy” from f_0 to f_1 .

Being homotopic is then the same as being homotopic relative to $\emptyset$. In general it is of course only possible to be homotopic relative to A if the two maps are already equal on A .

Lemma 1.1.4. *For two spaces X and Y , being homotopic is an equivalence relation on the set of maps $X \rightarrow Y$. The equivalence classes are called homotopy classes of maps.*

More generally, being homotopic relative to a subspace $A \subset X$ is an equivalence relation. The equivalence classes are called homotopy classes relative to A .

Proof. We leave reflexivity and symmetry to the reader and focus on transitivity, so let $f \simeq f' \simeq f''$. Then we may pick homotopies $h : I \times X \rightarrow Y$ from f to f' and $h' : I \times X \rightarrow Y$ from f' to f'' . Then define a homotopy $h'' : I \times X \rightarrow Y$ from f to f'' by

$$h''(s, x) = \begin{cases} h(2s, x) & \text{for } s \leq \frac{1}{2} \\ h'(2s - 1, x) & \text{for } s \geq \frac{1}{2} \end{cases}$$

(see Figure 2). This function $h'' : I \times X \rightarrow Y$ is well defined and continuous by the

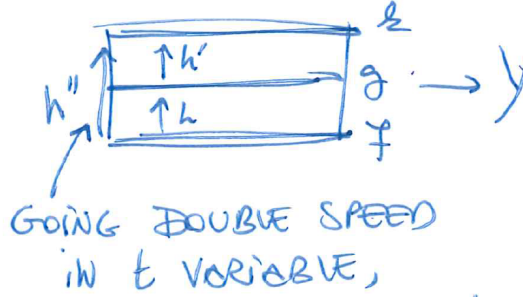


FIGURE 2. Transitivity of the equivalence relation “being homotopic”

gluing property for continuous maps: If $A = A_0 \cup A_1$ for closed subsets $A_0, A_1 \subset A$, then a function $f : A \rightarrow B$ is continuous if and only if both restrictions $f|_{A_0}$ and $f|_{A_1}$ are continuous. $\square$

Let X and Y be topological spaces. Recall that a map $f : X \rightarrow Y$ is a *homeomorphism* if there exists a map $g : Y \rightarrow X$ such that $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$. We can use homotopies to define the following weaker notion:

Definition 1.1.5. Let X and Y be topological spaces. A map $f : X \rightarrow Y$ is a *homotopy equivalence* if there exists a map $g : Y \rightarrow X$ such that $f \circ g \simeq \text{id}_Y$ and $g \circ f \simeq \text{id}_X$. A map g with this property is then called a *homotopy inverse* to f .

We say “a” (not “the”) homotopy inverse because, unlike the inverse of a bijection (eg., a homeomorphism), homotopy inverses, when they exist, are usually not unique.

Definition 1.1.6. Two spaces X and Y are *homotopy equivalent* if there exists a homotopy equivalence $X \rightarrow Y$. This shall also be denoted $X \simeq Y$.

Example 1.1.7. The inclusion $\{0\} \hookrightarrow \mathbb{R}^n$ is a homotopy equivalence.

Example 1.1.8. Let $S^{n-1} = \{x \in \mathbb{R}^n \mid |x| = 1\}$. Then the inclusion $S^{n-1} \hookrightarrow \mathbb{R}^n \setminus \{0\}$ is a homotopy equivalence. As a homotopy inverse we can use $x \mapsto x/|x|$.

Proof. If we write $f : S^{n-1} \rightarrow \mathbb{R}^n \setminus \{0\}$ for the inclusion and $g : \mathbb{R}^n \setminus \{0\} \rightarrow S^{n-1}$ for the putative homotopy inverse, then $gf = \text{id}_{S^{n-1}}$ so we may use the constant homotopy. The map $fg : \mathbb{R}^{n-1} \setminus \{0\} \rightarrow \mathbb{R}^{n-1} \setminus \{0\}$ sends x to $x/|x|$, and is homotopic to the identity using the straight line homotopy. Indeed, if $x \neq 0$ then the straight line from x to $x/|x|$ does not go through 0. $\square$

As the examples show, we can sometimes prove that two spaces are homotopy equivalent directly from the definition. We shall later prove that S^{n-1} is *not* homotopy equivalent to a one-point space, but (for $n > 1$) that requires more sophisticated tools.

Lemma 1.1.9. *Let $f_i : X \rightarrow Y$ and $g_i : Y \rightarrow Z$ be maps, for $i = 0, 1$. If $f_0 \simeq f_1$ and $g_0 \simeq g_1$, then $g_0 \circ f_0 \simeq g_1 \circ f_1$.*

More generally, if $A \subset X$ and $B \subset Y$ and $f_0 \simeq f_1 \text{ rel } A$, $g_0 \simeq g_1 \text{ rel } B$, and $f_0(A) = f_1(A) \subset B$, then $g_0 \circ f_0 \simeq g_1 \circ f_1 \text{ rel } A$.

Proof. Let $h : I \times X \rightarrow Y$ and $h' : I \times Y \rightarrow Z$ be homotopies from f_0 to f_1 and from g_0 to g_1 respectively. Then we can define a homotopy $H : I \times X \rightarrow Z$ by $H(t, x) = h'(t, h(t, x))$. This is a continuous function, because it can be written as the composition of the following continuous functions

$$I \times X \xrightarrow{\Delta \times \text{id}_X} I \times I \times X \xrightarrow{\text{id}_I \times h} I \times Y \xrightarrow{h'} Z,$$

where $\Delta : I \rightarrow I \times I$ denotes the diagonal. $\square$

Remark 1.1.10. We shall insist on always using $I \times X$ (and not $X \times I$) for the domain of homotopies. It is helpful to be consistent about this, because we shall soon need to consider homotopies in the case $X = I$. In this case a homotopy is a map $I \times I \rightarrow Y$, but we shall always use the *first* factor for the “homotopy parameter”, i.e. $h : I \times I \rightarrow Y$ is a homotopy from the path $s \mapsto h(0, s)$ to the path $s \mapsto h(1, s)$.

1.2. Definition of the fundamental group. The fundamental group is defined only for based spaces, i.e. spaces equipped with a chosen point:

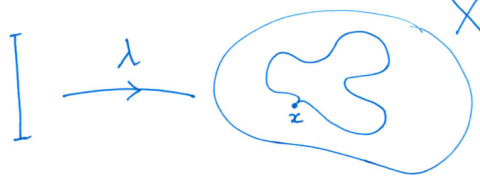
Definition 1.2.1. A *based space* is a pair (X, x) where X is a topological space and $x \in X$. A *based map* $(X, x) \rightarrow (Y, y)$ is a map $f : X \rightarrow Y$ such that $f(x) = y$.

The elements of the fundamental group will be homotopy classes of loops:

Definition 1.2.2. A *loop* in a based space (X, x) is a map $\lambda : I \rightarrow X$ such that $\lambda(0) = \lambda(1) = x$. We shall write

$$\pi_1(X, x) = \frac{\{\lambda : I \rightarrow X \mid \lambda(0) = \lambda(1) = x\}}{\simeq \text{ rel } \partial I}$$

for the set of loops modulo homotopy relative to ∂I .

FIGURE 3. Loop λ in X based at x **Definition 1.2.3.**

- (i) The *constant loop* in a based space (X, x) is the loop c_x defined by $c_x(t) = x$ for all $t \in I$.
- (ii) The *inverse* of a loop λ in (X, x) is the loop $\bar{\lambda}$ defined by $\bar{\lambda}(t) = \lambda(1-t)$, that is running the loop in the reverse direction.
- (iii) Let λ and λ' be loops in the same based space (X, x) . The *concatenation* is the loop $\lambda * \lambda'$ defined by

$$(\lambda * \lambda')(t) = \begin{cases} \lambda(2t) & \text{for } t \leq 1/2 \\ \lambda'(2t - 1) & \text{for } t \geq 1/2, \end{cases}$$

following the first loop at double speed, and then the second.

Concatenation of more than two loops is defined similarly. For example, if λ_0 , λ_1 , and λ_2 are loops, then we define a concatenated loop $\lambda_0 * \lambda_1 * \lambda_2$ given by $\lambda_0(3t)$ on $[0, 1/3]$, by $\lambda_1(3t - 1)$ on $[1/3, 2/3]$, and $\lambda_2(3t - 2)$ on $[2/3, 1]$.

The following lemma is the main input to show that concatenation defines a group structure on $\pi_1(X, x)$, where the neutral element is the constant loop at x and the inverse is given by taking the inverse loop in the above sense.

Lemma 1.2.4.

- (i) Let $\lambda_0, \lambda_1, \mu_0, \mu_1$ be loops in (X, x) . If $\lambda_0 \simeq \lambda_1 \text{ rel } \partial I$ and $\mu_0 \simeq \mu_1 \text{ rel } \partial I$, then $\lambda_0 * \mu_0 \simeq \lambda_1 * \mu_1 \text{ rel } \partial I$.
- (ii) Let $\lambda, \lambda',$ and λ'' be loops in (X, x) . Then $(\lambda * \lambda') * \lambda'' \simeq \lambda * \lambda' * \lambda'' \simeq \lambda * (\lambda' * \lambda'') \text{ rel } \partial I$.
- (iii) Let λ be a loop in (X, x) . Then $\lambda * \bar{\lambda} \simeq c_x \simeq \bar{\lambda} * \lambda \text{ rel } \partial I$.
- (iv) Let λ be a loop in (X, x) . Then $c_x * \lambda \simeq \lambda \simeq \lambda * c_x \text{ rel } \partial I$.
- (v) Let $f : (X, x) \rightarrow (Y, y)$ be a based map. Then $f \circ (\lambda * \mu) = (f \circ \lambda) * (f \circ \mu)$.

Proof sketch. To see (i) we pick homotopies h from λ_0 to λ_1 and h' from μ_0 to μ_1 , and define $H : I \times I \rightarrow X$ by $H(t, s) = h(t, 2s)$ for $s \leq 1/2$ and $H(t, s) = h'(t, 2s - 1)$ for $s \geq 1/2$. This function is continuous by the gluing property for continuous maps, and is a homotopy from $\lambda_0 * \mu_0$ to $\lambda_1 * \mu_1$ relative to ∂I .

The remaining four properties are proved in a similar way. $\square$

Definition 1.2.5. The binary operation $*$ on the set $\pi_1(X, x)$ is given by $[\lambda] * [\lambda'] = [\lambda * \lambda']$. Often we shall simply write $[\lambda][\lambda']$ for $[\lambda] * [\lambda']$.

This is then *well defined*: given two classes $\alpha, \beta \in \pi_1(X, x)$ we may choose loops λ, λ' with $\alpha = [\lambda]$ and $\beta = [\lambda']$ and define $\alpha * \beta = [\lambda * \lambda']$. Lemma 1.2.4(i) shows that the result is independent of choice of representatives λ and λ' . That lemma also implies the following.

Corollary 1.2.6. *This operation defines a group structure on the set $\pi_1(X, x)$. The identity element is $e = [c_x]$ and inverses are given by $[\lambda]^{-1} = [\bar{\lambda}]$.*

Definition 1.2.7. The *fundamental group* of (X, x) is the set $\pi_1(X, x)$ equipped with the group structure from Corollary 1.2.6. This group shall also be denoted $\pi_1(X, x)$.

Our first computations of fundamental groups will be done in Section 1.6.

1.3. Induced maps on the fundamental groups. Lemma 1.1.9 shows that the following is well defined.

Definition 1.3.1. For a map $f : (X, x) \rightarrow (Y, y)$ of based spaces, define a map of sets

$$f_* : \pi_1(X, x) \rightarrow \pi_1(Y, y)$$

by $f_*([\lambda]) = [f \circ \lambda]$.

Theorem 1.3.2.

- (i) *For a based map f as above, $f_* : \pi_1(X, x) \rightarrow \pi_1(Y, y)$ is a group homomorphism.*
- (ii) *If $f \simeq f' \text{ rel } \{x\}$, then $f_* = (f')_*$.*
- (iii) *If $\text{id}_X : (X, x) \rightarrow (X, x)$ is the identity map, then $(\text{id}_X)_* : \pi_1(X, x) \rightarrow \pi_1(X, x)$ is the identity homomorphism.*
- (iv) *If $f : (X, x) \rightarrow (Y, y)$ and $g : (Y, y) \rightarrow (Z, z)$, then the two homomorphisms $(g \circ f)_*$ and $g_* \circ f_* : \pi_1(X, x) \rightarrow \pi_1(Z, z)$ are equal.*

Proof. This follows from Lemma 1.2.4(v), Lemma 1.1.9, and the properties $\text{id}_X \circ \lambda = \lambda$ and $(g \circ f) \circ \lambda = g \circ (f \circ \lambda)$. $\square$

1.4. Paths and change of basepoint. It will sometimes be necessary to change basepoint, and to relate the fundamental group at different basepoints, we will use paths:

Definition 1.4.1. Let X be a topological space and $x_0, x_1 \in X$. A *path* from x_0 to x_1 is a map $\lambda : I \rightarrow X$ with $\lambda(i) = x_i$ for $i = 0, 1$. In that case the *inverse path* $\bar{\lambda}$ is the path from x_1 to x_0 defined by $\bar{\lambda}(t) = \lambda(1 - t)$. If μ is a path from x_1 to x_2 , then the *concatenation* $\lambda * \mu$ is the path from x_0 to x_2 defined by

$$(\lambda * \mu)(t) = \begin{cases} \lambda(2t) & \text{for } t \leq 1/2 \\ \mu(2t - 1) & \text{for } t \geq 1/2. \end{cases}$$

Concatenation of more than two paths is defined similarly. For example, if λ_0 , λ_1 , and λ_2 are paths with $\lambda_0(1) = \lambda_1(0)$ and $\lambda_1(1) = \lambda_2(0)$, then we define a concatenated path $\lambda_0 * \lambda_1 * \lambda_2$ given by $\lambda_0(3t)$ on $[0, 1/3]$, by $\lambda_1(3t - 1)$ on $[1/3, 2/3]$, and $\lambda_2(3t - 2)$ on $[2/3, 1]$.

A loop in (X, x) is then the special case of a path from x to x , and Lemma 1.2.4 may be generalized to paths in the following way. The only modification of the statements is that concatenation of paths is only defined when endpoints agree, which must be added as an extra assumption.

Lemma 1.4.2. *The following properties hold for the concatenation of paths.*

- (i) Let $\lambda_0, \lambda_1, \mu_0, \mu_1$ be paths in (X, x) such that $\lambda_0(1) = \mu_0(0)$ and $\lambda_1(1) = \mu_1(0)$. If $\lambda_0 \simeq \lambda_1 \text{ rel } \partial I$ and $\mu_0 \simeq \mu_1 \text{ rel } \partial I$, then $\lambda_0 * \mu_0 \simeq \lambda_1 * \mu_1 \text{ rel } \partial I$.
- (ii) Let $\lambda, \lambda',$ and λ'' be paths in X , with $\lambda(1) = \lambda'(0)$ and $\lambda'(1) = \lambda''(0)$. Then $(\lambda * \lambda') * \lambda'' \simeq \lambda * \lambda' * \lambda'' \simeq \lambda * (\lambda' * \lambda'') \text{ rel } \partial I$.
- (iii) Let λ be a path in X . Then $\lambda * \bar{\lambda} \simeq c_{\lambda(0)}$ and $\bar{\lambda} * \lambda \simeq c_{\lambda(1)} \text{ rel } \partial I$.
- (iv) Let λ be a path in X . Then $c_{\lambda(0)} * \lambda \simeq \lambda \simeq \lambda * c_{\lambda(1)} \text{ rel } \partial I$.
- (v) Let $f : X \rightarrow Y$ be a map and let λ, μ be paths in X with $\lambda(1) = \mu(0)$. Then $f \circ (\lambda * \mu) = (f \circ \lambda) * (f \circ \mu)$.

Proof. Similar to Lemma 1.2.4. □

Definition 1.4.3. Let $\mu : I \rightarrow X$ be a path from x_0 to x_1 and define $\mu_* : \pi_1(X, x_1) \rightarrow \pi_1(X, x_0)$ by

$$\mu_*([\lambda]) = [\mu * \lambda * \bar{\mu}].$$

See Figure 4 for an illustration. This is well defined by the previous lemma.

Lemma 1.4.4.

- (i) For $\mu : I \rightarrow X$ as above, $\mu_* : \pi_1(X, x_1) \rightarrow \pi_1(X, x_0)$ is a group homomorphism, depending only on the homotopy class of μ relative to ∂I .
- (ii) $(c_x)_* : \pi_1(X, x) \rightarrow \pi_1(X, x)$ is the identity homomorphism.
- (iii) If $\mu' : I \rightarrow X$ is a second path, from x_1 to x_2 , then the two homomorphisms $(\mu * \mu')_*$ and $\mu_* \circ \mu'_*$ from $\pi_1(X, x_2)$ to $\pi_1(X, x_0)$ are equal.

Proof. Follows from Lemma 1.4.2. □

Theorem 1.4.5. For any path $\mu : I \rightarrow X$ from x_0 to x_1 , the homomorphism $\mu_* : \pi_1(X, x_1) \rightarrow \pi_1(X, x_0)$ is an isomorphism.

Proof. Since $\bar{\mu} * \mu$ and $\mu * \bar{\mu}$ are both homotopic to constant paths relative to ∂I , Lemma 1.4.4 shows that $\bar{\mu}_* \circ \mu_* = (\bar{\mu} * \mu)_* = (c_{x_1})_* = 1 : \pi_1(X, x_1) \rightarrow \pi_1(X, x_1)$ and similarly $\mu_* \circ \bar{\mu}_* = 1 : \pi_1(X, x_0) \rightarrow \pi_1(X, x_0)$. Hence μ_* is an isomorphism with inverse $\bar{\mu}_*$. □

1.5. Homotopy invariance. We would like to show homotopy invariance of the fundamental group, or in other words that a homotopy equivalence $f : X \rightarrow Y$ induces an isomorphism between the fundamental groups $\pi_1(X, x)$ and $\pi_1(Y, f(x))$. Because a homotopy inverse g for f only satisfies that $g \circ f \simeq \text{id}_X$, we cannot assume that $g(f(x))$ equals x again, so a homotopy inverse will define a map from $\pi_1(Y, f(x))$ to $\pi_1(X, g(f(x)))$ which is the reason we will need to be careful with basepoints when we tackle the question of homotopy invariance. In the same way, because homotopies need not preserve basepoints, homotopic maps will only induce the same map on the fundamental groups up to basepoint issues, as made precise by the following proposition:

Proposition 1.5.1. Let $h : I \times X \rightarrow Y$ be a homotopy from f_0 to f_1 . Let $x \in X$ and define $\rho : I \rightarrow Y$ by $\rho(t) = h(t, x)$. Then ρ is a path from $y_0 = f_0(x)$ to $y_1 = f_1(x)$, and in this situation the homomorphism $(f_0)_* : \pi_1(X, x) \rightarrow \pi_1(Y, y_0)$ agrees with the composition

$$\pi_1(X, x) \xrightarrow{(f_1)_*} \pi_1(Y, y_1) \xrightarrow{\rho_*} \pi_1(Y, y_0),$$

i.e. $(f_0)_* = \rho_* \circ (f_1)_*$.

Proof. Let $H : I \times I \rightarrow Y$ be the map given by $H(t, s) = \rho(\max(0, s + (t - 1)))$. Then H is a homotopy from the constant path $c = c_{y_0}$ to ρ . The homotopy H is relative to $\{0\} \subset I$ but not relative to $\{1\}$; in fact $H(t, 1) = \rho(t)$. Then define $\bar{H}(t, s) = H(t, 1 - s)$, which is then a homotopy from c to $\bar{\rho}$, which is relative to $\{1\}$ but not $\{0\}$.

For a loop λ in (X, x) we may then define a homotopy from $c * (f_0 \circ \lambda) * \bar{c}$ to $\rho * (f_1 \circ \lambda) * \bar{\rho}$ by

$$(t, s) \mapsto \begin{cases} H(t, 3s) & \text{for } s \in [0, 1/3] \\ h(t, \lambda(3s - 1)) & \text{for } s \in [1/3, 2/3] \\ \bar{H}(t, 3s - 2) & \text{for } s \in [2/3, 1]. \end{cases}$$

This homotopy is relative to ∂I and proves that $\rho_* \circ (f_1)_*([\lambda]) = [\rho * (f_1 \circ \lambda) * \bar{\rho}] = [c * (f_0 \circ \lambda) * \bar{c}] = c_* \circ (f_0)_*([\lambda]) = (f_0)_*([\lambda])$. $\square$

Corollary 1.5.2. *In the situation of Proposition 1.5.1, the homomorphism $(f_0)_*$ is an isomorphism if and only if $(f_1)_*$ is an isomorphism.*

In particular, if $f : X \rightarrow X$ is homotopic to the identity, then $f_ : \pi_1(X, x) \rightarrow \pi_1(X, f(x))$ is an isomorphism for any $x \in X$.*

Theorem 1.5.3 (Homotopy invariance of the fundamental group). *Let $f : X \rightarrow Y$ be a homotopy equivalence. Then $f_* : \pi_1(X, x) \rightarrow \pi_1(Y, f(x))$ is an isomorphism.*

Proof. Write $y = f(x)$ and choose a homotopy inverse $g : Y \rightarrow X$. Then $g \circ f \simeq \text{id}_X$ implies that the composition

$$(1.1) \quad (g \circ f)_* = g_* \circ f_* : \pi_1(X, x) \rightarrow \pi_1(Y, y) \rightarrow \pi_1(X, g(y))$$

is an isomorphism and hence that f_* is injective and g_* is surjective.

Conversely, $f \circ g \simeq \text{id}_Y$ implies that

$$(f \circ g)_* = f_* \circ g_* : \pi_1(Y, y) \rightarrow \pi_1(X, g(y)) \rightarrow \pi_1(Y, f \circ g(y))$$

is an isomorphism and hence that $g_* : \pi_1(Y, y) \rightarrow \pi_1(X, g(y))$ is injective. But we just saw from the composition (1.1) that it is also surjective, so it must be bijective. But then using the composition (1.1) again shows that $f_* : \pi_1(X, x) \rightarrow \pi_1(Y, y)$ is also bijective. $\square$

The theorem may be used to show that two topological spaces are not homotopy equivalent, once we develop more tools for *calculating* fundamental groups. For example, if X and Y are path connected and the groups $\pi_1(X, x)$ and $\pi_1(Y, y)$ are not isomorphic for some $x \in X$ and $y \in Y$ (and hence not isomorphic for any x and y), then there cannot exist a homotopy equivalence $X \rightarrow Y$.

1.6. The fundamental group of the circle. We will in this section do our first computations of fundamental groups.

Definition 1.6.1. A topological space X is *contractible* if it is homotopy equivalent to a one-point space.

Proposition 1.6.2. *If X is contractible, then $\pi_1(X, x)$ is a trivial group for every $x \in X$.*

Proof. If we write $P = \{p\}$ for a one-point space, then there is only one loop in P and hence $\pi_1(P, p)$ must be a trivial group. If $f : X \rightarrow P$ is a homotopy equivalence and $x \in X$, then $f_* : \pi_1(X, x) \rightarrow \pi_1(P, p)$ is an isomorphism. $\square$

Definition 1.6.3. A space X is *simply connected* if it is path connected and if $\pi_1(X, x)$ is trivial for all $x \in X$ (equivalently, for one $x \in X$).

Don't forget that a path connected space is a *non-empty* space in which any two points can be connected by a continuous path. Hence we have implications

contractible $\Rightarrow$ simply connected $\Rightarrow$ path connected $\Rightarrow$ non-empty.

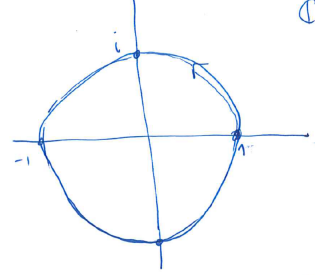
We shall soon see that S^1 is not simply connected, and later we show that S^2 is simply connected but not contractible.

Lemma 1.6.4. *Let X be simply connected and $x_0, x_1 \in X$. Then any two paths $\lambda, \lambda' : I \rightarrow X$ from x_0 to x_1 are homotopic relative to ∂I .*

Proof. The concatenation $\rho = \lambda' * \bar{\lambda}$ is a loop in (X, x_0) and hence by assumption is homotopic to c_{x_0} relative to ∂I . Since $\lambda' \simeq \rho * \lambda \text{ rel } \partial I$ we deduce $\lambda' \simeq c_{x_0} * \lambda \simeq \lambda \text{ rel } \partial I$. $\square$

Our first non-trivial calculation is that the fundamental group of a circle is isomorphic to $(\mathbb{Z}, +)$, and in particular the circle is not contractible. More precisely, we have the following.

Theorem 1.6.5. *Let $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$ be the circle. Then $\pi_1(S^1, 1)$ is an infinite cyclic group. A generator is given by the homotopy class rel ∂I of the loop $\lambda : I \rightarrow S^1$ given by $\lambda(t) = \exp(2\pi it)$.*



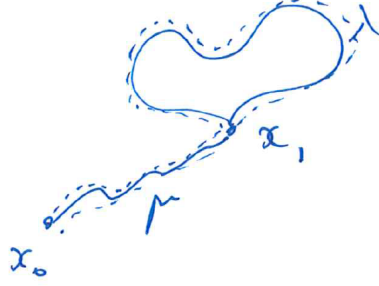
In order to make a rigorous proof of the isomorphism $\pi_1(S^1, 1) \cong \mathbb{Z}$, we first need to define a homomorphism $\pi_1(S^1, 1) \rightarrow \mathbb{Z}$. The key ingredient is the following result, of which you may have seen a version in a course on complex function theory in connection with defining *winding numbers*. It is a special case of a more general statement which we shall formulate and prove shortly (see Theorems 2.2.1 and 2.2.2).

Lemma 1.6.6.

- (i) *For any path $\lambda : I \rightarrow S^1$ with $\lambda(0) = 1$, there exists precisely one path $\mu : I \rightarrow \mathbb{R}$ with $\mu(0) = 0$ and $\lambda(t) = \exp(2\pi i\mu(t))$.*
- (ii) *For any homotopy of paths $h : I \times I \rightarrow S^1$ relative to ∂I with $h(t, 0) = 1$ for all t , there exists a unique lift to a homotopy $H : I \times I \rightarrow \mathbb{R}$ relative to ∂I , which has $h(t, s) = \exp(2\pi iH(t, s))$ and $H(t, 0) = 0$ for all t .*

We shall often say that the path μ is a “lift” of λ , and similarly that the homotopy H is a lift of h . The map $\mathbb{R} \rightarrow S^1$ taking x to $e^{2\pi ix}$ is represented in Figure 5, with the real line wrapping around the circle by projecting it to the circle as shown in the picture. The path λ is a path winding around the circle, while μ lives above it in the real line, starting at some point lying above the basepoint of the circle.

We will start by explaining how to use this lemma to calculate the fundamental group of the circle.

FIGURE 4. The map μ_* of Definition 1.4.3

Proof of Theorem 1.6.5 using 1.6.6. We wish to define a function

$$\phi : \pi_1(S^1, 1) \rightarrow \mathbb{Z}$$

by $\phi([\lambda]) = \mu(1)$, where $\mu : I \rightarrow \mathbb{R}$ is the path with $\mu(0) = 0$ and $\exp(2\pi i \mu(t)) = \lambda(t)$ whose existence and uniqueness is asserted in Lemma 1.6.6. We must prove that ϕ is well defined, i.e that $\mu(1)$ is independent of choice of loop λ representing $[\lambda] \in \pi_1(S^1, 1)$, so assume $[\lambda] = [\lambda'] \in \pi_1(S^1, 1)$ with $\mu'(0) = 0$ and $\lambda'(t) = \exp(2\pi i \mu'(t))$. Then there exists a homotopy h from λ to λ' relative to ∂I , so by the second part of the theorem there is a homotopy from μ to μ' relative to ∂I and in particular $\mu'(1) = \mu(1)$, as desired.

We then claim that ϕ is a homomorphism of groups. Let λ and λ' be two loops in $(S^1, 1)$ and write $\lambda(t) = \exp(2\pi i \mu(t))$ and $\lambda'(t) = \exp(2\pi i \mu'(t))$ with $\mu(0) = \mu'(0) = 0$. Then $\phi([\lambda]) = \mu(1)$ and $\phi([\lambda']) = \mu'(1)$. Now we may define a path $\mu'' : I \rightarrow \mathbb{R}$ from $\mu(1)$ to $\mu(1) + \mu'(1)$ as $\mu''(t) = \mu(1) + \mu'(t)$, and since $\mu(1) \in \mathbb{Z}$ we have $\exp(2\pi i \mu''(t)) = \lambda'(t)$ and hence by Lemma 1.2.4(v) we get

$$\exp(2\pi i (\mu * \mu'')(t)) = (\lambda * \lambda')(t),$$

so by definition we have $\phi([\lambda * \lambda']) = (\mu * \mu'')(1) = \mu''(1) = \mu(1) + \mu'(1) = \phi([\lambda]) + \phi([\lambda'])$.

To see that ϕ is injective, let $[\lambda] \in \text{Ker}(\phi)$ for a loop λ . That $\phi([\lambda]) = 0$ means that the lift μ of λ is in fact a *loop* in $(\mathbb{R}, 0)$ (not just a path in $\mathbb{R}$ starting at 0). Since $\mathbb{R}$ is contractible, the group $\pi_1(\mathbb{R}, 0)$ is trivial and hence $[\lambda] = p_*([\mu]) = e$. The loop $\lambda(t) = \exp(2\pi i t)$ lifts to $\mu(t) = t$ and hence maps to $1 \in \mathbb{Z}$ which is a generator, so ϕ is also surjective. $\square$

From the homotopy equivalence $S^1 \simeq \mathbb{R}^2 \setminus \{0\}$ we deduce the following.

Corollary 1.6.7. *For any $x \in \mathbb{R}^2 \setminus \{0\}$ the group $\pi_1(\mathbb{R}^2 \setminus \{0\}, x)$ is infinite cyclic.*

Corollary 1.6.8. *The spaces S^1 and $\mathbb{R}^2 \setminus \{0\}$ are not simply connected, and in particular not contractible.*

The following example shows the utility of associating not only groups to based spaces but also homomorphisms to maps of based spaces.

Corollary 1.6.9. *Let $A \subset \mathbb{R}^2$ be a non-empty bounded subset. Then $\mathbb{R}^2 \setminus A$ is not simply connected, and in particular not contractible.*

Proof. Write $X = \mathbb{R}^2 \setminus A$ and pick $y \in A$. Since A is bounded we may pick $R > 0$ such that A is a subset of the closed ball of radius R around y . Then define $\phi : \mathbb{R}^2 \setminus \{0\} \rightarrow X$ by $\phi(z) = y + \frac{|z|+R}{|z|}z$ and $\psi : X \rightarrow \mathbb{R}^2 \setminus \{0\}$ by $\psi(z) = z - y$. Then the composition $\psi \circ \phi : \mathbb{R}^2 \setminus \{0\} \rightarrow \mathbb{R}^2 \setminus \{0\}$ is the map given by $z \mapsto \frac{|z|+R}{|z|}z$,

which is homotopic to the identity using the straight-line homotopy. Therefore the composition

$$\pi_1(\mathbb{R}^2 \setminus \{0\}, (1, 0)) \xrightarrow{\phi_*} \pi_1(X, \phi(1, 0)) \xrightarrow{\psi_*} \pi_1(\mathbb{R}^2 \setminus \{0\}, \psi \circ \phi(1, 0))$$

is an isomorphism. Assuming for contradiction that X is simply connected, the group $\pi_1(X, \phi(1, 0))$ would be trivial, and hence the composition of the two group homomorphisms would be trivial. This is impossible, since we have seen above that the group $\pi_1(\mathbb{R}^2 \setminus \{0\}, (1, 0))$ is infinite cyclic. $\square$

2. COVERING SPACES (WEEKS 2-3)

We have last week defined the fundamental group of a space. Our next goal is to see how it relates to the notion of *covering maps* (or covering spaces).

The corresponding section in Hatcher is 1.3 and in May is various parts of his Chapter 3 (although the setting there is somewhat different).

2.1. Covering maps. The proof of Lemma 1.6.6 relies on compactness of I and $I \times I$, and is not too difficult. In order not to repeat the same argument twice, we shall work in more generality.

Definition 2.1.1. A map $p : X \rightarrow Y$ is a *trivial covering map* if there exists a discrete topological space F and a continuous map $q : X \rightarrow F$ such that the map $\phi : X \rightarrow Y \times F$ given by $\phi(x) = (p(x), q(x))$ is a homeomorphism.

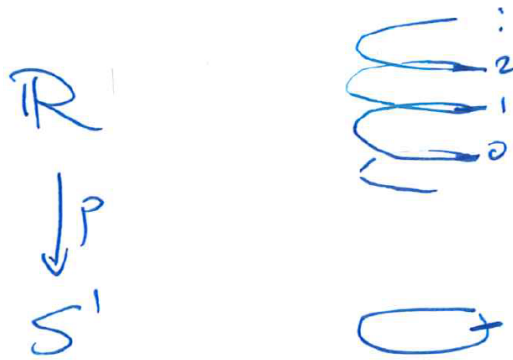


FIGURE 5. The map $\mathbb{R} \rightarrow S^1$ taking x to $e^{2\pi ix}$

A trivial covering map may admit more than one map $q : X \rightarrow F$ (exercise: find examples). The relationship between the maps is often summarized in the commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{\phi} & Y \times F \\ p \downarrow & & \downarrow \text{projection} \\ Y & \xlongequal{\quad} & Y. \end{array}$$

Remark 2.1.2. In practice, one often verifies that $p : X \rightarrow Y$ is a trivial covering map by exhibiting open sets $U_\alpha \subset X$, one for each $\alpha \in F$, such that

- $U_\alpha \cap U_\beta = \emptyset$ for $\alpha \neq \beta$, and
- $\bigcup_{\alpha \in F} U_\alpha = X$, and
- $p|_{U_\alpha} : U_\alpha \rightarrow Y$ is a homeomorphism for all $\alpha \in F$.

In that case p is a trivial covering map because we may set $q(x) = \alpha$ if $x \in U_\alpha$. (Exercise: fill in the details!)

Definition 2.1.3. Let $p : X \rightarrow Y$ be a map. Then a subset $A \subset Y$ is called *evenly covered* if the restriction

$$p|_{p^{-1}(A)} : p^{-1}(A) \rightarrow A$$

is a trivial covering map.

The map p is a *covering map* if any $y \in Y$ admits a neighborhood which is an evenly covered subset.

Recall that a “neighborhood of $y \in Y$ ” is a subset $A \subset Y$ such that there exists an open subset $U \subset Y$ with $y \in U \subset A$. It is easy to see that any subset of an evenly covered $A \subset Y$ is also evenly covered. Therefore, if y admits a neighborhood which is evenly covered, then it also admits an open neighborhood which is evenly covered.

Some textbooks speak of “covering spaces” instead of covering maps, but we wish to emphasize that it is a condition on the map $p : X \rightarrow Y$ and not only a condition on its domain.

Lemma 2.1.4. *The map $p : \mathbb{R} \rightarrow S^1$ given by $p(t) = \exp(2\pi it)$ is a covering map.*

(See Figure 5 for an illustration of that map.)

Proof. We claim that if $a < b$ and $|b - a| < 1$, then the subspace $A_{a,b} = p([a, b]) \subset S^1$ is evenly covered. That will prove the claim since any element of S^1 will be contained in the interior of either $A_{0,1/2}$, $A_{1/4,3/4}$, $A_{1/2,1}$, or $A_{3/4,5/4}$. (These subsets correspond to the “top half”, “left half”, “bottom half” and “right half” of the circle, respectively.)

To prove the claim, we first observe that two points in $\mathbb{R}$ are mapped to the same point in S^1 if and only if they differ by an integer, from which we deduce

$$p^{-1}(A_{a,b}) = \bigcup_{n \in \mathbb{Z}} [a + n, b + n].$$

Since $|a - b| < 1$, these intervals are mutually disjoint, and each restriction $p|_{[a+n, b+n]} : [a + n, b + n] \rightarrow A_{a,b} \subset S^1$ is a continuous bijection and hence a homeomorphism since $[a, b]$ is compact and S^1 is Hausdorff. Each interval $[a + n, b + n]$ is also open in $p^{-1}(A_{a,b})$, being the intersection with $(a + n - \varepsilon, b + n + \varepsilon) \subset \mathbb{R}$ for sufficiently small ε . Hence $p|_{p^{-1}(A_{a,b})} : p^{-1}(A_{a,b}) \rightarrow A_{a,b}$ is a trivial covering map by Remark 2.1.2. $\square$

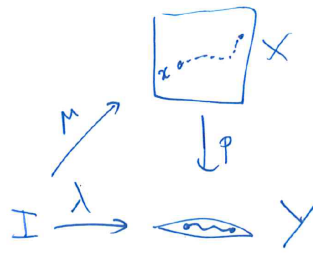
Remark 2.1.5. If we remove the condition in Definition 2.1.1 that the space F have the discrete topology, we get the notion of a *trivial fiber bundle*. A *locally trivial fiber bundle* is a map $p : X \rightarrow Y$ with the property that every $y \in Y$ has a neighborhood $A \subset Y$ such that $p|_{p^{-1}(A)} : p^{-1}(A) \rightarrow A$ is a trivial fiber bundle.

Clearly any covering map is a locally trivial fiber bundle. In fact a locally trivial fiber bundle $p : X \rightarrow Y$ is a covering map if and only if the subsets $p^{-1}(y) \subset X$ are discrete (i.e. the subspace topology from Y agrees with the discrete topology) for all $y \in Y$.

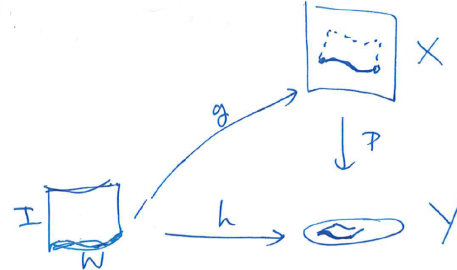
2.2. Lifting paths and homotopies. Our goal is to prove the following two important results for covering maps, generalizing Lemma 1.6.6 (i) and (ii), and hence finishing the calculation of $\pi_1(S^1, 1)$.

Theorem 2.2.1. *Let $p : X \rightarrow Y$ be a covering map, $\lambda : I \rightarrow Y$ a path and let $x \in X$ be such that $p(x) = \lambda(0)$. Then there exists precisely one path $\mu : I \rightarrow X$ with $p \circ \mu = \lambda$ and $\mu(0) = x$.*

Theorem 2.2.2. *Let $p : X \rightarrow Y$ be a covering map, W a topological space and $h : I \times W \rightarrow Y$ a homotopy. Let $g : W \rightarrow X$ be such that $p \circ g = h(0, -)$. Then there exists precisely one homotopy $H : I \times W \rightarrow X$ with $p \circ H = h$ and $H(0, -) = g$. Moreover, if h is a homotopy relative to a subspace $V \subset W$, then so is H .*



Path lifting (Theorem 2.2.1)



Homotopy lifting (Theorem 2.2.2)

The path or homotopy H is often called a “lift” of h , and written in a diagram

$$(2.1) \quad \begin{array}{ccc} \{0\} \times W & \xrightarrow{g} & X \\ \text{inclusion} \downarrow & \nearrow H & \downarrow p \\ I \times W & \xrightarrow{h} & Y \end{array}$$

In this diagram, the solid arrows are the “input” and the dashed arrow is the “output”: the theorem asserts that for any given solid arrows, forming a commutative square diagram and with p a covering map, there is precisely one way to fill in the dashed arrow to form a commutative diagram.

Remark 2.2.3. Before giving the proof, we make the following evident but important remarks. Let $\phi : X \rightarrow X'$ and $\psi : Y \rightarrow Y'$ be homeomorphisms, and let $p' = \psi \circ p \circ (\phi^{-1}) : X' \rightarrow Y'$. Then the theorems hold for p if and only if they hold for p' . More precisely, there exists a lift in diagram (2.1) if and only if there exists one after replacing X by X' , Y by Y' , p by p' , g by $\phi \circ g$, and h by $\psi \circ h$, and the lift is unique if and only if it is so after these replacements.

Similarly, given a homeomorphism $\eta : A \rightarrow I \times W$ we may replace $I \times W$ by A , H by $H \circ \eta$, $\{0\} \times W$ by $A_0 := \eta^{-1}(\{0\} \times W)$, and g by $g \circ \eta|_{A_0}$.

We start by showing that it is enough to prove the non-relative version of Theorem 2.2.2:

Lemma 2.2.4. *The relative part of Theorem 2.2.2 is a consequence of the non-relative part of the statement.*

Proof. Suppose that $h : I \times W \rightarrow Y$ is a homotopy relative to $V \subset W$ and $H : I \times W \rightarrow X$ is the unique lift of h given by the first part of Theorem 2.2.2. Then the restriction of H to $I \times V$ is the unique lift of the restriction of h to $I \times V$. But the constant lift $H'(t, v) = g(v)$ is a valid lift, and hence must be equal to H on $I \times V$ by uniqueness of the lift. It follows that H is also a homotopy relative to V . $\square$

Lemma 2.2.5. *Theorems 2.2.1 and 2.2.2 hold when p is a trivial covering map.*

Proof. By Remark 2.2.3 we may assume $X = Y \times F$ for a discrete topological space F , and $p(x, \alpha) = x$. This case is obvious: we have $g(0, w) = (h(0, w), q(w))$ for

some continuous $q : W \rightarrow F$, and may set $H(t, w) = (h(t, w), q(w))$. This is the only possibility, since for each $w \in W$ the composition

$$I \xrightarrow{(-, w)} I \times W \xrightarrow{H} Y \times F \xrightarrow{\text{proj}} F$$

must be a continuous function $I \rightarrow F$ with value $q(w)$ at $0 \in I$.

This proves Theorem 2.2.2, and Theorem 2.2.1 is the special case where W is a one-point space $\{w\}$ (at least up to the obvious homeomorphism $I \times \{w\} \cong I$, cf. Remark 2.2.3). $\square$

Lemma 2.2.6. *The theorems above hold in the case where $h(I \times W) \subset Y$ is evenly covered.*

Proof. By the same reasoning as in Remark 2.2.3, we may replace Y by $A = h(I \times W)$, X by $p^{-1}(A)$, and p by its restriction to $p^{-1}(A)$, which reduces to the case of a trivial cover. $\square$

Corollary 2.2.7. *In the situation of Theorem 2.2.2, suppose in addition given a subinterval $[a, b] \subset I$ such that $h([a, b] \times W) \subset Y$ is evenly covered, and a map $H_a : [0, a] \times W \rightarrow X$ with $p \circ H_a = h|_{[0, a] \times W}$. Then there exists precisely one map $H_b : [0, b] \times W \rightarrow X$ with $H_b|_{[0, a] \times W} = H_a$ and $p \circ H_b = h|_{[0, b] \times W}$.*

In the context of diagram (2.1), H_a is a “partial lift”, since it is only defined on part of $I \times W$. The corollary gives an extension to a slightly larger part of $I \times W$.

Proof. By the gluing property for continuous functions, it is equivalent to ask for the restriction $H' = H_b|_{[a, b] \times W} : [a, b] \times W \rightarrow X$, with $p \circ H' = h|_{[a, b] \times W}$ and $H'(a, w) = H_a(a, w)$ for all $w \in W$. Then apply Lemma 2.2.6 with $I \times W$ replaced by the homeomorphic space $[a, b] \times W$ and g replaced by $H_a(a, -)$. $\square$

Lemma 2.2.8. *Let $\mathcal{U}$ be an open cover of $[0, 1]$ (i.e. a family of open subsets of $[0, 1]$ whose union is all of $[0, 1]$). Then there exists a natural number n such that for each $i = 1, \dots, n$ there exists an element $U_i \in \mathcal{U}$ such that $[(i-1)/n, i/n] \subset U_i$.*

You may have seen a version of this lemma in an analysis class (under the name that $\varepsilon = 1/n$ is a “Lebesgue number” for the covering).

Proof. Since the sets $U \in \mathcal{U}$ are open in $[0, 1]$ and cover, we may for each $x \in [0, 1]$ find $\varepsilon_x > 0$ and $U_x \in \mathcal{U}$ such that $[0, 1] \cap (x - \varepsilon_x, x + \varepsilon_x) \subset U_x$. As $x \in [0, 1]$ varies, the sets $V_x = [0, 1] \cap (x - \varepsilon_x/2, x + \varepsilon_x/2)$ form an open cover of $[0, 1]$ so by compactness there exists finitely many $x_1, \dots, x_N \in [0, 1]$ with $V_{x_1} \cup \dots \cup V_{x_N} = [0, 1]$. Let $\varepsilon = \min\{\varepsilon_{x_j} \mid 1 \leq j \leq N\}$. For any $y \in [0, 1]$ we then have $y \in V_{x_j}$ for some j , and then

$$(y - \varepsilon/2, y + \varepsilon/2) \subset (y - \varepsilon_{x_j}/2, y + \varepsilon_{x_j}/2) \subset (x_j - \varepsilon_{x_j}, x_j + \varepsilon_{x_j}),$$

and hence $[0, 1] \cap (y - \varepsilon/2, y + \varepsilon/2) \subset U_{x_j}$. If we then let n be a natural number large enough that $1/n < \varepsilon/2$, we have for all $i = 1, \dots, n$ that $[(i-1)/n, i/n] \subset (i/n - \varepsilon/2, i/n + \varepsilon/2)$ and hence $[(i-1)/n, i/n] \subset U_{x_j}$ for some j . $\square$

Proof of Theorem 2.2.1. Since each $\lambda(t) \in Y$ admits a neighborhood which is evenly covered, we may find a cover of $[0, 1]$ by open sets whose image in Y is evenly covered. Hence by Lemma 2.2.8 we may find a natural number n such that $\lambda([(i-1)/n, i/n]) \subset Y$ is evenly covered for all $i = 1, \dots, n$.

Let us now prove uniqueness of the lift $\mu : I \rightarrow X$ with $\mu(0) = x$ and $p \circ \mu = \lambda$. If μ' is another lift we shall prove by induction that $\mu|_{[0, i/n]} = \mu'|_{[0, i/n]}$ for all $i = 0, \dots, n$. The initial case $i = 0$ is clear because both maps must send 0 to x . If $\mu|_{[0, (i-1)/n]} = \mu'|_{[0, (i-1)/n]}$, we may use the uniqueness part of Corollary 2.2.7 with $a = (i-1)/n$ and $b = i/n$ (and W a one-point space) to see that $\mu|_{[0, i/n]} = \mu'|_{[0, i/n]}$.

Existence is similar: we inductively construct maps $\mu_i : I \rightarrow X$ with $\mu_i(0) = x$ and $p \circ \mu_i = \lambda|_{[0, i/n]}$, at the end of which we may set $\mu = \mu_n$. For $i = 0$ we simply let $\mu_0(0) = x$. If μ_{i-1} has been constructed, we may use the existence part of Corollary 2.2.7 with $a = (i-1)/n$ and $b = i/n$ and $U = W$ both being a one-point space to construct μ_i with the required properties. $\square$

Proof of Theorem 2.2.2. By Lemma 2.2.4, it is enough to prove the non-relative part of the statement. So let $h : I \times W \rightarrow Y$ be a homotopy.

Uniqueness of lifts H of h is now easy: for each $w \in W$ the path $\mu : H(-, w) : I \rightarrow X$ is a lift of $\lambda : h(-, w) : I \rightarrow Y$ with prescribed starting point $f(w)$, and by Theorem 2.2.1 such lifts are unique.

For existence we first consider the special case where each $t \in [0, 1]$ admits an open neighborhood $J_t \subset [0, 1]$ such that $h(J_t \times W) \subset Y$ is evenly covered. If we pick such J_t for each t they form an open cover of $[0, 1]$ so by Lemma 2.2.8 we may find a natural number n such that $h([(i-1)/n, i/n] \times W) \subset Y$ is evenly covered for each $i = 1, \dots, n$. Then lifts $H_i : [0, i/n] \times W \rightarrow X$ with $H_i(0, -) = g$ and $p \circ H_i = h|_{[0, i/n] \times W}$ exist by induction on i , using Corollary 2.2.7.

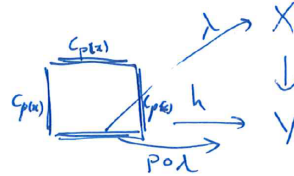
For the general case of existence, we first fix $w \in W$ and find for each $t \in I$ open neighborhoods $J_t \subset I$ of t and $U_t \subset W$ of w such that $h(J_t \times U_t) \subset Y$ is evenly covered. This is possible because p is a covering map and h is continuous, and $I \times W$ has the product topology. By compactness, the interval will be covered by finitely many $J_{t_1}, \dots, J_{t_N}$ and we let $U_w = \bigcap_{i=1}^N U_{t_i}$. Then the case of existence already considered may be applied to the restriction $h|_{I \times U_w} : I \times U_w \rightarrow Y$, which may therefore be lifted to a homotopy $H_w : I \times U_w \rightarrow X$ with $H_w(0, -) = g|_{U_w}$ and $p \circ H_w = h|_{I \times U_w}$. This proves that h may be lifted to a continuous homotopy “locally in W ”.

Finally, uniqueness implies that two local solutions H_w and $H_{w'}$ agree on $I \times (U_w \cap U_{w'})$, so they are the restrictions of a single continuous homotopy $H : I \times W \rightarrow X$ with the required properties. $\square$

We shall explore the consequences of these “path lifting” and “homotopy lifting” theorems more fully later, but let us give two simple applications (which we shall use later).

Theorem 2.2.9. *Let (X, x) and (Y, y) be based spaces and $p : X \rightarrow Y$ a covering map. Then the homomorphism $p_* : \pi_1(X, x) \rightarrow \pi_1(Y, y)$ is injective.*

Proof. Let $\alpha \in \pi_1(X, x)$ have $p_*(\alpha) = e \in \pi_1(Y, y)$. If we write $\alpha = [\lambda]$ for a loop λ in (X, x) , then by assumption the loop $p \circ \lambda$ in (Y, y) is homotopic to c_y rel ∂I . If $h : I \times I \rightarrow Y$ is a homotopy relative to ∂I with $h(0, s) = (p \circ \lambda)(s)$ and $h(1, s) = c_y(s)$ for all s , then Theorem 2.2.2 implies that there exists a homotopy $H : I \times I \rightarrow X$ relative to ∂I with $p \circ H = h$ and $H(0, s) = \lambda(s)$ for all s . Then $s \mapsto H(1, s)$ is a lift of c_y starting at $x \in X$ so uniqueness gives $H(1, s) = c_x(s)$ for all s . Hence H is a homotopy from λ to c_x relative to ∂I , so $\alpha = [f] = [c_x] = e \in \pi_1(X, x)$. $\square$



Corollary 2.2.10. *Let λ be a loop in (Y, y) . Then there exists a loop μ in (X, x) with $\lambda = p \circ \mu$ if and only if the homotopy class $[\lambda] \in \pi_1(Y, y)$ is in the image of $p_* : \pi_1(X, x) \rightarrow \pi_1(Y, y)$.*

Proof. Clearly if $\lambda = p \circ \mu$ for a loop μ in (X, x) , then $[\lambda] = p_*([\mu])$.

Conversely, if $[\lambda] = p_*(\mu)$ we may conclude that $p \circ \mu \simeq \lambda \text{ rel } \partial I$. Pick a homotopy $h : I \times I \rightarrow Y$ relative to ∂I and write it as $h = p \circ H$ for a homotopy $H : I \times I \rightarrow X$ with $H(0, s) = \mu(s)$ for all $s \in I$. This is possible by Theorem 2.2.2. Then $p \circ H(1, s) = \lambda(s)$, so $s \mapsto H(1, s)$ is a path in X lifting λ and starting at x . Since H is relative to ∂I we have $H(1, 1) = H(1, 0) = \mu(1) = x$, so this lift is indeed a loop in (X, x) . $\square$

It is instructive to compare Corollary 2.2.10 with Theorem 2.2.1, which implies that λ may be lifted to a *path* μ starting at x without any assumptions on the homotopy class of λ ; in general this path may not end at x of course.

2.3. Right action on the fiber. We shall generalize the method we used to calculate $\pi_1(S^1, 1) \cong \mathbb{Z}$, using the covering map $p : \mathbb{R} \rightarrow S^1$ with $p(t) = \exp(2\pi it)$.

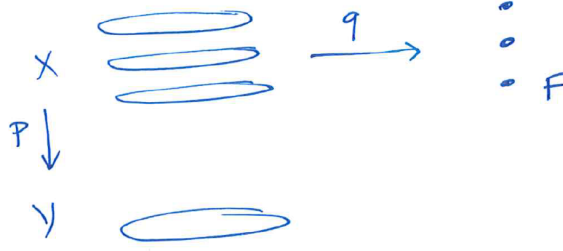


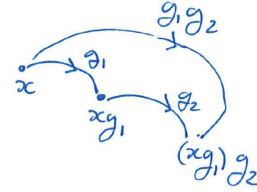
FIGURE 6. Example of a trivial covering

A good way to “determine” a group G is to find a suitable *action* of the group G on a set F . For convenience of notation we shall study *right actions*:

Definition 2.3.1. A *right action* of a group G on a set F is a function

$$\alpha : F \times G \rightarrow F$$

satisfying that $\alpha(x, e) = x$ and $\alpha(\alpha(x, g_1), g_2) = \alpha(x, g_1 g_2)$ for all $x \in F$ and all $g_1, g_2 \in G$. It is common to write actions as $x.g = \alpha(x, g)$, in which case the axioms read $x.e = x$ and $x.(g_1 g_2) = (x.g_1).g_2$.



Example 2.3.2. Set $F = \mathbb{R}$ and $G = \mathbb{Z}$. Then the translation map

$$\begin{aligned} T : \mathbb{R} \times \mathbb{Z} &\longrightarrow \mathbb{R} \\ (r, z) &\mapsto r + z \end{aligned}$$

defines a right action of $\mathbb{Z}$ on $\mathbb{R}$.

Definition 2.3.3. Let G be a group, F a set, and $\alpha : F \times G \rightarrow F$ a right action. We write again $x.g = \alpha(x, g)$.

- (i) The *orbit* of $x \in F$ is $x.G = \{y \in F \mid \exists g \in G \text{ with } y = x.g\} \subset F$.
- (ii) The *stabilizer* of $x \in F$ is the set (subgroup) $\{g \in G \mid x.g = x\} \leq G$.
- (iii) The action is *transitive* if it has precisely one orbit. In other words, F must be non-empty and for all $x, y \in F$ there must exist $g \in G$ with $x.g = y$.
- (iv) The action is *free* if the stabilizer of any element of F is the trivial group. In other words for all $x \in F$ and $g \in G \setminus \{e\}$ we must have $x.g \neq x$.

Example 2.3.4. The action of $\mathbb{Z}$ by translation on $\mathbb{R}$ of Example 2.3.2 is free but not transitive. For $x \in \mathbb{R}$, the orbit of x is the set $x.\mathbb{Z} = \{x + z \mid z \in \mathbb{Z}\}$.

Lemma 2.3.5. Let $\alpha : F \times G \rightarrow F$ be a right action, and let $x \in F$. Then the function

$$\begin{aligned} \alpha_x : G &\longrightarrow F \\ g &\mapsto \alpha(x, g) \end{aligned}$$

is injective if and only if α is free and surjective if and only if α is transitive.

Proof. Follows directly from definitions. $\square$

If the action is both free and transitive, we therefore get a *bijection* $\alpha_x : G \xrightarrow{\cong} F$ for any choice of element $x \in F$. We will use such identifications $G \cong F$ to compute fundamental groups G . In order to apply this strategy, we must construct useful actions of fundamental groups on sets. This is where covering maps enter the game.

Definition 2.3.6. Let (Y, y) be a based space and $p : X \rightarrow Y$ a covering map. The *fiber* of p is the set $F = p^{-1}(y) \subset X$.

Definition 2.3.7. Let (Y, y) , $p : X \rightarrow Y$, and $F \subset X$ be as above. For $[\lambda] \in \pi_1(Y, y)$, for a loop λ in (Y, y) , and $x \in F$, let $x.[\lambda] \in F$ be the element defined in the following way: let $\mu : I \rightarrow X$ be the (uniquely determined, by Theorem 2.2.1) path with $p \circ \mu = \lambda$ and $\mu(0) = x$. Then $p(\mu(1)) = \lambda(1) = y$, so $\mu(1) \in F$, and we let $x.[\lambda] = \mu(1)$.

It is not obvious that this “definition” really works: we must check that the formula for $x.[\lambda]$ indeed only depends on the homotopy class of λ . In fact we have the following.

Lemma 2.3.8. The above definition of $x.[\lambda]$ is independent of choice of loop $\lambda : I \rightarrow Y$ representing the homotopy class $[\lambda] \in \pi_1(Y, y)$, and the resulting map

$$\begin{aligned} \alpha : F \times \pi_1(Y, y) &\longrightarrow F \\ (x, [\lambda]) &\mapsto x.[\lambda] \end{aligned}$$

is a right action of the group $\pi_1(Y, y)$ on the set F .

Proof. We first check independence of choice of representative. If $\lambda' : I \rightarrow Y$ is another loop representing the same class $[\lambda'] = [\lambda] \in \pi_1(Y, y)$, then we may pick a homotopy $h : I \times I \rightarrow Y$ relative to ∂I from λ to λ' . By Theorem 2.2.2 there then exists a homotopy $H : I \times I \rightarrow X$ relative to ∂I with $p \circ H = h$ and $H(0, s) = \mu(s)$ for μ the unique lift of λ at x . In particular $p(H(1, s)) = \lambda'(s)$ and $H(1, 0) = x$,

so the (unique) lift of λ' starting at x is defined by $\mu'(s) = H(1, s)$. Since H is relative to ∂I , we have $H(t, 1) = H(0, 1) = \mu(1)$ for all $t \in I$. In particular $\mu'(1) = H(1, 1) = H(0, 1) = \mu(1)$, so the two candidates for $x.[\lambda]$ are equal.

To see that α is a (right) group action, we must verify that $x.e = x$ and that $x.([\lambda_1] * [\lambda_2]) = (x.[\lambda_1]).[\lambda_2]$ for all $x \in F$. For the first, we may represent e by $c_y : I \rightarrow Y$, which we may lift to $c_x : I \rightarrow X$ and hence get $x.e = c_x(1) = x$. To verify the second property, we start from the definition of $(x.[\lambda_1]).[\lambda_2]$: let $\lambda_1, \lambda_2 : I \rightarrow Y$ be loops in (Y, y) representing the two homotopy classes, and let $\mu_1 : I \rightarrow X$ be the unique path with $p \circ \mu_1 = \lambda_1$ and $\mu_1(0) = x$. Hence $x.[\lambda_1] = \mu(1)$. Then let $\mu_2 : I \rightarrow Y$ be the unique path with $p \circ \mu_2 = \lambda_2$ and $\mu_2(0) = x.[\lambda_1] = \mu_1(1)$. By definition of the action, $((x.[\lambda_1]).[\lambda_2]) = \mu_2(1)$. But then the concatenation $\mu_1 * \mu_2 : I \rightarrow Y$ is a path with $(\mu_1 * \mu_2)(0) = x$ and $p \circ (\mu_1 * \mu_2) = \lambda_1 * \lambda_2$, and hence $x.([\lambda_1] * [\lambda_2]) = x.([\lambda_1 * \lambda_2]) = (\mu_1 * \mu_2)(1) = \mu_2(1)$, as we needed. (See the figure.) $\square$

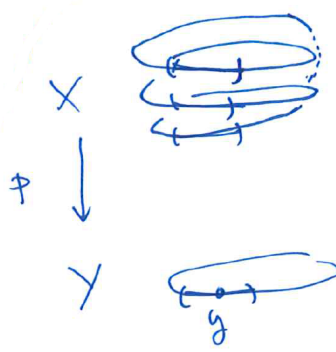


FIGURE 7. Example of an evenly covered neighborhood of $y \in Y$

Theorem 2.3.9. *Let (Y, y) be a path connected based space and $p : X \rightarrow Y$ a covering map. Let $\pi_1(Y, y)$ act on $F = p^{-1}(y) \subset X$ as defined above.*

- (i) *The action is transitive if and only if X is path connected.*
- (ii) *The action is free if and only if $\pi_1(X, x)$ is trivial for all $x \in F$.*

We point out that if X is path connected, then the groups $\pi_1(X, x)$ with $x \in F$ are all isomorphic by Theorem 1.4.5, so $\pi_1(X, x)$ is trivial for all $x \in F$ if and only if it is trivial for one $x \in F$.

Proof. We first treat (i). Assume X is path connected and $x, x' \in F$. If we pick a path $\lambda : I \rightarrow X$ from x to x' , then $[p \circ \lambda] \in \pi_1(Y, y)$ and $x.[\lambda] = x'$. It remains(!) to see that F is non-empty. By definition, a path connected space is non-empty, so we may pick an $x \in X$. Since Y is path connected we may choose a path $\gamma : I \rightarrow Y$ from $p(x)$ to y , and by path lifting we may write $\gamma = p \circ \rho$ for a path $\rho : I \rightarrow X$ with $\rho(0) = x$. Then $p(\rho(1)) = y$ so $\rho(1) \in F$ and in particular F is not empty.

Conversely, assume the action is transitive. We must prove that any two $x, x' \in X$ can be connected by a path, and consider first the case $x, x' \in F$. By assumption, we may pick $[\lambda] \in \pi_1(Y, y)$ with $x.[\lambda] = x'$. But that means that $\lambda = p \circ \mu$ with $\mu : I \rightarrow X$ a path with $\mu(0) = x$ and $\mu(1) = x'$. Now suppose $x \in X$ is arbitrary. Since Y is path connected we may find a path $\gamma : I \rightarrow Y$ from $p(x)$ to y and by Theorem 2.2.1 we may write $\gamma = p \circ \rho$ for a path $\rho : I \rightarrow X$ with $\rho(0) = x$. Then

$p(\rho(1)) = y$ so $\rho(1) \in F$ and hence any $x \in X$ admits a path to a point in F , but since any two points in F may be connected by a path in X we deduce that in fact any two points in X may be connected by a path in X . Since the action is transitive, F and hence X is non-empty, so we have proved X is path connected.

We then treat (ii). Assume $\pi_1(X, x)$ is trivial for all x , let $[\lambda] \in \pi_1(Y, y)$ and $x \in F$, and assume $x.[\lambda] = x$. This means $\lambda = p \circ \mu$ with $\mu : I \rightarrow X$ a path from x to x . But then $[\mu] \in \pi_1(X, x)$ represents the trivial element so $\mu \simeq c_x \text{ rel } \partial I$ and hence also $\lambda = p \circ \mu \simeq c_y \text{ rel } \partial I$, so $[\lambda] = e \in \pi_1(Y, y)$. This proves freeness of the action.

Conversely, suppose the action of $\pi_1(Y, y)$ on $F = p^{-1}(y)$ is free, let $x \in F$ and $[\mu] \in \pi_1(X, x)$ for a loop μ in (X, x) . Then $[p \circ \mu] \in \pi_1(Y, y)$ and by definition $x.[p \circ \mu] = x$. But since the action is assumed free we must have $[p \circ \mu] = e$, but then Theorem 2.2.9 implies that p_* is injective, so in fact $[\mu] = e$. $\square$

Definition 2.3.10. Let Y be a path connected space. A covering map $f : X \rightarrow Y$ is *universal* if X is simply connected.

With this terminology Theorem 2.3.9 implies the following.

Corollary 2.3.11. *If Y is path connected, $p : X \rightarrow Y$ is a universal covering map, $x \in X$ and $y = p(x) \in Y$, then the map*

$$\begin{aligned} \alpha_x : \pi_1(Y, y) &\longrightarrow F = p^{-1}(y) \\ [\lambda] &\mapsto \alpha(x, [\lambda]) = x.[\lambda] \end{aligned}$$

is a bijection.

The value of this corollary is that the set $F = p^{-1}(y)$ is a *subset* of X which in practice may often easily be determined. In contrast, the set underlying the group $\pi_1(Y, y)$ is defined as a *quotient* of the enormous and unwieldy set of all loops, by the rather coarse equivalence relation “ $\simeq \text{ rel } \partial I$ ”. Therefore, the bijection in the corollary “determines” the set underlying the group $\pi_1(Y, y)$. Once the underlying set is determined, it is in practice often rather easy to determine the group structure on that set, as shown for example in the case of $\pi_1(S^1, 1) \cong \mathbb{Z}$. We will see other examples next week.

2.4. Covering maps from group actions. We now discuss a different appearance of group actions in the theory of covering maps and fundamental groups.

Definition 2.4.1. Let G be a group, X a topological space. A *continuous left action* of G on X is a map $\alpha : G \times X \rightarrow X$ which is continuous (when G is given the discrete topology) and also a left action on the underlying set, i.e. $\alpha(g_2 g_1, x) = \alpha(g_2, \alpha(g_1, x))$ for all $x \in X$ and $g_1, g_2 \in G$. We shall also write $g.x$ for $\alpha(g, x)$, in which case the axioms read $e.x = x$ and $(g_2 g_1).x = g_2.(g_1.x)$.

Remark 2.4.2. In section 2.3 we considered *right actions* of $\pi_1(Y, y)$ on $F = p^{-1}(y) \subset X$ for covering maps $p : X \rightarrow Y$. The group actions we discuss here play a different role and it is on purpose that we act on the left this time.

Any left action $\alpha : G \times X \rightarrow X$ can be converted to a right action $\bar{\alpha} : X \times G \rightarrow X$ by setting $\bar{\alpha}(x, g) = \alpha(g^{-1}, x)$ and vice versa, so in principle all actions could be converted to left actions. We take the attitude that it is better and less confusing to let groups act from the side they *want* to act from.

For any continuous left action of G on X we shall write X/G for the space of orbits, equipped with the quotient topology from X . Then the quotient map

$$p : X \rightarrow X/G$$

defined by sending a point $x \in X$ to its orbit $G.x \in X/G$, is continuous. In some cases p will be a covering map (for example if $X = \mathbb{R}$ and $G = \mathbb{Z}$ acts by translation) and in some cases it will not (for example if $G = \mathbb{Q}$ acts by translation on $\mathbb{R}$).

Proposition 2.4.3. *Let $G \times X \rightarrow X$ be a continuous left action such that any $x \in X$ has a neighborhood $V \subset X$ such that $V \cap gV = \emptyset$ for all $g \in G \setminus \{e\}$. Then the natural map $p : X \rightarrow X/G$ is a covering map.*

Before embarking on the proof, let us point out that for any group action $\alpha : G \times X \rightarrow X$, the map $x \mapsto \alpha(g, x) = g.x$ is a homeomorphism of X for all $g \in G$. Indeed, the axioms for a group action imply that $x \mapsto (g^{-1}).x$ is an inverse.

Proof. For any open set $V \subset X$ we have

$$p^{-1}(p(V)) = \bigcup_{g \in G} g.V,$$

and since each $g \in G$ acts by homeomorphisms on X , this is a union of open sets and hence open. By definition of the quotient topology, $p(V) \subset X/G$ is therefore open for any open $V \subset X$, i.e. p is an open map.

Now suppose $V \subset X$ is open and disjoint from $g.V$ for all $g \in G \setminus \{e\}$, as in the assumption of the lemma. This implies $g_1.V \cap g_2.V = \emptyset$ for $g_1 \neq g_2$, since otherwise V and $(g_1^{-1}g_2).V$ wouldn't be disjoint. Finally, if we let $U = p(V)$, then each restriction

$$p|_{g.V} : g.V \rightarrow U$$

is injective, again since V does not contain two distinct elements in the same orbit, is surjective by definition, continuous since U has the subspace topology from the inclusion $U \subset X/G$, and its inverse is continuous because p is an open map. Hence, p is a covering map by Remark 2.1.2. $\square$

2.5. The fundamental group of a quotient space. Suppose given an action $G \times X \rightarrow X$ satisfying the assumption of Proposition 2.4.3 and assume in addition that X is simply connected. Then $p : X \rightarrow X/G$ is a universal covering and hence we may use it to calculate the fundamental group of $Y = X/G$, as in Corollary 2.3.11. Given a point $x \in X$ we may let $y = p(x)$. Then $F = p^{-1}(y) \subset X$ is the orbit of x and we have bijections

$$(2.2) \quad \begin{aligned} G &\rightarrow p^{-1}(y) \\ \pi_1(X/G, y) &\rightarrow p^{-1}(y), \end{aligned}$$

given by acting on x , i.e. $g \mapsto g.x$ and $\alpha \mapsto x.\alpha$. The second is a bijection by Theorem 2.3.9 and Lemma 2.3.5. The first is a bijection by the obvious version of Lemma 2.3.5 for left actions, together with the observation that the action of G on $G.x$ is free and transitive. This proves most of the following.

Theorem 2.5.1. *Let X be simply connected and let $G \times X \rightarrow X$ be an action satisfying the assumptions of Proposition 2.4.3. Let $x \in X$. Then $x \in X$ gives rise to a group isomorphism $\phi_x : G \rightarrow \pi_1(Y, p(x))$ defined by*

$$(2.3) \quad \phi_x(g) = [p \circ \rho_g],$$

where $\rho_g : I \rightarrow X$ is any path from x to $g.x$.

Proof. Let us write $y = p(x)$. Composing the two bijections (2.2) and spelling out their definition precisely gives (2.3), so that formula defines a (well defined) bijection of underlying sets.

It remains to see that $\phi_x(gg') = \phi_x(g)\phi_x(g')$ for all g and g' . Choose paths λ_g from x to $g.x$ and $\lambda_{g'}$ from x to $g'.x$. We may define a path $g.\lambda_{g'} : I \rightarrow X$ by $(g.\lambda_{g'})(t) = g(\lambda_{g'}(t))$; then $g.\lambda_{g'}$ is a path from $g.x$ to $(gg').x$, and hence the concatenation $\lambda_g * (g.\lambda_{g'})$ is a path from x to $(gg').x$. Use this as a choice of $\lambda_{gg'}$. Then

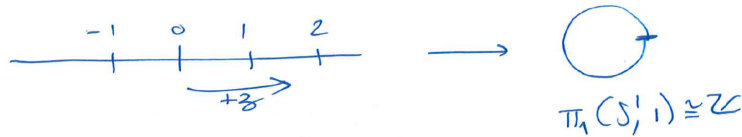
$$\phi_x(gg') = [p \circ (\lambda_g * (g.\lambda_{g'}))] = [p \circ \lambda_g][p \circ (g.\lambda_{g'})] = [p \circ \lambda_g][p \circ \lambda_{g'}] = \phi_x(g)\phi_x(g').$$

□

2.6. Examples of covering maps from group actions. Theorem 2.5.1 gives an effective method for calculating fundamental groups of a path connected space Y , provided one is able to *guess* a homeomorphism (or just a homotopy equivalence) $Y \approx X/G$ for a continuous action of a group G on a space X satisfying the assumptions in Proposition 2.4.3 and Theorem 2.5.1. In practice that is often possible. Let us give a few examples.

2.6.1. Circle. We can take $G = \mathbb{Z}$ acting on $X = \mathbb{R}$ by translation. Then for any $x \in \mathbb{R}$ the open subset $V = (x - \frac{1}{2}, x + \frac{1}{2})$ is disjoint from all its G -translates, so $p : X \rightarrow X/G$ is a covering map. Then X/G is homeomorphic to S^1 , so we reproduce the isomorphism $\pi_1(S^1, 1) \cong \mathbb{Z}$.

2.6.2. Torus. Similarly we can take $G = \mathbb{Z}^2$ acting on $\mathbb{R}^2$ by translation. The sets V may be taken as the open squares $(x - \frac{1}{2}, x + \frac{1}{2}) \times (y - \frac{1}{2}, y + \frac{1}{2})$, which are disjoint from their G -translates. Hence the quotient map $q : X \rightarrow T = X/G$ is a covering map and we deduce $\pi_1(T, y) \cong \mathbb{Z}^2$.



In fact T is homeomorphic to a *torus*, as we shall somewhat explain (a rigorous proof would presuppose a precise definition of the torus). The map $i : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by

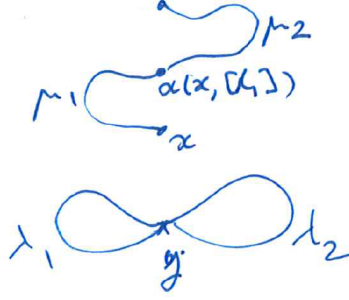
$$i(u, v) = (2 + \cos(2\pi u))(\cos(2\pi v), \sin(2\pi v), 0) + (0, 0, \sin(2\pi u))$$

clearly satisfies $i(u+n, v+m) = i(u, v)$, so it factors over a continuous map $X/G \rightarrow \mathbb{R}^3$. In fact this may be seen to be injective (exercise). Now X/G is the image of the compact set $[0, 1] \times [0, 1] \subset X$ so it is compact, and since $\mathbb{R}^3$ is Hausdorff we deduce that the map $X/G \rightarrow \mathbb{R}^3$ induced by i is injective. The formula parametrizes the surface of revolution obtained from the circle in the (x, y) -plane of radius 1 and center $(2, 0)$ around the z -axis.

Alternatively, check that the inclusion $[0, 1]^2 \subset X$ induces a homeomorphism from quotient of the square $[0, 1]^2$ by the equivalence relation generated by $(0, t) \sim (1, t)$ and $(t, 0) \sim (t, 1)$ for all $t \in I$ to X/G . In a class on point-set topology you may have seen a proof that the quotient of the square by these identifications on the boundary is homeomorphic to the torus.

Of course, the space $\mathbb{R}^n/\mathbb{Z}^n$ may be handled in a completely similar way.

2.6.3. *Klein bottle.* The set of homeomorphisms $f : X \rightarrow X$ forms a group with respect to composition, and this group acts on X in the obvious way: $f.x = f(x)$. Any (finite or infinite) set of homeomorphisms $X \rightarrow X$ generates a subgroup of this group of all homeomorphisms, which then acts in the same way. In this example we shall construct a group action on $X = \mathbb{R}^2$ in this way.



Let G be the group of transformations of $X = \mathbb{R}^2$ generated by $S : (x, y) \mapsto (x, y + 1)$ and $T : (x, y) \mapsto (x + \frac{1}{2}, -y)$. It is easy to verify that $ST = TS^{-1}$, and hence any element of G may be written as $T^a S^b$ for $a, b \in \mathbb{Z}$ and hence the function

$$\begin{aligned} \mathbb{Z}^2 &\rightarrow G \\ (a, b) &\mapsto T^a S^b \end{aligned}$$

is surjective. From the formula $T^a S^b(0, 0) = T^a(0, b) = (a/2, (-1)^a b)$ we deduce that it is also injective. In particular $TS \neq TS^{-1}$ since they come from different elements $(1, 1)$ and $(1, -1) \in \mathbb{Z}^2$, so $ST = TS^{-1} \neq TS$ and in particular G is not commutative. (And therefore the bijection $\mathbb{Z}^2 \rightarrow G$ cannot be a homomorphism.)

The action again satisfies the assumptions of 2.4.3, since for any $(x, y) \in \mathbb{R}^2$ the set $V = (x - \frac{1}{4}, x + \frac{1}{4}) \times (y - \frac{1}{2}, y + \frac{1}{2})$ is disjoint from $T^a S^b V$ if $(a, b) \neq (0, 0)$: if $a \neq 0$ then these sets have no points with same first coordinate, and if $a = 0$ and $b \neq 0$ then they have no points with same second coordinate. Hence the projection $p : \mathbb{R}^2 \rightarrow K = \mathbb{R}^2/G$ is again a covering map and the basepoint $x = (0, 0) \in X$ gives an isomorphism $\phi_x : G \rightarrow \pi_1(K, y)$. This example shows that the fundamental group need not be abelian.

In fact this space is homeomorphic to the Klein bottle: it may be identified with the quotient of the rectangle $[0, \frac{1}{2}] \times [-\frac{1}{2}, \frac{1}{2}]$ by the equivalence relation generated by $(t, -\frac{1}{2}) \simeq (t, \frac{1}{2})$ and $(0, t) \sim (\frac{1}{2}, -t)$.

2.6.4. *Real projective space.* The *real projective plane* $\mathbb{R}P^2$ is defined as the quotient of S^2 by the equivalence relation $x \sim -x$. We may calculate its fundamental group by “guessing” the homeomorphism $\mathbb{R}P^2 \cong X/G$, where $X = S^2$ and G is the cyclic group of order two whose generator acts on S^2 by $x \mapsto -x$. We shall show later that S^n is simply connected for $n \geq 2$, and hence Theorem 2.5.1 implies $\pi_1(\mathbb{R}P^2, y)$ is cyclic of order two for any basepoint y .

Similarly one defines $\mathbb{R}P^n$ as the quotient of S^n by $x \sim -x$. The fundamental group is cyclic of order 2 for any $n > 1$. (But $\mathbb{R}P^1$ is homeomorphic to S^1 so has infinite cyclic fundamental group.)

2.6.5. *Thrice punctured sphere.* Let $X = \{z \in \mathbb{C} \mid \text{Im}(z) > 0\}$ be the upper half plane, and let G be the group of transformation of X generated by $S(z) = z + 2$ and $T(z) = 1/((1/z) + 2)$. It can be shown that the quotient map $p : X \rightarrow Y = X/G$ is a covering map, and that Y is homeomorphic to S^2 with three points removed. The group G is a so-called *free group* on two generators. (We shall not give a

proof of the homeomorphism asserted here, but we define what “free group on two generators” means in Section 3.4 below.)

2.7. More lifting theorems. Now that we have better tools for calculating the fundamental group of a space, let us discuss one possible use for it.

Let (X, x) , (Y, y) , and (W, w) be based spaces, and let $p : X \rightarrow Y$ be a covering map with $p(x) = y$. We shall study the question of whether a given map $f : W \rightarrow Y$ with $f(w) = y$ admits a *lift*, i.e. a map $g : W \rightarrow X$ with $g(w) = x$ and $p \circ g = f$, and whether such lifts are unique. The question can be summarized in the diagram

$$\begin{array}{ccc} & & (X, x) \\ & \nearrow g & \downarrow p \\ (W, w) & \xrightarrow{f} & (Y, y). \end{array}$$

As we shall see in the exercises, this type of lifting question has applications to questions which do not explicitly concern fundamental groups and covering maps. For example, it may be used with $X = \mathbb{C}$, $Y = \mathbb{C} \setminus \{0\}$ and $p(z) = e^z$ to study when a complex-valued function has a logarithm. Similarly, with $X = \mathbb{C} \setminus \{0\} = Y$ and $p(z) = z^2$ it may be used to study which complex-valued functions have a square root, etc.

Definition 2.7.1. A space W is *locally path connected* if for any $z \in W$ and any neighborhood $V \subset W$ of z , there exists a smaller neighborhood $A \subset V$ of z which is path connected.

This condition is equivalent to the condition that any path component of any open subset of W is itself open. It is often paraphrased as “any $z \in W$ has arbitrarily small path connected neighborhoods”.

Theorem 2.7.2. Assume W be path connected and locally path connected. Then a lift g of f exists if and only if

$$(2.4) \quad f_*(\pi_1(W, w)) \subset p_*(\pi_1(X, x)),$$

and in that case the lift is unique.

We will use this theorem in the classification of coverings in Section 2.8.

Proof. We first prove “only if”, which shall not use any of the assumptions on W . Indeed, if $f = p \circ g$, then we have two equal homomorphisms

$$f_* = p_* \circ g_* : \pi_1(W, w) \rightarrow \pi_1(Y, y),$$

which then have equal image. It follows that $f_*(\pi_1(W, w)) = p_*(g_*(\pi_1(W, w))) \subset p_*(\pi_1(X, x))$.

We then prove uniqueness of g , which shall use only path connectedness of W . Suppose for contradiction that $g, g' : W \rightarrow X$ are two different lifts, and choose $z \in W$ with $g(z) \neq g'(z)$. Since W is path connected we may choose a path $\lambda : I \rightarrow W$ from w to z , in which case the paths $g \circ \lambda$ and $g' \circ \lambda$ are two lifts of the same path $f \circ \lambda : I \rightarrow Y$, with equal starting point x but distinct endpoints $g(z) \neq g'(z)$. By Theorem 2.2.1, this is not possible.

To prove “if”, let us first point out a consequence of condition (2.4). Given any two paths $\lambda, \lambda' : I \rightarrow W$ with $\lambda(0) = \lambda'(0) = w$ and the same endpoint $\lambda(1) = \lambda'(1) = z$ we can by Theorem 2.2.1 write $f \circ \lambda = p \circ \mu$ and $f \circ \lambda' = p \circ \mu'$ for two paths $\mu, \mu' : I \rightarrow X$ with $\mu(0) = \mu'(0) = x$. Then (2.4) implies that also $\mu(1) = \mu'(1)$, since the loop $\rho = \lambda' * \bar{\lambda} : I \rightarrow W$ in (W, w) maps to the loop

$f \circ \rho$ in (Y, y) which by Corollary 2.2.10 lifts to a loop $\tau : I \rightarrow X$ in (X, x) since $[f \circ \rho] = f_*[\rho] \in p_*(\pi_1(X, x))$. With $f \circ \rho = p \circ \tau$ we may calculate

$$\begin{aligned} p \circ \mu' &= f \circ \lambda' \simeq (f \circ \rho) * (f \circ \lambda) = (p \circ \tau) * (p \circ \mu) \\ &= p \circ (\tau * \mu) \text{ rel } \partial I, \end{aligned}$$

and by lifting a homotopy (i.e. by Theorem 2.2.2) we see that $\mu' \simeq \tau * \mu \text{ rel } \partial I$ and in particular $\mu(1) = \mu'(1)$. That is, the endpoint $\mu(1) \in X$ depends only on $z \in W$, not on the chosen path $\lambda : I \rightarrow W$ from w to z .

Assuming still (2.4), we may therefore define a function $g : W \rightarrow X$ by sending $z \in W$ to $\mu_z(1)$, where $\mu_z : I \rightarrow X$ is a path with $\mu_z(0) = x$ and $p \circ \mu_z = f \circ \lambda_z$ for a path $\lambda_z : I \rightarrow W$ from w to z . We have shown this is well defined (independent of choice of path λ_z), and it is easy to see that $g(w) = x$, but we must show that g is continuous, and this is where local path connectedness of W shall be used. The local path connectedness of W implies that any $z \in W$ admits an open neighborhood $A \subset W$ such that $f(A) \subset Y$ is contained in an evenly covered subset: indeed, $f(z) \in Y$ admits some evenly covered open neighborhood, whose inverse image is then an open neighborhood of $z \in W$, but then we have assumed existence of a smaller path connected neighborhood A . Since continuity may be verified locally, it suffices to verify continuity of $g|_A$ for all such A .

To study $g|_A$ with $A \subset W$ path connected and open, and $f(A)$ evenly covered, let us use that $f(A)$ is evenly covered to pick homeomorphism

$$h = (p|_{p^{-1}(f(A))}, q) : p^{-1}(f(A)) \rightarrow f(A) \times F$$

for some discrete space F . Then $g|_A$ has image in $p^{-1}(f(A))$, and since h is a homeomorphism it suffices to show that $h \circ g|_A$ is continuous, which we show by considering each “coordinate” separately: the first coordinate is a function $A \rightarrow f(A)$ and the second is a function $A \rightarrow F$. The first coordinate of $h \circ g|_A$ is $p \circ g|_A = f|_A$, which is continuous by assumption. We claim that the second coordinate is constant, and hence continuous. To see this, we first pick $\lambda_z : I \rightarrow W$ from w to one particular $z \in A$, and then for all other $z' \in A$ choose $\lambda_{z'} = \lambda_z * \eta_{z'}$ for a path $\eta_{z'} : I \rightarrow A$ from z to z' , then the path $\mu_{z'}$ lifting $f \circ \lambda_{z'} = (f \circ \lambda_z) * (f \circ \eta_{z'})$ will be the concatenation of μ_z , lifting $f \circ \lambda_z$, and the lift of $f \circ \eta_{z'} : I \rightarrow f(A)$ which starts at $\mu_z(1)$. By continuity, any lift of $f \circ \eta_{z'}$, after composing with the homeomorphism h , will have constant second coordinate. Hence the second coordinate of $h \circ g|_A$ is indeed constant. $\square$

2.8. Classification of covering maps. Let us fix a path connected based space (Y, y) . We would like to classify all possible path connected coverings of Y , that is give a description of the collection of triples (X, x, p) , where (X, x) is a path connected based space and $p : X \rightarrow Y$ is a covering map with $p(x) = y$, up to an appropriate notion of equivalence/isomorphism. In this case, we will say that (X, x, p) and (X', x', p') are isomorphic if there exists a homeomorphism $h : X \rightarrow X'$ with $h(x) = x'$ and $p' \circ h = p$:

$$\begin{array}{ccc} (X, x) & \xrightarrow[\quad h \quad]{\cong} & (X', x') \\ & \searrow p \quad \swarrow p' & \\ & (Y, y) & \end{array}$$

We have already seen in Theorem 2.2.9 that the induced map on fundamental groups

$$p_* : \pi_1(X, x) \longrightarrow \pi_1(Y, y)$$

is injective, and hence associates a subgroup $H = p_*\pi_1(X, x) < \pi_1(Y, y)$ ¹. The classification theorem will need niceness assumptions on Y : it should be locally path connected as in Definition 2.7.1, as well as semi-locally simply connected, in the following sense:

Definition 2.8.1. A space Y is *semi-locally simply connected* if any $u \in Y$ admits a path connected neighborhood $U \subset Y$ such that $\pi_1(U, u) \rightarrow \pi_1(Y, u)$ is the trivial homomorphism.

Remark 2.8.2. Most spaces encountered in nature satisfy this technical condition of being semi-locally simply connected. In fact most spaces satisfy the stronger condition of being *locally contractible*: any point admits a contractible neighborhood. For example this holds for any open subset of $\mathbb{R}^n$.

The classification theorem is the following:

Theorem 2.8.3 (Classification of covering spaces). *Let (Y, y) be a based space such that Y is path connected, locally path connected, and semi-locally simply connected. Then there is a bijection of sets*

$$\Phi: \left\{ (X, x, p) \left| \begin{array}{l} X \text{ path connected} \\ p: X \rightarrow Y \text{ covering} \\ x \in X, p(x) = y \end{array} \right. \right\} /_{\text{iso}} \xrightarrow{\cong} \{ H < \pi_1(Y, y) \mid H \text{ subgroup} \}$$

between the set of isomorphism classes of (X, x, p) , where X is path connected space, $x \in X$, and $p: X \rightarrow Y$ is a covering map with $p(x) = y$ up to the above described equivalence relation, and the set of subgroups of $\pi_1(Y, y)$. In particular, the universal covering exists (corresponding to $H = \{e\}$) and is unique up to isomorphism.

We will use the rest of Section 2 to prove this theorem. We need to show that the map Φ in the statement is

- (1) well-defined, i.e., independent of (X, x, p) in its isomorphism class;
- (2) injective, i.e., non-isomorphic (X, x, p) 's give non-isomorphic H 's;
- (3) surjective, i.e., every subgroup H is in the image.

We start by (1) and (2). The assumption that Y is locally path connected will be used for (2), and that of being semi-locally simply connected will be used for (3).

Lemma 2.8.4. *Let $p: X \rightarrow Y$ be a covering map and assume Y is locally path connected. Then X is also locally path connected.*

Proof. Suppose given an open neighborhood $V \subset X$ of $x \in X$, for which we wish to find a smaller open neighborhood $V' \subset X$. Let $A \subset Y$ be an evenly covered neighborhood of $p(x)$ and pick a homeomorphism $h = (p|_{p^{-1}(A)}, q) : p^{-1}(A) \rightarrow A \times F$ for a discrete space F . Without loss of generality we may assume $A \subset Y$ is path connected and open (otherwise, use local path connectedness of Y to replace A with a smaller path connected open neighborhood of $p(x)$). Hence for each $f \in F$, the subset $A \times \{f\} \subset A \times F$ is path connected and open. For appropriate f it will contain $h(x)$, so $x \in h^{-1}(A \times \{f\})$ is a path connected open neighborhood. After possibly replacing $A \subset Y$ by a smaller path connected open neighborhood of $p(x)$, $V' = h^{-1}(A \times \{f\})$ will be contained in V . $\square$

¹We use the notation $H < G$ for H a subgroup of G , where we allow $H = G$.

We may now use Theorem 2.7.2 above to prove that the map Φ in Theorem 2.8.3 is both well-defined and injective, that is proving the uniqueness part of the classification theorem:

Proposition 2.8.5. *Let Y be a path connected and locally path connected space, and let $y \in Y$. Let (X, x, p) and (X', x', p') be two path connected based covering maps of (Y, y) . Then there exists a homeomorphism $h : X \rightarrow X'$ with $h(x) = x'$ and $p' \circ h = p$ if and only if the subgroups $H = p_*(\pi_1(X, x))$ and $H' = p'_*(\pi_1(X', x'))$ of $G = \pi_1(Y, y)$ are equal. In that case the homeomorphism h is unique.*

Proof. By the Lemma, X and X' must both be locally path connected. To get the map h we apply Theorem 2.7.2 with $(W, w) = (X, x)$ and to get its inverse we swap the roles of (X, x, p) and (X', x', p') . This gives two based maps $h : (X, x) \rightarrow (X', x')$ and $h' : (X', x') \rightarrow (X, x)$. The composition $h' \circ h$ is a based map $(X, x) \rightarrow (X, x)$ with $p \circ (h' \circ h) = p$, but the identity map is another such based map so by the uniqueness part of Theorem 2.7.2 $h' \circ h$ must be the identity. Similarly, $h \circ h'$ is the identity map of (X', x') , so h and h' are both homeomorphisms and are inverse to each other. $\square$

2.9. Existence of covering maps. It remains to treat the existence part of the classification Theorem 2.8.3: given a subgroup $H < \pi_1(Y, y)$ does it correspond to a covering map from a path connected space? The biggest possible subgroup $H = \pi_1(Y, y)$ corresponds to the identity map (which is a covering), so this is always realized. Since $p_* : \pi_1(X, x) \rightarrow \pi_1(Y, y)$ is injective, the smallest possible subgroup $H = \{e\}$ must correspond to a simply connected X , i.e. a universal cover of Y . We will start the section by treating that case. We will later use the universal cover to construct the other coverings.

2.9.1. Existence of universal covers. We want to show that any path connected space Y without too nasty local pathologies admits a universal cover. Let $G = \pi_1(Y, y)$. The proof is by explicit construction of a simply connected space X , a continuous action $G \times X \rightarrow X$ satisfying the assumptions of Proposition 2.4.3, and a homeomorphism $X/G \rightarrow Y$. For time reasons, we shall not give the full proof of the following result in the lectures. You can find a more complete proof in the appendix Section 3.3.

Theorem 2.9.1. *Let Y be a path connected and semi-locally simply connected space (as in Definition 2.8.1). Let $y \in Y$. Then there exists a simply connected space X , a continuous left action of $G = \pi_1(Y, y)$ on X satisfying the assumption of Proposition 2.4.3, and a homeomorphism $Y \cong X/G$.*

Proof sketch. The proof of this result amounts to a construction of X and the homeomorphism h , and a proof that they satisfy what is asserted. The full proof is rather lengthy, and we shall not give all details in this class, but be content with defining the underlying set of X , the topology on X , the action of G on X , and the map $p : X \rightarrow Y$. See the appendix for more details.

If we write $G = \pi_1(Y, y)$, the set X is defined by

$$X = \left\{ \lambda : I \rightarrow Y \mid \begin{array}{l} \lambda \text{ continuous} \\ \lambda(0) = y \end{array} \right\} / \sim \text{ homotopy relative to } \partial I.$$

The map $p : X \rightarrow Y$ is defined by $p([\lambda]) = \lambda(1)$ and the left action of G on X defined by concatenation of paths $[\rho] \cdot [\lambda] = [\rho * \lambda]$ for ρ a loop in (Y, y) . Clearly $p([\rho] \cdot [\lambda]) = (\rho * \lambda)(1) = \lambda(1) = p([\lambda])$, so if we let X/G denote the set of orbits for

the action (“set” because we have not yet defined a topology on X), then p induces a well defined map

$$\begin{aligned}\bar{p} : X/G &\rightarrow Y \\ G.[\lambda] &\mapsto p([\lambda]) = \lambda(1),\end{aligned}$$

where we have written $G.[\lambda]$ for the orbit containing $[\lambda]$. The assumption that Y be path connected implies that p and $\bar{p}$ are surjective.

The action of $G = \pi_1(Y, y)$ on the set X is free, because if $[\rho][\lambda] = [\rho'][\lambda]$ then $\rho * \lambda \simeq \rho' * \lambda \text{ rel } \partial I$, but then concatenating by $\bar{\lambda}$ on the right gives $\rho' \simeq \rho' * \lambda * \bar{\lambda} \simeq \rho * \lambda * \bar{\lambda} \simeq \rho \text{ rel } \partial I$, so $[\rho'] = [\rho]$. By a similar argument, if $p([\lambda]) = p([\lambda'])$, then $[\lambda'] = [\rho].[\lambda]$ where $\rho = \lambda' * \bar{\lambda}$.

To summarize, we have proved that the action $G \times X \rightarrow X$ is free and that $p : X \rightarrow Y$ factors through a bijection $\bar{p} : X/G \rightarrow Y$. It remains to define a topology on X such that the action satisfies the assumption of Proposition 2.4.3 and then prove that the bijection $\bar{p}$ is a homeomorphism. See the appendix Section 3.3 for more details on this. $\square$

2.9.2. More quotient constructions. Let X be a simply connected space, and let $G \times X \rightarrow X$ be a left action of a group G satisfying the assumption in Proposition 2.4.3. If $H < G$ is a subgroup, we may also apply Proposition 2.4.3 to the action of H and deduce that $p_H : X \rightarrow X/H$ is a covering map. We have maps

$$(2.5) \quad \begin{array}{ccc} X & \xrightarrow{p_H} & X/H \\ & \searrow p_G & \downarrow p' \\ & & X/G, \end{array}$$

where $p' : X/H \rightarrow X/G$ sends the H -orbit of x to the G -orbit of x . If $x \in X$ is a basepoint we obtain basepoints $p_H(x) \in X/H$ and $p_G(x) \in X/G$, and we apply Theorem 2.5.1 in both cases, to get isomorphisms

$$\begin{array}{ccc} H & \xrightarrow[\cong]{\phi_x} & \pi_1(X/H, p_H(x)) \\ \text{inclusion} \downarrow & & \downarrow (p')_* \\ G & \xrightarrow[\cong]{\phi_x} & \pi_1(X/G, p_G(x)), \end{array}$$

which are easily seen to be compatible (i.e. the diagram above commutes). In particular we conclude that the image of $(p')_* : \pi_1(X/H, p_H(x)) \rightarrow \pi_1(X/G, p_G(x)) \cong G$ is identified with the given subgroup H .

Lemma 2.9.2. *In the above situation $p' : X/H \rightarrow X/G$ is a covering map.*

Proof sketch. Very similar to the proof of Proposition 2.4.3. $\square$

We can now assemble all the above results to prove the classification theorem.

2.9.3. Proof of Theorem 2.8.3.

Proof. Proposition 2.8.5 show that the map Φ is well-defined and injective. Theorem 2.9.1 show that the trivial group is always in the image of Φ , namely the image of the universal cover. Let X denote the universal cover. We have $X/G \cong Y$. One can get any subgroup $H < G = \pi_1(Y, y)$ in the image using the covering $X/H \rightarrow X/G \cong Y$ as in Section 2.9.2. Hence Φ is also surjective. $\square$

In some cases, it is possible to list all possible subgroups of $\pi_1(Y, y)$, and guess corresponding covering maps.

Example 2.9.3. For $(Y, y) = (S^1, 1)$ we have $\pi_1(Y, y) = \mathbb{Z}$, whose subgroups are precisely $n\mathbb{Z}$ with $n = 0, 1, 2, \dots$. For $n = 0$ the subgroup $0\mathbb{Z} = \{0\}$ corresponds to the universal cover $p : \mathbb{R} \rightarrow S^1$ sending $t \mapsto e^{2\pi it}$. For $n > 0$ we may instead use the map $S^1 \rightarrow S^1$ given by $z \mapsto z^n$.

3. FREE GROUPS AND VAN KAMPEN THEOREM (WEEK 3)

In this section, we give the van Kampen theorem, that allows to compute the fundamental group of a space “piece by piece”. We start with a simpler version of the theorem that will be the only one completely covered in the course. The full version of the theorem needs the notion of free groups that is explained in a series of appendices in this section, for the interested reader, that are not part of the curriculum for this course.

3.1. van Kampen lite. The van Kampen theorem is a different method for “calculating” fundamental groups of a space, not directly related to the method using covering maps. Instead one tries to compute the fundamental group of a space X using the decomposition of $X = U \cup V$ as the union of two subspaces. The theorem then gives a “formula” for $\pi_1(U \cup V, x)$ in terms of $\pi_1(U, x)$, $\pi_1(V, x)$, and $\pi_1(U \cap V, x)$.

We start with a simplified version of the theorem, that is already really useful.

Theorem 3.1.1. *Let (X, x) be a based space, $U, V \subset X$ open subspaces containing x such that $U \cup V = X$ and $U \cap V$ is path connected. Let $i : (U, x) \rightarrow (X, x)$ and $j : (V, x) \rightarrow (X, x)$ denote the inclusion maps. Then the group $\pi_1(X, x)$ is generated by the images of the two homomorphisms*

$$\begin{aligned} i_* : \pi_1(U, x) &\rightarrow \pi_1(X, x) \\ j_* : \pi_1(V, x) &\rightarrow \pi_1(X, x). \end{aligned}$$

Corollary 3.1.2. *Let (X, x) be a based space, $U, V \subset X$ open subspaces containing x such that U and V are simply connected, $U \cap V$ is path connected, and $X = U \cup V$. Then X is simply connected.*

Example 3.1.3. For all $n \geq 2$, the n -sphere $S^n = \{x \in \mathbb{R}^{n+1} \mid |x| = 1\}$ is simply connected.

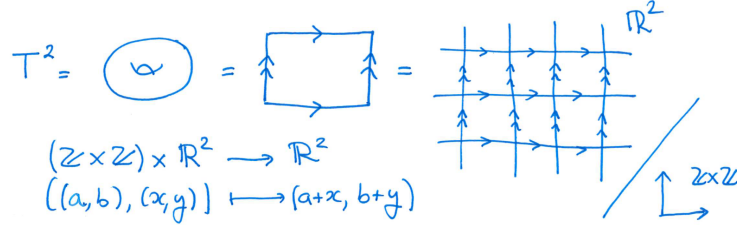
Indeed, if we pick three distinct points $x, y, z \in S^n$, we may let $U = S^n \setminus \{y\}$ and $V = S^n \setminus \{z\}$. By “stereographic projection”, U and V are homeomorphic to $\mathbb{R}^n$ and hence simply connected. The intersection $U \cap V$ is homeomorphic to $\mathbb{R}^n \setminus \{0\}$ and hence path connected unless $n = 0$. Together with Theorem 1.6.5, this completes the calculation of the fundamental group of S^n for $n \geq 1$.

Proof of Theorem 3.1.1. Let $[\lambda] \in \pi_1(X, x)$, represented by a loop λ . It suffices to show that $\lambda \simeq \lambda_1 * \dots * \lambda_n$ (relative to ∂I), where all λ_j are loops in (X, x) that either lie completely in U or in V .

The interval I is covered by the open sets $\{\lambda^{-1}(U), \lambda^{-1}(V)\}$. By compactness, a positive integer n such that for each $1 \leq i \leq n$ the subinterval $[(i-1)/n, i/n]$ is contained in either $\lambda^{-1}(U)$ or $\lambda^{-1}(V)$. Let $\{i_1, \dots, i_k\}$ be the set of indices such that $\lambda(i_j) \in U \cap V$. For each i_j , pick a path δ_j from $\lambda(i_j)$ to x , which is possible as $U \cap V$ is assumed to be path connected. Then

$$\begin{aligned} \lambda &= \lambda[0, \frac{i_1}{n}] * \lambda[\frac{i_1}{n}, \frac{i_2}{n}] * \dots * \lambda[\frac{i_k}{n}, 1] \\ &\simeq \lambda[0, \frac{i_1}{n}] * \delta_1 * \bar{\delta}_1 * \lambda[\frac{i_1}{n}, \frac{i_2}{n}] * \delta_2 * \bar{\delta}_2 * \dots * \bar{\delta}_k * \lambda[\frac{i_k}{n}, 1]. \end{aligned}$$

Now by construction, $\lambda[0, \frac{i_1}{n}] * \delta_1$, each $\bar{\delta}_{j-1} * \lambda[\frac{i_{j-1}}{n}, \frac{i_j}{n}] * \delta_j$ and $\delta_k * \lambda[\frac{i_k}{n}, 1]$ are loops based at x , either completely in U or completely in V . $\square$



Example 3.1.4. Let $X = \mathbb{C} \setminus \{1, -1\}$ with basepoint $x = 0$, and cover this space by the open sets

$$U = \{z \in X \mid \operatorname{Re}(z) < 1\} \quad \text{and} \quad V = \{z \in X \mid \operatorname{Re}(z) > -1\}$$

and use the basepoint $0 \in U \cap V$. Then $U \cap V$ is path connected, so $\pi_1(X, x)$ is generated by the images of $\pi_1(U, x)$ and $\pi_1(V, x)$.

It is not hard to see that U and V are both homotopy equivalent to a circle and therefore have infinite cyclic fundamental groups. In fact, the homotopy classes of $t \mapsto -1 + \exp(2\pi it)$ and $t \mapsto 1 - \exp(2\pi it)$ may be taken as generators. If we write $\alpha \in \pi_1(U, x)$ and $\beta \in \pi_1(V, x)$ we have shown that any element $\gamma \in \pi_1(X, x)$ may be written as

$$\gamma = \alpha_1^{n_1} \beta_1^{m_1} \dots \alpha_k^{n_k} \beta_k^{m_k}$$

for some $n_i, m_i \in \mathbb{Z}$.

If we write out the powers (e.g. replace $\alpha^4 \beta^2$ by $\alpha \alpha \alpha \alpha \beta \beta$), then we see that any element of $\pi_1(X, 0)$ may be written as a “word” in the “alphabet” $\{\alpha, \beta, \alpha^{-1}, \beta^{-1}\}$. If the “letter” α appears next to α^{-1} we may cancel both, and deduce inductively that any element of $\pi_1(X, 0)$ may be written as a “reduced word”: one in which α never appears right next to α^{-1} and similarly for β . In fact this is in some sense a calculation of $\pi_1(X, 0)$, or at least its underlying set: it is possible to prove that two reduced words represent the same element of $\pi_1(X, 0)$ if and only if they are “equal words”, i.e. the equal as formal strings of letters in the alphabet. A better statement is that $\pi_1(X, 0)$ is a *free group on two generators*; see the appendix for the definition and some important properties of this notion.

3.2. van Kampen. The more general van Kampen theorem, in the same setting as Theorem 3.1.1, gives a formula for $\pi_1(X, x)$ in terms of the groups $\pi_1(U, x)$, $\pi_1(V, x)$, and the homomorphisms i_* and j_* . The precise statement uses the notion of “free groups” alluded to in the example above, as well as some related notions. We will here only give an overview of the necessary concepts so that we can at least state the general theorem. This is covered rigorously in the appendix section 3.4.

Given groups $G_1, \dots, G_k$, one can define a new group $G = G_1 * \dots * G_k$, called the *free product* of the groups. This is the group “generated by the groups $G_1, \dots, G_k$ with no additional relation than those already in the groups”. Explicitly, the free product can be explicitly constructed as the group of words $g_1 g_2 \dots g_r$ in elements $g_j \in G_{i_j}$ for some i_j , with the product given by concatenation of words, and with relations those of the G_i ’s together with the relations coming from “reducing a word”, by removing a unit, or multiplying elements that can be multiplied. (See section 3.4 for a detailed definition.)

This group has the following “universal property” (think “defining property”): given homomorphisms $\phi_i : G_i \rightarrow H$ for all i , there exists a unique homomorphism $\phi : G = G_1 * \dots * G_k \rightarrow H$ such that ϕ restricts to ϕ_i on G_i for each i . (See also Theorem 3.4.7.)

Example 3.2.1. The group $\mathbb{Z} * \mathbb{Z} = F_2$ is the free group on two generators. Its elements can be written as words $a^{m_1}b^{n_1}a^{m_2}\dots b^{m_k}$. Note that $ab \neq ba$ in this group; it is non-abelian.

Returning to van Kampen's theorem, suppose that $X = U_0 \cup U_1$ for open subsets $U_0, U_1 \subset X$ such that U_0 , U_1 , and $U_0 \cap U_1$ are all path connected. Let $x \in U_0 \cap U_1$. Then the inclusions $i_0 : U_0 \rightarrow X$ and $i_1 : U_1 \rightarrow X$ induce together a group homomorphism

$$(3.1) \quad \pi_1(U_0, x) * \pi_1(U_1, x) \rightarrow \pi_1(X, x),$$

by the universal property of the free product discussed above. Theorem 3.1.1, which asserts that any element of $\pi_1(X, x)$ may be written as a product of elements in the image of $(i_0)_*$ and elements in the image of $(i_1)_*$, may now be rephrased as saying that the homomorphism (3.1) is *surjective*. In general it is not injective, although we shall see that it is injective when $U_0 \cap U_1$ is simply connected, so in this case we have “calculated” $\pi_1(X, x)$ as the free product of $\pi_1(U_0, x)$ and $\pi_1(U_1, x)$.

Let us write $U_{01} = U_0 \cap U_1$ and $j_0 : U_{01} \rightarrow U_0$ and $j_1 : U_{01} \rightarrow U_1$ for the inclusions. Then for any element $\alpha \in \pi_1(U_{01}, x)$ we have elements $(j_0)_*(\alpha) \in \pi_1(U_0, x)$ and $(j_1)_*(\alpha^{-1}) \in \pi_1(U_1, x)$. Both of these two elements may be interpreted as living in the free product $\pi_1(U_0, x) * \pi_1(U_1, x)$, where we may multiply them to get the element

$$(3.2) \quad ((j_0)_*(\alpha))((j_1)_*(\alpha^{-1})) \in \pi_1(U_0, x) * \pi_1(U_1, x).$$

Here we have omitted the injective homomorphisms $\phi_i : \pi_1(U_i, x) \rightarrow \pi_1(U_0, x) * \pi_1(U_1, x)$ from the notation.

The element (3.2) is in the kernel of the homomorphism (3.1), since $i_0 \circ j_0 = i_1 \circ j_1$. We get one of these elements for each $\alpha \in \pi_1(U_0 \cap U_1, x)$, and since the kernel is a normal subgroup, any product of conjugates of elements of this form will also be in the kernel. The van Kampen theorem asserts that this is *precisely* what the kernel of (3.1) is.

Theorem 3.2.2. *Let $N < \pi_1(U_0, x) * \pi_1(U_1, x)$ be the smallest normal subgroup containing the elements (3.2) for all $\alpha \in \pi_1(U_0 \cap U_1, x)$. Then, under the assumptions of Theorem 3.1.1, the kernel of (3.1) is precisely N .*

Thus we have a canonical isomorphism

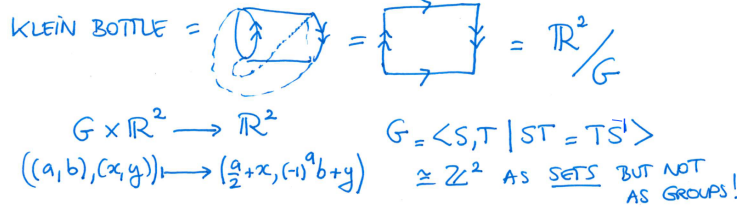
$$\frac{\pi_1(U_0, x) * \pi_1(U_1, x)}{N} \xrightarrow{\cong} \pi_1(X, x),$$

giving the desired “formula” for $\pi_1(X, x)$ in terms of $\pi_1(U_0, x)$ and $\pi_1(U_1, x)$, and the two homomorphisms $(j_0)_*$ and $(j_1)_*$. We have defined rigorously what the domain of this homomorphism is, and the homomorphism itself. Theorem 3.1.1 asserts that it is surjective, but we have not proved injectivity, i.e. Theorem 3.2.2. We refer to e.g. Section 1.1.20 of Hatcher's proof for details of that proof, but let us mention that it is similar in spirit to surjectivity: if a word is in the kernel it means that some concatenation is null homotopic; then use compactness of $I \times I$ to cut that homotopy into finitely many pieces, each of which takes place entirely in either U_0 or U_1 ; the pieces of the homotopy taking place in $U_0 \cap U_1$ give rise to relations of the form (3.2).

An immediate corollary is the following:

Corollary 3.2.3. *Suppose $X = U \cup V$ with U, V open such that $U \cap V$ is simply connected. Then $\pi_1(X, x) \cong \pi_1(U, x) * \pi_1(V, x)$.*

Example 3.2.4. Let $X = S^1 \vee S^1 = (S^1 \sqcup S^1)/1 \sim 1$ be the wedge of two circles (as in the figure). Let U be X minus an arc in one circle and V be X minus an arc in the other circle. Then X , U and V satisfy the hypotheses of the corollary and we obtain that $\pi_1(X) \cong \pi_1(S^1) * \pi_1(S^1) \cong \mathbb{Z} * \mathbb{Z}$.



KLEIN BOTTLE =  = $\mathbb{R}^2 / G$

$G \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$
 $(a, b), (x, y) \mapsto (\frac{a}{2} + x, (-1)^a b + y)$

$G = \langle S, T \mid ST = TS^{-1} \rangle$
 $\cong \mathbb{Z}^2$ AS SETS BUT NOT AS GROUPS!

Let us collect a few constructions and proofs which may be mentioned in class, but for which we shall not have time to give full details.

3.3. Appendix: proof of Theorem 2.9.1. We have already constructed the set X , the free action $G \times X \rightarrow X$, and a map $p : X \rightarrow Y$ inducing a bijection $\bar{p} : X/G \rightarrow Y$. It remains to define the topology on X , prove that the action is continuous and satisfies the assumption of Proposition 2.4.3, and prove that $\bar{p}$ is a homeomorphism.

Let us construct for any $x \in X$, and any path connected open neighborhood $U \subset Y$ of $p(x)$ such that $\pi_1(U, p(x)) \rightarrow \pi_1(Y, p(x))$ is trivial, a map

$$j_{U,x} : U \rightarrow X,$$

as follows. The point x is represented by a path $\lambda : I \rightarrow Y$ with $\lambda(0) = y$ and $\lambda(1) = p(x)$. For any $v \in U$ we may use path connectedness to choose a path $\lambda_v : I \rightarrow U$ with $\lambda_v(0) = p(x)$ and $\lambda_v(1) = v$ and set $j_{U,x}(v) = [\lambda * \lambda_v]$. The condition on $\pi_1(U, p(x)) \rightarrow \pi_1(Y, p(x))$ implies that any loop in $(U, p(x))$ is homotopic to a constant loop, when regarded as a loop in $(Y, p(x))$, from which it follows that $[\lambda * \lambda_v] \in X$ is independent of choice of λ_v , so $j_{U,x}$ is well defined. Let us point out further properties, the (easy) proofs of which we leave to the reader

- (i) for all $v \in U$, $p \circ j_{U,x}(v) = v$,
- (ii) for $p(x) \in U' \subset U$, the map $j_{U',x}$ is the restriction of $j_{U,x}$,
- (iii) for $x' \in j_{U,x}(U)$, $j_{U,x} = j_{U,x'}$,
- (iv) $j_{U,g.x}(v) = g \cdot (j_{U,x}(v))$.

Now define a topology on X by declaring a subset $A \subset X$ open if and only if $j_{U,x}^{-1}(A)$ is a neighborhood of $p(x)$ for all such $x \in A$ and some sufficiently small open neighborhood U of $p(x) \in Y$ (if this holds for some U , then it holds for all smaller U as well, by (ii)).

Property (iii) then implies that the image $j_{U,x}(U)$ is then an open subset of X . If $V \subset U$ is an open path connected subset, then $j_{U,x}(V) = j_{V,x'}(V)$ for some $x' \in V \subset U$, so $j_{U,x}$ takes open path connected sets to open subsets of X . Since X , and hence U , is locally path connected, it follows that $j_{U,x} : U \rightarrow X$ is an open map. By definition of the topology it is also continuous, so it is a homeomorphism $U \rightarrow j_{U,x}(U)$.

Now we may deduce that the map $X \rightarrow X$ given by $x \mapsto g.x$ restricts to a homeomorphism $j_{U,x}(U) \rightarrow j_{U,g.x}(U)$, since property (iv) implies that it is the composition of $j_{U,g.x}$ and the inverse of $j_{U,x}$, regarded as homeomorphisms onto their image. It follows that the action $G \times X \rightarrow X$ is continuous.

Any $x \in X$ has a neighborhood of the form $V = j_{U,x}(U)$, but then $gV = j_{U,g.x}(U)$, which we claim is disjoint from V if $g \neq e$. Indeed, if $z \in V \cap gV$, then

$z = g.z'$ for some $z' \in V$, which must satisfy $p(z) = p(g.z') = p(z')$. If we write $z = j_{U,x}(v)$ and $z' = j_{U,x}(v')$, then we have $v = p \circ j_{U,x}(v) = p \circ j_{U,x}(v') = v'$, so in fact $z = z'$, but then $z = g.z'$ contradicts that the action is free. Hence we have proved that the action satisfies the assumption of Proposition 2.4.3.

Then we show that $p : X \rightarrow Y$ is continuous. Since the $j_{U,x}$ are homeomorphisms onto open subsets covering X , it suffices to see that $p \circ j_{U,x}$ is continuous for all U and x , but $p \circ j_{U,x}$ is just the inclusion $U \rightarrow Y$. Hence p and therefore $\bar{p} : X/G \rightarrow Y$ is continuous.

The map $j_{U,x} : U \rightarrow X$ composed with the quotient map $X \rightarrow X/G$ provides an inverse to $\bar{p}$, locally on the set U , and since both $j_{U,x}$ and the quotient map are continuous, then so is $\bar{p}^{-1}|_U$, but since the open sets U cover Y we have shown $\bar{p}^{-1}$ is also continuous. $\square$

3.4. Appendix: words, free monoids, and free groups. In order to state the general van Kampen theorem, we first need to discuss a bit of algebra, namely *coproducts of groups*, in the literature often called “free products”. Most of this section is purely algebraic, developing the algebra necessary to formulate the general van Kampen theorem.

Let J be a set and let G_j be a group for each $j \in J$. In this section we shall associate a new group

$$G = \bigast_{j \in J} G_j,$$

called the *coproduct* of the G_j , which contain all $G_j \subset G$ as subgroups (actually, more precisely there are specified isomorphisms between the G_j and subgroups of G), such that $G_i \cap G_j = \{e\}$ for $i \neq j$. This group G has the “universal property” that given any group H and any set of homomorphisms $\psi_i : G_i \rightarrow H$, there is precisely one homomorphism $\psi : G \rightarrow H$ with $\psi|_{G_j} = \psi_j$ for all $j \in J$.

The idea of the construction is that elements of G should be “formal expressions” $w = m_1 m_2 \dots m_n$ where each m_i belongs to $G_{\sigma(i)}$ for some $\sigma(i) \in J$, and that these formal expressions should be subject only to the relations which follow from the group axioms. In the rest of this appendix we make this idea precise, and use it to formulate the general van Kampen theorem.

3.4.1. Appendix: coproducts of monoids. It is convenient to first work in the more general context of *monoids*. Recall that a monoid is a set M with a product $M \times M \rightarrow M$ which is associative and unital. Unital means that there exists an element $e \in M$ such that $em = m = me$ for all $m \in M$; such an e is necessarily unique. A homomorphism of monoids is a function $f : M \rightarrow M$ such that $f(mm') = f(m)f(m')$ for all $m, m' \in M$ and such that $f(e) = e'$. (The second property is automatic if M' is a group, but not in general.)

Words. Fix a set J and monoids M_j as above. A *word* is a triple (n, σ, w) , where $n \in \mathbb{N}$ is a non-negative integer, $\sigma : \{1, \dots, n\} \rightarrow J$ is a function, and

$$w = (w_1, \dots, w_n) \in \prod_{k=1}^n M_{\sigma(k)},$$

i.e. a k -tuple whose k th coordinate is an element of $M_{\sigma(k)}$.

Let us write W for the set of words (the set W of course depends on J and the M_j , but we omit that from the notation). If (n', σ', w') is another word, with $w' = (w'_1, \dots, w'_{n'})$, then the *concatenation* of the two words is $(n + n', \sigma * \sigma', w * w')$, where

$$(\sigma * \sigma')(k) = \begin{cases} \sigma(k) & \text{if } k \leq n \\ \sigma'(k - n) & \text{if } n < k \leq n + n' \end{cases}$$

and

$$w * w' = (w_1, \dots, w_n, w'_1, \dots, w'_{n'}).$$

This indeed defines an element of W : the k th coordinate is an element of $M_{(\sigma * \sigma')(k)}$.

Informally, we shall sometimes write elements of W simply as $w = (w_1, \dots, w_n)$ and omit the function $\sigma : \{1, \dots, n\} \rightarrow J$ from the notation, but we emphasize that σ is nevertheless an important part of the data. Notice also that $n = 0$ is allowed, giving rise to an element denoted $() \in W$.

Lemma 3.4.1. *Concatenation defines a monoid structure on W .*

Proof. It is easy to see associativity: if $w = (w_1, \dots, w_n)$, $w' = (w'_1, \dots, w'_{n'})$, and $w'' = (w''_1, \dots, w''_{n''})$, then $(w * w') * w''$ and $w * (w' * w'')$ are both equal to

$$(w_1, \dots, w_n, w'_1, \dots, w'_{n'}, w''_1, \dots, w''_{n''}).$$

The 0-tuple $() \in W$ is a unit for concatenation. $\square$

The number n is called the *length* of the word w and shall be denoted $\ell(w)$. The unit for the monoid has length 0 and is the only word of length 0.

Reducing words. Let us say that a word $w' \in W$ is an *elementary multiplication reduction* of another word $w \in W$ provided there exist a $j \in J$, words $w_0, w_1 \in W$, and elements $m_0, m_1 \in M_j$ such that

$$\begin{aligned} w &= w_0 * \varphi_i(m_0) * \varphi_i(m_1) * w_1 \\ w' &= w_0 * \varphi_i(m_0 m_1) * w_1. \end{aligned}$$

We say that w' is an *elementary unit reduction* of w provided there exist a $j \in J$ and words $w_0, w_1 \in W$ such that

$$\begin{aligned} w &= w_0 * \varphi_i(e) * w_1 \\ w' &= w_0 * w_1. \end{aligned}$$

We say that w' is an *elementary reduction* of w if it is either an elementary multiplication reduction or an elementary unit reduction. Notice that in either case we have $\ell(w') = \ell(w) - 1$. We say that w' is a *reduction* of w if there exist an $n \in \mathbb{N}$ and words $w_0, \dots, w_n \in W$ such that each w_i is an elementary reduction of w_{i-1} and such that $w_0 = w$ and $w_n = w'$. In this case $n = \ell(w) - \ell(w')$ is a non-negative integer, and is strictly positive unless $w = w'$.

Definition 3.4.2. Let us write $w \geq w'$ if w' is a reduction of w . A word $w \in W$ is *reduced* if $w \geq w'$ implies $w = w'$, i.e. if w admits no elementary reductions.

Let us also write $\sim$ for the equivalence relation on W generated by $\geq$.

Lemma 3.4.3. *If $w_0 \sim w'_0$ and $w_1 \sim w'_1$, then $w_0 w_1 \sim w'_0 w'_1$.*

Proof. It is clear from the definition that if w'_0 is an elementary reduction of w_0 , then $w'_0 w_1$ is an elementary reduction of $w_0 w_1$ and in particular $w'_0 w_1 \sim w_0 w_1$. Since $\sim$ is generated by elementary reduction we deduce that $w_0 \sim w'_0$ implies $w_0 w_1 \sim w'_0 w_1$. By a similar argument, if $w_1 \sim w'_1$, then $w'_0 w_1 \sim w'_0 w'_1$. $\square$

Coproducts of monoids. Let $\bar{\varphi}_i : M_i \rightarrow W$ be given by $\varphi_i(m) = (m)$. (More precisely, $n = 1$ and $\sigma(1) = i$.) The functions $\bar{\varphi}_i : M_i \rightarrow W$ are not homomorphisms: for example, they do not send identity element to identity element. We wish to pass to a quotient of W in order to “force” the φ_i to be homomorphisms: this is precisely what the equivalence relation $\sim$ is for.

Definition 3.4.4. Let

$$M = \bigstar_{j \in J} M_j = W / \sim,$$

and for two elements $[w]$ and $[w'] \in M$, define $[w][w'] = [w * w']$. This is well defined by Lemma 3.4.3, makes M into a monoid, and the quotient map $q : W \rightarrow M$ into a homomorphism. Let $\varphi_i : M_i \rightarrow \bigstar_{j \in J} M_j$ be the homomorphisms given by the composition

$$\varphi_i : M_i \xrightarrow{\bar{\varphi}_i} W \xrightarrow{q} M.$$

The functions ϕ_j are indeed homomorphisms: the elements $\bar{\varphi}_i(mm')$ and $\bar{\varphi}_i(m)\bar{\varphi}_i(m')$ of W differ by an elementary reduction and hence become equal in M . Similarly, $\bar{\varphi}_i(e)$ and $()$ differ by an elementary reduction so become equal in M .

The class $q(w)$ of a word $w = (w_1, \dots, w_n)$ with $w_i \in M_{\sigma(i)}$ is then the product

$$(3.3) \quad q(w_1, \dots, w_n) = \varphi_{\sigma(1)}(w_1) \dots \varphi_{\sigma(n)}(w_n).$$

The monoid $\bigstar_{j \in J}$ is called the *coproduct* of the monoids M_j , and the following result is known as the “universal property” of this construction.

Theorem 3.4.5. *Let J and M_j be as above, and suppose given a further monoid N and homomorphisms $\psi_j : M_j \rightarrow N$ for each j . Then there exists a unique homomorphism $\psi : \bigstar_{j \in J} M_j \rightarrow N$ such that $\psi_j = \psi \circ \varphi_j$ for all $j \in J$.*

Proof. The formula (3.3) shows that the class of a word $w = (w_1, \dots, w_n) \in W$ with $w_i \in M_{\sigma(i)}$ must have

$$\psi([w]) = \psi_{\sigma(1)}(w_1) \dots \psi_{\sigma(n)}(w_n) \in N,$$

which proves uniqueness. To see existence, it suffices to see that if we change one of the w_i by an elementary reduction, then the right hand side of this equation does not change. This follows from the assumption that each ψ_i preserves multiplication and units. $\square$

In the case $J = \{0, 1\}$, the coproduct of M_0 and M_1 is often written as $M_0 * M_1$, and the universal property may be paraphrased by a diagram

$$\begin{array}{ccccc} M_0 & & & & \\ & \searrow \varphi_0 & & \searrow \psi_0 & \\ & & M_0 * M_1 & \cdots \cdots \cdots & N \\ & \nearrow \varphi_1 & & \nearrow \psi_1 & \\ M_1 & & & & \end{array}$$

Namely, the theorem asserts that given all the solid arrows, there is precisely one dotted arrow which makes both triangles commute.

3.4.2. Appendix: coproducts of groups. Since groups are special cases of monoids, we may form a monoid $\bigstar_{j \in J} G_j$ for a family of groups G_j , indexed by $j \in J$.

Lemma 3.4.6. *Let J be a set, and G_j a group for each $j \in J$. Then the monoid $\bigstar_{j \in J} G_j$ is a group.*

Proof. An arbitrary element $x \in \ast_{j \in J} G_j$ may be represented by a word $w = (g_1, \dots, g_n)$ for some $n \geq 0$ and $g_i \in G_{\sigma(i)}$. Then the word $w' = (g_n^{-1}, \dots, g_1^{-1})$ represents an inverse. $\square$

Restating the universal property of $\ast_{j \in J} G_j$ in this special case, we deduce the following.

Theorem 3.4.7. *Let J and G_j be as above, and suppose given a further group H and homomorphisms $\psi_j : G_j \rightarrow H$ for each j . Then there exists a unique homomorphism $\psi : \ast_{j \in J} G_j \rightarrow H$ such that $\psi_j = \psi \circ \varphi_j$ for all $j \in J$.* $\square$

Remark 3.4.8. In particular we may form the coproduct $G_0 \ast G_1$ of two groups G_0 and G_1 . (This is the same as what is known as the “free product” of G_0 and G_1 in some other textbooks.)

3.4.3. Appendix: reduced words. The following result gives a firmer grasp on the underlying set of the coproduct $M = \ast_{j \in J} M_j$. Recall from Definition 3.4.2 that a word is reduced if it admits no elementary reductions.

Theorem 3.4.9. *Let $R \subset W$ be the set of reduced words, and $q : W \rightarrow M = W/\sim$ be the quotient map. Then the composition*

$$R \hookrightarrow W \xrightarrow{q} M$$

is a bijection of sets.

Corollary 3.4.10. *The monoid homomorphisms $\varphi_i : M_i \rightarrow M$ are injective, and we have*

$$\varphi_i(M_i) \cap \varphi_j(M_j) = \{e\}$$

if $i \neq j$.

Proof of the Corollary, using the Theorem. If $m, m' \in M_i \setminus \{e\}$, then the reduced words representing $\varphi_i(m)$ and $\varphi_i(m')$ are (m) and (m') respectively. These are equal only if $m = m'$. If $m \neq e$ and $m' = e$, then $\varphi_i(m) \neq \varphi_i(m')$ since the reduced words (m) and $()$ are different (they even have different length). Similarly, if $m \in M_i \setminus \{e\}$ and $m' \in M_j \setminus \{e\}$ with $i \neq j$, then (m) and (m') are distinct reduced words. $\square$

In this sense, the monoid M “contains” the monoids M_j : we may use φ_j to identify M_j with the submonoid $\varphi_j(M_j) \subset M$. Under this identification, the submonoids $M_j \subset M$ are also disjoint except of course that they share the identity element.

The following lemma is the essential step in the proof of the theorem.

Lemma 3.4.11. *For any $w \in W$ there is precisely one reduced word $r(w)$ with $w \geq r(w)$.*

Proof. Existence is easily proved by induction on $\ell(w)$. If $\ell(w) = 0$ then $w = ()$ and we (must) set $r(w) = w$. Suppose inductively we know existence for all words of length strictly smaller than $\ell(w)$. If w is already reduced then we may set $r(w) = w$, otherwise we may pick an elementary reduction w' of w and set $r(w) = r(w')$.

To see uniqueness, we again proceed by induction on $\ell(w)$. For $w = ()$, clearly $r(w) = ()$ is the only option. Suppose uniqueness is proved for all words of length

strictly smaller than $\ell(w)$. If $w \geq a$ and $w \geq b$ for reduced words a and b , then there exists a string of elementary reductions

$$\begin{aligned} w &= w_0 \geq w_1 \geq \cdots \geq w_n = a \\ w &= w'_0 \geq w'_1 \geq \cdots \geq w'_{n'} = b. \end{aligned}$$

If w is already reduced, then $n = n' = 0$ and $a = w = b$. Otherwise, n and n' are both positive, so we have two elementary reductions w_1 and w'_1 of the same word w . If $w_1 = w'_1$, then a and b are two reductions of a word of length $\ell(w) - 1$ and hence $a = b$, by the induction hypothesis. Otherwise, there exists a word w'' , which is an elementary reduction of both w_1 and w'_1 : for example if w_1 and w'_1 are obtained from w by removing an identity from w in two different entries, then w'' is obtained from w by removing both; similarly if w_1 and/or w'_1 is instead obtained by multiplying two adjacent entries, then w'' is obtained from w by performing both of the two elementary reductions.

Since $\ell(w_1) = \ell(w'_1) = \ell(w) - 1$ and $\ell(w'') = \ell(w) - 2$, these three words reduce to unique reduced words $r(w_1)$, $r(w'_1)$, and $r(w'')$, by the induction hypothesis. Since a is a reduced word which is a reduction of w_1 we must have $r(w_1) = a$, and similarly $r(w'_1) = b$. Since w'' is an elementary reduction of w_1 we must have $r(w_1) = r(w'')$, and similarly $r(w'_1) = r(w'')$. Hence $a = r(w_1) = r(w'') = r(w'_1) = b$, establishing the induction step. $\square$

Corollary 3.4.12. *For any $w \in W$ there exists a unique reduced word $r(w)$ with $w \sim r(w)$.*

Proof. Let $R \subset W$ denote the set of reduced words. The lemma gives a well defined retraction $r : W \rightarrow R$, and it is clear that $w \geq w'$ implies $r(w) = r(w')$. Since $\geq$ generates $\sim$, it follows that $r : W \rightarrow R$ is constant on equivalence classes. Hence there cannot exist a zig-zag of reductions connecting two distinct reduced words. $\square$

3.4.4. Appendix: free groups on a set of generators. Let us discuss the special case where all G_j are infinite cyclic groups on a specified generator g_j so

$$G_j = \langle g_j \rangle = \{g_j^n \mid n \in \mathbb{Z}\}.$$

Let $G = \ast_{j \in J} G_j$ be the coproduct, which comes with the (injective) homomorphisms $\phi_i : G_i \rightarrow G$. By a slight abuse of notation we shall write also $g_j \in G$ for the (distinct, by Theorem 3.4.9) elements $\phi_j(g_j)$.

The resulting group $G = \ast_{j \in J} \langle g_j \rangle$ is called the *free group* on generators $g_j \in G$. Its elements may be written as

$$g = g_{\sigma(1)}^{a_1} g_{\sigma(2)}^{a_2} \cdots g_{\sigma(n)}^{a_n}$$

for indices $\sigma(1), \dots, \sigma(n) \in J$ and integers $a_1, \dots, a_n \in \mathbb{Z}$.

An interesting special case is when $J = \{0, 1\}$ in which case we will write $x = g_0$ and $y = g_1$. The resulting group G is called the “free group on two generators” and its elements may be written as $g = x^{a_1} y^{b_1} \cdots x^{a_n} y^{b_n}$ for $n \geq 0$ and integers a_i, b_i . It is not commutative, since Theorem 3.4.9 implies that for example $xy \neq yx$. The general van Kampen theorem shows that this is also isomorphic to the fundamental group of the plane with two points removed.

The free group on the generators $g_j \in G$ has the universal property that if H is any other group and there are specified elements $h_j \in H$ for all $j \in J$, then there is one and only one homomorphism $\psi : G \rightarrow H$ with $\psi(g_j) = h_j$.

4. CATEGORIES AND FUNCTORS (WEEK 4)

We give here a short introduction to the notions of *categories* and *functors*. This language is used throughout mathematics.

See also Hatcher, section 2.3.

4.1. Categories. Essentially all areas of mathematics study some kind of objects (vector spaces, groups, topological spaces, ...) and there is a notion of “map” between these objects (linear maps, group homomorphisms, continuous maps, ...). All these examples share some very basic features which are formalized by *category theory*; for example the maps may be composed and composition of maps is associative. Here is the formal definition.

Definition 4.1.1. A *category* C consists of the following data.

- (i) A collection $\text{ob}(C)$. Elements of this collection are called “objects of C ”.
- (ii) For each pair of objects $x, y \in \text{Ob}(C)$ a set $C(x, y)$. Elements of this set are called “morphisms from x to y ”.
- (iii) For each triple of objects $x, y, z \in \text{Ob}(C)$ a function $C(y, z) \times C(x, y) \rightarrow C(x, z)$. This function is called “composition” and denoted $(g, f) \mapsto g \circ_C f$ (or $g \circ f$ or just gf).
- (iv) For each object $x \in \text{Ob}(C)$ a morphism $\text{id}_x \in C(x, x)$. This morphism is called the “identity of x ”.

This data is subject to the following axioms.

- (v) For each 4-tuple of objects $x, y, z, w \in \text{Ob}(C)$ and morphisms $f \in C(x, y)$, $g \in C(y, z)$, and $h \in C(z, w)$ we have $(h \circ_C g) \circ_C f = h \circ_C (g \circ_C f)$.
- (vi) For all objects $x, y \in \text{Ob}(C)$ and morphisms $f \in C(x, y)$ we have $f \circ \text{id}_x = f = \text{id}_y \circ f$.

Example 4.1.2.

- (i) There is a category **Sets** whose objects are sets, morphisms are functions, and whose composition is composition of functions: $(f \circ g)(x) = f(g(x))$.
- (ii) For a field k there is a category **Vect** $_k$ whose objects are k -vector spaces, whose morphisms are the linear maps, and whose composition is composition of linear maps.
- (iii) There is a category **Top** whose objects are topological spaces, whose morphisms are continuous maps, and whose composition is composition of continuous functions.
- (iv) Etc.

It is traditional that the name of a category refers mostly to what the objects are, but we emphasize that the two other pieces of data, namely the morphism sets $C(x, y)$ for all x, y , and composition functions $\circ_C : C(y, z) \times C(x, y) \rightarrow C(x, z)$, are equally important parts of the data.

Sketch of a proof that the above examples actually define categories. The important case to treat is **Sets**. The fact that composition is associative is well known and follows from the definition: to check that $h \circ (g \circ f) = (h \circ g) \circ f$ we check that these functions have the same value on any element: $(h \circ (g \circ f))(x) = h((g \circ f)(x)) = h(g(f(x))) = (h \circ g)(f(x)) = ((h \circ g) \circ f)(x)$. For each set x , the identity map of x satisfies the requirements for id_x .

All the other examples are “sets with extra structure”, and the morphisms are “functions which preserve the structure”. Therefore associativity of composition is

inherited from **Sets**. We ought to explain why composition is actually well defined, but this is done in elementary courses (composition of linear functions is again linear, composition of continuous maps is again continuous, etc). We should also explain that the identity map is in the category, but that is also elementary (the identity map of a vector spaces is linear, the identity map of a topological space is continuous, etc). $\square$

Remark 4.1.3. In many *examples* of categories, e.g. **Sets** the objects have also “elements”, but it is important to emphasize that this is not part of the formal structure of a category. (As a simple analogy, many groups consist of bijections of a certain set, for example symmetric groups, alternating groups, general linear groups, etc., but the axioms for an abstract group does not involve any bijections of a set.)

Remark 4.1.4. The identity morphisms id_x are unique if they exist. Indeed, it follows from the axioms for id_x that if $\text{id}'_x \in C(x, x)$ is another morphism satisfying $1'_x \circ f = f$ for all f , then $\text{id}_x = \text{id}'_x \circ \text{id}_x = \text{id}'_x$. (Thus, instead of considering the morphisms id_x as *data* when specifying a category C , we would have required their existence for all $x \in \text{Ob}(C)$ as an axiom.)

There are also many constructions which make new categories out of old ones. A very simple one is the *opposite category* C^{op} . It has the same objects as C , but morphism sets are defined as $C^{\text{op}}(x, y) = C(y, x)$ and if composition in C is denoted $\circ_C$ then composition in C^{op} is defined as $f \circ_{C^{\text{op}}} g = g \circ_C f$.

A useful category of spaces defined using the equivalence relation of homotopy is the following:

Definition 4.1.5. Let $\mathcal{H}$ be the category whose objects are topological spaces, and whose morphisms are *homotopy classes* of maps, i.e.

$$\mathcal{H}(X, Y) = \frac{f : X \rightarrow Y \mid f \text{ is continuous}}{\simeq},$$

and where composition is defined as $[g] \circ_{\mathcal{H}} [f] = [g \circ f]$.

We have seen a while ago that this composition is well defined, i.e. that if $[f] = [f']$ and $[g] = [g']$, then $[g] \circ [f] = [g'] \circ [f']$. We should also verify existence of identity morphisms and that composition is associative, but that is inherited from **Top**.

4.2. Isomorphisms. An important notion in a category, that you have already met in various disguises elsewhere, is that of isomorphism:

Definition 4.2.1. Let C be a category and $x, y \in \text{Ob}(C)$. A morphism $f \in C(x, y)$ is an *isomorphism* if there exists a morphism $g \in C(y, x)$ such that $f \circ g = \text{id}_y$ and $g \circ f = \text{id}_x$.

Two objects $x, y \in \text{Ob}(C)$ are *isomorphic* if there exists an isomorphism $f \in C(x, y)$.

In particular categories this spells out to well known notions: the isomorphisms in **Sets** are precisely the *bijections*, the isomorphisms in **Vect** $_k$ are precisely the k -linear isomorphisms, and the isomorphisms in **Top** are precisely the homeomorphisms. (Not the continuous bijections!)

It is not hard to see that for fixed f there is at most one morphism g with $f \circ g = \text{id}_y$ and $g \circ f = \text{id}_x$. In contrast, if x and y are isomorphic, then there are usually many different isomorphisms between them.

Lemma 4.2.2. *Let $\mathcal{H}$ be the homotopy category from Definition 4.1.5. Then a morphism $[f] \in \mathcal{H}(X, Y)$ is an isomorphism if and only if a representative $f : X \rightarrow Y$ is a homotopy equivalence.*

Proof. Suppose f is a homotopy equivalence and let $g : Y \rightarrow X$ be a homotopy inverse. Then the equations $g \circ f \simeq \text{id}_X$ and $f \circ g \simeq \text{id}_Y$ imply $[g] \circ [f] = [g \circ f] = [\text{id}_X]$ and similarly $[f] \circ [g] = [\text{id}_Y]$. Since $[\text{id}_X]$ and $[\text{id}_Y]$ are identity morphisms in $\mathcal{H}$, we have proved that $[f]$ is an isomorphism in $\mathcal{H}$.

Conversely, if $[f] \in \mathcal{H}(X, Y)$ is an isomorphism we may (by definition of isomorphism) pick $[g] \in \mathcal{H}(Y, X)$ with $[f] \circ [g] = [\text{id}_Y]$ and $[g] \circ [f] = [\text{id}_X]$. But then $[f \circ g] = [f] \circ [g] = [\text{id}_Y]$ which means $f \circ g \simeq \text{id}_Y$, and $g \circ f \simeq \text{id}_X$ is deduced similarly. $\square$

4.3. Functors. We now define the notion of functor, which are the maps between categories.

Definition 4.3.1. Let C and D be categories. A *functor* $F : C \rightarrow D$ consists of the following data.

- (i) For each $x \in \text{Ob}(C)$ an object $Fx \in \text{Ob}(D)$.
- (ii) For each morphism $f \in C(x, y)$ a morphism $Ff \in D(Fx, Fy)$.

This data is subject to the following requirements.

- (iii) $F(\text{id}_x) = \text{id}_{Fx}$ for all $x \in \text{Ob}(C)$.
- (iv) For all $g \in C(y, z)$ and $f \in C(x, y)$ we have $F(g \circ_C f) = Fg \circ_D Ff \in D(Fx, Fz)$.

Example 4.3.2. For any category C we have the *identity functor* $\iota : C \rightarrow C$ defined on objects as $\iota x = x$ and on morphisms as $\iota f = f$.

Example 4.3.3. For any two categories C and D and an object $d \in \text{Ob}(D)$, there is a *constant functor* $c : C \rightarrow D$ defined on objects as $c(x) = d$ and $c(f) = \text{id}_d$ for all $x, y \in \text{Ob}(C)$ and all $f \in C(x, y)$.

Example 4.3.4. The category $\mathbf{Top}_*$ has objects the based spaces (X, x) . Morphisms from (X, x) to (Y, y) are the continuous maps $f : X \rightarrow Y$ with $f(x) = y$. Composition is composition of maps. Then we have seen that the fundamental group defines a functor

$$\pi_1 : \mathbf{Top}_* \rightarrow \mathbf{Groups}.$$

Example 4.3.5. Let $\mathbf{Groups}$ be the category whose objects are groups, morphisms are group homomorphisms, and composition is composition of homomorphisms. Let $\mathbf{Ab}$ be the category whose objects are the abelian groups, morphisms are group homomorphisms, and composition is composition of homomorphisms.

Then we may define a functor $\mathbf{Groups} \rightarrow \mathbf{Ab}$ as follows. For a group G the *commutator subgroup* $G' < G$ is the subgroup generated by all commutators $[x, y] = xyx^{-1}y^{-1}$. The commutator subgroup is automatically normal, and we may define $\text{ab}(G) = G/G'$. Then all commutators in $\text{ab}(G)$ give the neutral element, and hence $\text{ab}(G)$ is abelian. If $\phi : H \rightarrow G$ is a group homomorphism, then it sends a commutator $[x, y]$ to the commutator $[\phi(x), \phi(y)]$ and hence $\phi(H') \subset G'$ so ϕ induces a map of quotient groups $\text{ab}(\phi) : \text{ab}(H) \rightarrow \text{ab}(G)$. It is easy to check that this operation on groups and group homomorphisms define a functor $\text{ab} : \mathbf{Groups} \rightarrow \mathbf{Ab}$.

Example 4.3.6. If C, C' and C'' are categories, and $F : C \rightarrow C'$ and $G : C' \rightarrow C''$ are functors, then we may form the composition $GF : C \rightarrow C''$. It is defined on objects as $(GF)(x) = G(F(x))$ and on morphisms as $(GF)(f) = G(F(f))$.

For example, we may define a functor $\mathbf{Top}_* \rightarrow \mathbf{Ab}$ by composing π_1 and ab .

Most important constructions in algebraic topology are functors, and in particular may be applied to morphisms as well as to objects. The source category for the functors is often $\mathbf{Top}$ or some variation, while the target category is of a more algebraic nature (groups, vector spaces, ...).

A useful result about functors is that they preserve isomorphisms:

Theorem 4.3.7. *Let $F : C \rightarrow D$ be a functor, let $x, y \in \text{Ob}(C)$, and let $f \in C(x, y)$ be an isomorphism. Then $Ff \in D(Fx, Fy)$ is also an isomorphism.*

Proof. Let $g \in C(y, x)$ satisfy $g \circ f = \text{id}_x$ and $f \circ g = \text{id}_y$. Then use the axioms for a functor to deduce that

$$\text{id}_{Fx} = F(\text{id}_x) = F(g \circ f) = Fg \circ Ff$$

and similarly $\text{id}_{Fy} = Ff \circ Fg$. But that says that $Ff \in D(Fx, Fy)$ is an isomorphism with inverse Fg . $\square$

4.4. Some more exotic looking examples. Let us give one slightly more exotic looking example. These will not be directly relevant for this class, but may help illustrate other aspects of categories.

Example 4.4.1. For a group G , let C_G be the following category.

- (1) $\text{Ob}(C_G)$ consists of a single object, which we shall denote $*$.
- (2) $C_G(*, *) = G$.
- (3) The composition map $C_G(*, *) \times C_G(*, *) \rightarrow C_G(*, *)$ is the multiplication map $G \times G \rightarrow G$ of the group.
- (4) The morphism $\text{id}_* \in C_G(*, *) = G$ is the neutral element of G .

We leave it to the reader to verify that this satisfies the axioms for a category (they will notice that the proof does not use inverses, and in fact we could have assumed only that G is a monoid).

We also leave it to the reader to explain why a group homomorphism $\phi : H \rightarrow G$ induces a functor $C_H \rightarrow C_G$ given on morphisms by ϕ , and that all functors $C_H \rightarrow C_G$ are of this form.

Example 4.4.2. Let C_G be as above and let $F : C_G \rightarrow \mathbf{Sets}$ be a functor. Then $X = F(*)$ is a set, and for each $g \in G = C_G(*, *)$ the functor F assigns a map $Fg : X \rightarrow X$ of sets. This assignment must send the neutral element of G to the identity map and must send group multiplication to composition of maps $x \rightarrow x$. This is just the axioms for a group action! (I.e. if we set $gx = (Fg)(x)$ for $g \in G$ and $x \in X$ then $e.x = x$ and $(gg').x = g.(g'.x)$.) Conversely, any (left) group action $G \times X \rightarrow X$ gives rise to a functor $F : C_G \rightarrow \mathbf{Sets}$ by setting $F(*) = X$ and $(Fg)(x) = g.x$.

In this way, a functor $F : C_G \rightarrow \mathbf{Sets}$ is precisely the same data as that of a set X together with a left action of G on X .

Example 4.4.3. A *poset* is a pair $(P, \leq)$ where P is a set and $\leq$ is a relation on P satisfying $x \leq x$ for all $x \in P$ and that $x \leq z$ for all $x, y, z \in P$ with $x \leq y$ and $y \leq z$.

Let $\{*\}$ be a one-element set. To a poset $(P, \leq)$ we may associate a category $C = C_{(P, \leq)}$ with $\text{Ob}(C) = P$ and with

$$C(x, y) = \begin{cases} \{*\} & \text{if } x \leq y \\ \emptyset & \text{if not } x \leq y. \end{cases}$$

Composition $C(y, z) \times C(x, y) \rightarrow C(x, z)$ is defined by sending $(*, *) \mapsto *$ if $x \leq y \leq z$ and as the unique function $\emptyset \rightarrow \emptyset$ if not $x \leq y \leq z$.

4.5. Appendix: Functor categories. The following definition gives a notion of maps between functors:

Definition 4.5.1. Let C and D be categories and $F, G : C \rightarrow D$ be two functors. A *natural transformation* $T : F \Rightarrow G$ is the data of a morphism $Tx \in D(Fx, Gx)$ for each object x of C , subject to the axiom that the diagram

$$\begin{array}{ccc} Fx & \xrightarrow{T_x} & Gx \\ \downarrow Ff & & \downarrow gf \\ Fy & \xrightarrow{T_y} & Gy \end{array}$$

is commutative for all objects x and y of C , and all morphisms $f \in C(x, y)$.

Example 4.5.2. For any functor $F : C \rightarrow D$ there is the *identity natural transformation* $\iota : F \Rightarrow F$, given by $\iota x = \text{id}_{Fx}$ for all objects x of C .

Definition 4.5.3. Let C and D be categories, let F, G , and H be functors $C \rightarrow D$, and let $S : F \Rightarrow G$ and $T : G \Rightarrow H$ be natural transformations. Then the *composition* $T \circ S : F \Rightarrow H$ is the natural transformation given by

$$(T \circ S)x = (Tx) \circ (Sx) \in D(Fx, Hx).$$

Lemma 4.5.4. *Composition of natural transformations is transitive, in the sense that $(T_0 \circ T_1) \circ T_2 = T_0 \circ (T_1 \circ T_2)$, and satisfies $T \circ \iota = T = \iota \circ T$.*

Proof. This is inherited from the corresponding properties of composition in D . $\square$

The lemma essentially asserts that we may define a category $\mathbf{Fun}(C, D)$ whose objects are the functors $C \rightarrow D$, and whose morphisms from F to G are the natural transformations $F \Rightarrow G$. However, in this generality it may not satisfy the requirement that the morphisms from F to G form a set. This nuisance is related to the fact that we do not require the collection of objects of C to be a set. (You may have heard that set theory does not allow for a “set of all sets”; but we do want a “category of all sets”.)

Definition 4.5.5. A *small category* C consists of a set $\text{Ob}(C)$, for each pair of objects $x, y \in \text{Ob}(C)$ a set $C(x, y)$, for each triple of objects $x, y, z \in \text{Ob}(C)$ a function $C(y, z) \times C(x, y) \rightarrow C(x, z)$, and for each object $x \in \text{Ob}(C)$ a morphism $\text{id}_x \in C(x, x)$, satisfying the requirements in Definition 4.1.1.

Lemma 4.5.6. *Let C and D be small categories. Then there is a (small) category $\mathbf{Fun}(C, D)$ whose set of objects is the set of all functors $C \rightarrow D$, whose morphisms from F to G is the set of natural transformations $F \Rightarrow G$, whose composition is composition of natural transformations, and whose identities are the identity natural transformations.*

Definition 4.5.7. A natural transformation $T : F \Rightarrow G$ is a *natural isomorphism* if there exists a natural transformation $S : G \Rightarrow F$ such that $S \circ T$ is equal to ι_F and $T \circ S$ is equal to ι_G .

In case C and D are small, this condition is easily seen to agree with isomorphism in the category $\mathbf{Fun}(F, G)$.

Lemma 4.5.8. *A natural transformation $T : F \Rightarrow G$ is a natural isomorphism if and only if each morphism $Tx \in D(Fx, Gx)$ is an isomorphism in D .*

Proof. If all Tx are isomorphisms, we may define S by letting $Sx \in D(Gx, Fx)$ be the inverse of Tx for each x . The reverse implication is similar. $\square$

5. SINGULAR HOMOLOGY (WEEK 4)

We now turn to the second major topic in this class, *singular homology*. The first task is to define the singular chain groups $C_p(X)$ of a topological space X for all $p \in \mathbb{N}$, and the boundary homomorphism $\partial : C_p(X) \rightarrow C_{p-1}(X)$.

The reference in Hatcher for this topic is Section 2.1.

5.1. Free abelian group. Let X be a set. Informally, the *free abelian group* with basis X is the of formal linear combinations of elements of X with coefficients in $\mathbb{Z}$, i.e. expressions

$$c = \sum_{x \in X} c_x x$$

where all $c_x \in \mathbb{Z}$ and that $c_x = 0$ for all but finitely many $x \in X$. Two such expressions are equal if and only if the coefficients n_x agree for all $x \in X$. Addition is defined the obvious way: if $c' = \sum_{x \in X} c'_x x$, then

$$c + c' = \sum_{x \in X} (c_x + c'_x) x.$$

Definition 5.1.1. Let $\mathbb{Z}X$ be the group of formal linear combinations $\sum_{x \in X} c_x x$ with addition defined as above.

(In case you worry about what “formal expression” might mean, let’s say the official definition is the set of functions $c : X \rightarrow \mathbb{Z}$ such that $\{x \in X \mid c(x) \neq 0\}$ is finite.)

For each $x_0 \in X$ there is a corresponding element of $\mathbb{Z}X$, viz. the linear combination $\sum_{x \in X} c_x x$ with $c_{x_0} = 1$ and $c_x = 0$ for all $x \neq x_0$. We shall use the same notation $x_0 \in \mathbb{Z}X$ for this element associated to $x_0 \in X$.

Lemma 5.1.2. *Let A be an abelian group. Then two homomorphisms $\phi, \psi : \mathbb{Z}X \rightarrow A$ agree if and only if $\phi(x) = \psi(x)$ for all $x \in X$.*

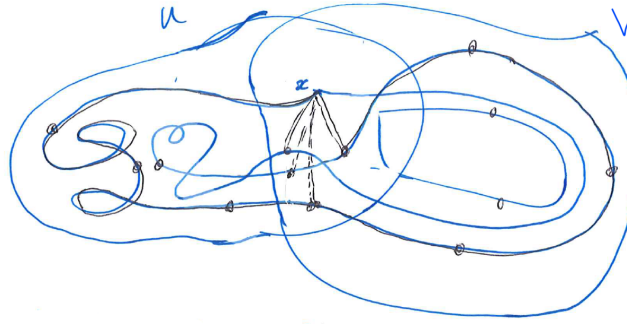
For any function $f : X \rightarrow A$ there exists a homomorphism $\phi : \mathbb{Z}X \rightarrow A$ with $\phi(x) = f(x)$ for all $x \in X$.

Proof. Since any linear element of $\mathbb{Z}X$ is a $\mathbb{Z}$ -linear combination of elements of X , and homomorphisms preserve linear combinations, two homomorphisms from $\mathbb{Z}X$ must agree if they agree X .

Given $f : X \rightarrow A$, define ϕ by $\phi(c) = \sum c_x f(x)$ if $c = \sum c_x x$. This is well defined because the numbers c_x are uniquely determined by c , and clearly satisfies $\phi(x) = f(x)$ for $x \in X$. It is a homomorphism by the definition of addition in $\mathbb{Z}X$. $\square$

Said differently, the lemma asserts that the function $\mathbf{Ab}(\mathbb{Z}X, A) \rightarrow \mathbf{Sets}(X, A)$, which sends $\phi \mapsto \phi|_X$, is a bijection for all sets X and all abelian groups A .

In practice, this Lemma is often used to define homomorphisms out of free abelian groups: for example, we may define a homomorphism $\phi : \mathbb{Z}X \rightarrow A$ by giving some formula for $\phi(x)$ when $x \in X$, and then letting ϕ be the unique extension to a homomorphism. For example, we may write “Let $\phi : \mathbb{Z}X \rightarrow \mathbb{Z}$ be the homomorphism given as $\phi(x) = 1$ for all $x \in X$ ”; this would then mean that $\phi(\sum c_x x) = \sum c_x$, i.e. ϕ is the homomorphism that sends a linear combination of basis elements to the sum of the coefficients.



5.2. Standard simplices and their faces. Singular homology is defined using simplices, as we define now.

Definition 5.2.1. For an integer $p \geq 0$, let Δ^p be the space

$$\Delta^p = \{(t_0, \dots, t_p) \in \mathbb{R}^{p+1} \mid \sum_{i=0}^p t_i = 1, \forall i : t_i \geq 0\},$$

equipped with the subspace topology from $\mathbb{R}^{p+1}$ in its usual Euclidean topology.

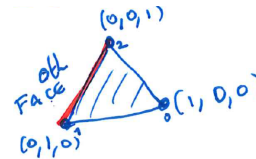
For example, Δ^0 is a one-point space, Δ^1 is homeomorphic to an interval, Δ^2 is a solid triangle, Δ^3 is a tetrahedron. In general, the topological space Δ^p is called the *standard p -simplex*.

We shall write $e_0, \dots, e_p \in \mathbb{R}^{p+1}$ for the *standard basis*, e.g. $e_0 = (1, 0, \dots, 0)$, $e_1 = (0, 1, 0, \dots, 0)$, etc. Then the point $t = (t_0, \dots, t_p) \in \Delta^p$ may be written as the linear combination $t = t_0 e_0 + \dots + t_p e_p$.

Definition 5.2.2. For $0 \leq i \leq p$, let $\delta^i : \Delta^{p-1} \rightarrow \Delta^p$ be the map defined as

$$\delta^i(t_0, \dots, t_{p-1}) = (t_0, \dots, t_{i-1}, 0, t_i, \dots, t_{p-1}).$$

The image $\delta^i(\Delta^{p-1}) \subset \Delta^p$ is called the *i th face* of Δ^p , and the union $\cup_{i=0}^p \delta^i(\Delta^{p-1})$ is called the *boundary* of Δ^p and denoted $\partial \Delta^p \subset \Delta^p$.



It is instructive to draw the faces for small values of p : the two faces of Δ^1 are the two endpoints of the interval, the three faces of Δ^2 are the three edges in a triangle, the four faces of Δ^3 are the triangles in the boundary of a solid tetrahedron, etc.

Lemma 5.2.3. For $0 \leq i < j \leq p$, we have $\delta^j \delta^i = \delta^i \delta^{j-1} : \Delta^{p-2} \rightarrow \Delta^p$.

Proof. Both maps send $(t_0, \dots, t_{p-2}) \in \Delta^{p-2}$ to

$$(t_0, \dots, t_{i-1}, 0, t_i, \dots, t_{j-2}, 0, t_{j-1}, \dots, t_{p-2}).$$

□

5.3. Singular chains and singular homology. We now use simplices to define singular chains, from which singular homology will finally be defined.

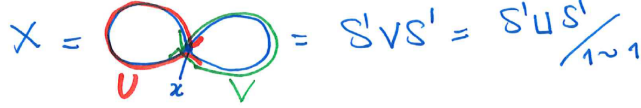
Definition 5.3.1. For a topological space X , let $\text{Sin}_p(X)$ denote the set of continuous maps $\sigma : \Delta^p \rightarrow X$. For $0 \leq i \leq p$, define a function

$$d_i : \text{Sin}_p(X) \rightarrow \text{Sin}_{p-1}(X)$$

by $d_i(\sigma) = \sigma \circ \delta^i$.

Elements of $\text{Sin}_p(X)$ are called *singular p -simplices* in X , or simply p -simplices. If σ is a p -simplex in X , the $(p-1)$ -simplex $d_i(\sigma)$ is called the i th *face* of σ .

For example, a 0-simplex in X is the same thing as a point in X . Using a homeomorphism $\Delta^1 \approx I$, a 1-simplex in X becomes the same thing as a *path* in X . In that case the 0-simplices $d_0(\sigma)$ and $d_1(\sigma)$ are just the two endpoints of the path. (See Figure 8.)



Corollary 5.3.2. If $\sigma \in \text{Sin}_p(X)$ and $0 \leq i < j \leq p$, then $d_i d_j(\sigma) = d_{j-1} d_i(\sigma)$.

Proof. Follows by composing σ and $\delta^j \circ \delta^i = \delta^i \circ \delta^{j-1}$. $\square$

Definition 5.3.3. For $p \geq 0$, let $C_p(X) = \mathbb{Z}\text{Sin}_p(X)$ be the free abelian group with basis $\text{Sin}_p(X)$. Elements of $C_p(X)$ are called *singular p -chains* in X .

For $p < 0$ we define $\text{Sin}_p(X) = \emptyset$ and hence $C_p(X)$ is a trivial group.

Thus any p -simplex (i.e. continuous map $\sigma : \Delta^p \rightarrow X$) is a p -chain, and a general p -chain is a $\mathbb{Z}$ -linear combination of p -simplices.

With these preparations, we can make the following important definition.

Definition 5.3.4. Let $\partial : C_p(X) \rightarrow C_{p-1}(X)$ be the homomorphism given by

$$\partial\sigma = \sum_{i=0}^p (-1)^i d_i(\sigma)$$

when $\sigma \in \text{Sin}_p(X)$.

Of course, this formula is extended to all of $C_p(X)$ using Lemma 5.1.2: on a general chain $c = \sum_{\sigma} c_{\sigma} \sigma$ we have $\partial c = \sum_{\sigma} \sum_i (-1)^i c_{\sigma} d_i(\sigma)$.

It is customary to use the same notation ∂ for all p , although it might have been better to denote it $\partial_p : C_p(X) \rightarrow C_{p-1}(X)$.

Lemma 5.3.5. The composition $\partial^2 = \partial \circ \partial : C_p(X) \rightarrow C_{p-2}(X)$ is the trivial homomorphism.

Proof. The claim is that $\partial \circ \partial$ takes any element of $C_p(X)$ to $0 \in C_{p-2}(X)$. It suffices to prove that $\partial \partial \sigma = 0$ for $\sigma \in \text{Sin}_p(X)$. We calculate

$$\partial^2 \sigma = \partial \left(\sum_{j=0}^p (-1)^j d_j(\sigma) \right) = \sum_{j=0}^p (-1)^j \partial(d_j(\sigma)) = \sum_{j=0}^p \sum_{i=0}^{p-1} (-1)^i (-1)^j d_i(d_j(\sigma)).$$

Now if $i < j$ we have

$$(-1)^{i+j}d_i(d_j(\sigma)) = -(-1)^{i+(j-1)}d_{j-1}(d_i(\sigma)),$$

and $j - 1 \geq i$. In this way, the $p(p+1)/2$ many terms in the formula for $\partial^2\sigma$ in which $i < j$ precisely cancel with the terms in which $i \geq j$, and hence $\partial^2\sigma = 0$. $\square$

Corollary 5.3.6. *The image of the homomorphism $\partial : C_{p+1}(X) \rightarrow C_p(X)$ is contained in the kernel of the homomorphism $\partial : C_p(X) \rightarrow C_{p-1}(X)$*

Proof. Any element $c \in C_p(X)$ in the image of ∂ may be written $c = \partial(c')$ for some $c' \in C_{p+1}(X)$, but then $\partial(c) = \partial(\partial(c')) = 0$, and hence c is in the kernel of $\partial : C_p(X) \rightarrow C_{p-1}(X)$. $\square$

Definition 5.3.7. Let $Z_p(X) \subset C_p(X)$ be the kernel of $\partial : C_p(X) \rightarrow C_{p-1}(X)$. Its elements are called *p-cycles* in X .
Let $B_p(X) \subset C_p(X)$ be the image of $\partial : C_{p+1}(X) \rightarrow C_p(X)$. Its elements are called *boundaries*.

Thus we have proved $B_p(X) \subset Z_p(X)$. Since both are subgroups of $C_p(X)$ and $C_p(X)$ is an abelian group, in fact $B_p(X)$ is a normal subgroup of $Z_p(X)$ and we may form the quotient group of $Z_p(X)$ by $B_p(X)$. This is the *singular homology groups* of X .

Definition 5.3.8. Let

$$H_p(X) = Z_p(X)/B_p(X) = \frac{\text{Ker}(\partial : C_p(X) \rightarrow C_{p-1}(X))}{\text{Im}(\partial : C_{p+1}(X) \rightarrow C_p(X))}.$$

This abelian group is called the *p*th singular homology group of X .

We shall say that two chains $c, c' \in C_p(X)$ are *homologous* if $c - c' \in B_p(X)$, i.e. if there exists an $\alpha \in C_{p+1}(X)$ with $\partial\alpha = c - c'$. This is an equivalence relation, and $H_p(X)$ is the group of *p-cycles* modulo the relation of being homologous.

5.4. The zeroth singular homology group. Like the fundamental group, the definition of $H_p(X)$ is of the form “very big set modulo very coarse equivalence relation”, and it is rarely possible to calculate it directly from the definition without first developing more tools. As an exception, it is in fact not too difficult to determine $H_0(X)$ up to isomorphism. Let us first recall that we defined $C_{-1}(X) = 0$, so any $c \in C_0(X)$ is a cycle and represents a class $[c] \in H_0(X)$. For a point $x \in X$, we shall use the same notation $x \in C_0(X)$ for the cycle corresponding to the map $\Delta^0 \rightarrow X$ sending the unique point $(1) \in \Delta^0$ to x . It represents a class $[x] \in H_0(X)$.

Proposition 5.4.1. *Let X be a path connected space. For any two points $x, x' \in X$ we have $[x] = [x']$. The homomorphism*

$$\begin{aligned} \mathbb{Z} &\rightarrow H_0(X) \\ n &\mapsto n[x] \end{aligned}$$

is an isomorphism.

Proof. Since $\Delta^1 \approx I$, we may use path-connectedness of X to pick a 1-simplex $\sigma : \Delta^1 \rightarrow X$ with $d_0\sigma = x$ and $d_1\sigma = x'$, and hence $\partial\sigma = x - x'$, but that proves $[x] = [x']$. This also proves surjectivity of the homomorphism, since any element of H_0 is represented by a linear combination of points.

To see injectivity, we define a homomorphism $\varepsilon : Z_0(X) = C_0(X) \rightarrow \mathbb{Z}$ by letting $\varepsilon(x) = 1$ for $x \in \text{Sin}_0(X) = X$. For a 1-simplex $\sigma : \Delta^1 \rightarrow X$, we have

$\partial\sigma = d_0\sigma - d_1\sigma$. Since $d_0\sigma$ and $d_1\sigma$ are both 0-simplices in X , we have $\varepsilon(d_0\sigma) = \varepsilon(d_1\sigma) = 1$, and hence $\varepsilon(\partial(\sigma)) = 0$. Since this holds for all simplices σ , we see that $B_0(X) \subset \text{Ker}(\varepsilon)$, and we get a well defined homomorphism

$$\begin{aligned}\phi : H_0(X) = Z_0(X)/B_0(X) &\rightarrow \mathbb{Z} \\ [c] &\mapsto \varepsilon(c).\end{aligned}$$

Clearly $\phi(n[x]) = n$, proving that $n \mapsto n[x]$ is injective as desired. $\square$

Remark 5.4.2. For a general X , not necessarily path connected, we may write $\pi_0(X)$ for the set of path components of X , i.e. the quotient of X by the equivalence relation with $x \sim x'$ when there exists a path from x to x' . The same argument as above gives a well defined map of sets

$$\begin{aligned}\pi_0(X) &\rightarrow H_0(X) \\ [x] &\mapsto [x],\end{aligned}$$

where the first “[x]” denotes the path component containing x and the second denotes the homology class of the map $\Delta^0 \rightarrow X$ sending the point to x . The domain $\pi_0(X)$ is not a group, but by Lemma 5.1.2 there is an induced homomorphism of abelian groups

$$\mathbb{Z}\pi_0(X) \rightarrow H_0(X).$$

This is in fact an isomorphism for any space X , by a variation of the proof of Proposition 5.4.1 which we leave as an exercise.

6. CHAIN COMPLEXES, HOMOLOGY, AND FIRST COMPUTATIONS (WEEK 5)

This week we will see some important properties of chain complexes and homology: the relationship between π_1 and H_1 , short and long exact sequences and the 5-lemma, homotopy invariance and excision. We will then use these properties to compute the homology groups of spheres. The proof (or sketch of proof) of homotopy invariance and excision will be postponed to week 6.

See Hatcher, Section 2.A for the relation to fundamental group, and Section 2.1 for the rest.

6.1. Relationship between $\pi_1(X, x)$ and $H_1(X)$. Let us write $k : \Delta^1 \rightarrow I$ for the homeomorphism given by $(t_0, t_1) \mapsto t_1$, with inverse $t \mapsto (1 - t, t)$. This gives a bijection between paths in X and singular 1-simplices, sending a path $\lambda : I \rightarrow X$ to the 1-simplex $\lambda \circ k : \Delta^1 \rightarrow X$.

Lemma 6.1.1. *If $\lambda : I \rightarrow X$ is a loop, then $\lambda \circ k : \Delta^1 \rightarrow X$ is a 1-cycle.*

Proof. The chain $\sigma = \lambda \circ k$ has $\partial\sigma = \sigma(e_1) - \sigma(e_0) = \lambda(1) - \lambda(0) = 0$. $\square$

Lemma 6.1.2. *If $\lambda \simeq \lambda' : I \rightarrow X$ rel ∂I , then the cycles $\lambda \circ k$ and $\lambda' \circ k$ are homologous.*

Proof. Let $\rho, \rho' : \Delta^2 \rightarrow I \times I$ be given by

$$\begin{aligned}\rho(t_0, t_1, t_2) &= t_0(0, 0) + t_1(1, 0) + t_2(1, 1) \\ \rho'(t_0, t_1, t_2) &= t_0(0, 0) + t_1(0, 1) + t_2(1, 1).\end{aligned}$$

These parametrize the two triangles obtained by cutting the square $I \times I$ along the diagonal. We see that $\rho \circ \delta^1 = \rho' \circ \delta^1 : \Delta^1 \rightarrow I^2$, both parametrizing the diagonal in the square.

For any map $h : I \times I \rightarrow X$ we have elements $h \circ \rho, h \circ \rho' \in \text{Sin}_2(X)$, and may form the 2-chain

$$c_h = h \circ \rho' - h \circ \rho.$$

We calculate $\partial h \circ \rho = h \circ \rho \circ \delta^0 - h \circ \rho \circ \delta^1 + h \circ \rho \circ \delta^2$ and $\partial h \circ \rho' = h \circ \rho' \circ \delta^0 - h \circ \rho' \circ \delta^1 + h \circ \rho' \circ \delta^2$. Since $\rho \circ \delta^1 = \rho' \circ \delta^1$, these terms cancel when calculating ∂c_h , and we obtain

$$\partial c_h = h \circ \rho' \circ \delta^0 + h \circ \rho' \circ \delta^2 - h \circ \rho \circ \delta^0 - h \circ \rho \circ \delta^2.$$

Now $h \circ \rho' \circ \delta^0(t_0, t_1) = h(t_1, 1)$, and similarly $h \circ \rho' \circ \delta^2(t_0, t_1) = h(0, t_1)$, $h \circ \rho \circ \delta^0(t_0, t_1) = h(1, t_1)$, and $h \circ \rho \circ \delta^2(t_0, t_1) = h(t_1, 0)$.

Hence if λ and λ' are loops in (X, x) and h is a homotopy from λ to λ' rel ∂I , then $h(0, t_1) = h(1, t_1) = x$ for all t_1 , so $h \circ \rho' \circ \delta^2 = h \circ \rho \circ \delta^0$ are both the constant map to x . Moreover, in that case $h \circ \rho' \circ \delta^0(t_0, t_1) = h(t_1, 1) = \lambda'(t_1)$ and $h \circ \rho \circ \delta^2(t_0, t_1) = h(t_1, 0) = \lambda(t_1)$, so we see that in this case

$$\partial c_h = \lambda' \circ k - \lambda \circ k,$$

and hence $\lambda' \circ k - \lambda \circ k \in B_1(X)$. $\square$

The following is now well defined.

Definition 6.1.3. Let (X, x) be a pointed space. Let $k : \Delta^1 \rightarrow I$ be the homeomorphism above, i.e. $k(t_0, t_1) = t_1$. Let $\phi : \pi_1(X, x) \rightarrow H_1(X)$ be the function which sends the homotopy class of a path $\lambda : I \rightarrow X$ to the homology class of the cycle $\lambda \circ k : \Delta^1 \rightarrow X$.

Lemma 6.1.4. *The function $\phi : \pi_1(X, x) \rightarrow H_1(X)$ is a homomorphism.*

Proof. Let $\lambda_0, \lambda_2 : I \rightarrow X$ be loops in (X, x) and let $\lambda_1 = \lambda_2 * \lambda_0$ be the concatenation. Define a continuous map $\sigma : \Delta^2 \rightarrow X$ by $\sigma(t_0, t_1, t_2) = \lambda_1(t_1/2 + t_2)$. Then check that $d_0\sigma = \lambda_0 \circ k$, $d_1\sigma = \lambda_1 \circ k$, and $d_2\sigma = \lambda_2 \circ k$. Therefore $\lambda_1 \circ k$ is homologous to $\lambda_0 \circ k + \lambda_2 \circ k$, which proves $\phi([\lambda_1]) = \phi([\lambda_0]) + \phi([\lambda_2])$. $\square$

Since $H_1(X)$ is abelian, any commutator $[\alpha, \beta] = \alpha\beta\alpha^{-1}\beta^{-1}$ between elements $\alpha, \beta \in \pi_1(X, x)$ must have $\phi([\alpha, \beta]) = 0$. Recall that the *commutator subgroup* of $\pi_1(X, x)$ is the subgroup generated by all commutators, that this subgroup is always normal, and that the *abelianization* $\text{ab}(\pi_1(X, x))$ is defined as the quotient group of $\pi_1(X, x)$ by the commutator subgroup. Thus ϕ induces a homomorphism

$$\text{ab}(\pi_1(X, x)) \rightarrow H_1(X).$$

Theorem 6.1.5. *If X is path connected, the homomorphism $\text{ab}(\pi_1(X, x)) \rightarrow H_1(X)$ induced by $\phi : \pi_1(X, x) \rightarrow H_1(X)$ is an isomorphism.*

Proof. See e.g. Theorem 2A.1 in Hatcher's book. $\square$

For example, it follows from this theorem that $H_1(S^1) \cong \pi_1(S^1, 1) \cong \mathbb{Z}$, since $\pi_1(S^1, 1)$ is already abelian. Similarly the torus $S^1 \times S^1$ has $H_1(S^1 \times S^1) \approx \mathbb{Z}^2$, etc. We saw that the Klein bottle K has non-abelian fundamental group; in fact its abelianization is isomorphic to $\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$, so this calculates the first singular homology group of the Klein bottle.

6.2. Chain complexes. The following widespread notion in mathematics generalizes singular chains.

Definition 6.2.1. A *chain complex* consists of an abelian group C_p for all $p \in \mathbb{Z}$ and a homomorphism $\partial_p : C_p \rightarrow C_{p-1}$ for all $p \in \mathbb{Z}$, such that $\partial_{p-1} \circ \partial_p = 0$ for all $p \in \mathbb{Z}$.

The data of all the abelian groups $C_p, p \in \mathbb{Z}$ shall often be abbreviated C_* when referring to all of them. We shall often write the same symbol $\partial = \partial_p : C_p \rightarrow C_{p-1}$ for all the homomorphisms. The collection of all the groups and all the homomorphisms shall often be abbreviated as (C_*, ∂) . For example, we may say “let (C_*, ∂) be a chain complex”.

Definition 6.2.2. Let (B_*, ∂^B) and (C_*, ∂^C) be chain complexes. A *chain map* $f : (B_*, \partial^B) \rightarrow (C_*, \partial^C)$ consists of homomorphisms $f_p : B_p \rightarrow C_p$ for all p , such that $\partial^C \circ f_p = f_{p-1} \circ \partial^B$ for all p . In other words, the diagram

$$\begin{array}{ccc} B_p & \xrightarrow{f_p} & C_p \\ \partial^B \downarrow & & \downarrow \partial^C \\ B_{p-1} & \xrightarrow{f_{p-1}} & C_{p-1} \end{array}$$

should be commutative for all p .

Lemma 6.2.3. *Let (A_*, ∂^A) , (B_*, ∂^B) , and (C_*, ∂^C) be chain complexes, and let $f : (A_*, \partial^A) \rightarrow (B_*, \partial^B)$ and $g : (B_*, \partial^B) \rightarrow (C_*, \partial^C)$ be chain maps. Then the composition $g \circ f : (A_*, \partial^A) \rightarrow (C_*, \partial^C)$, defined by the compositions $g_p \circ f_p : A_p \rightarrow C_p$, is also a chain map.*

Proof. Exercise. □

Note that composition of chain maps is associative and has identity maps. Therefore we have defined a category whose objects are chain complexes, whose morphisms are chain maps, and whose composition is composition of chain maps. We shall call this the *category of chain complexes* and occasionally denote it **Ch**.

Definition 6.2.4. Let (C_*, ∂) be a chain complex. Its *homology groups* $H_p(C_*, \partial)$ are defined as

$$H_p(C_*, \partial) = \frac{\text{Ker}(\partial : C_p \rightarrow C_{p-1})}{\text{Im}(\partial : C_{p+1} \rightarrow C_p)}.$$

If $f : (B_*, \partial) \rightarrow (C_*, \partial)$ is a chain map, we define the *induced* homomorphism $H_p(f) : H_p(B_*, \partial) \rightarrow H_p(C_*, \partial)$ by the formula

$$(6.1) \quad [c] \mapsto [f_p(c)]$$

for $c \in \text{Ker}(\partial : B_p \rightarrow B_{p-1})$. We shall also often write $f_* : H_p(B_*, \partial) \rightarrow H_p(C_*, \partial)$ for this induced homomorphism.

It is clear that the definition of $H_p(C_*, \partial)$ makes sense, but we should explain why f_* is well defined: why is $f(c) \in \text{Ker}(\partial' : C'_p \rightarrow C'_{p-1})$, and why does the class $[f(c)]$ depend only on the class $[c]$ and not on the choice of representative c ? Both follow from the fact that f is a chain map: $\partial f(c) = f(\partial c) = 0$ since $c \in \text{Ker}(\partial)$, and if $[c] = [d]$, then $d = c + \partial x$ so $f(d) = f(c) + f(\partial x) = f(c) + \partial(f(x))$ and hence $[f(d)] = [f(c)]$. Finally we should explain why $f_* : H_p(B_*, \partial) \rightarrow H_p(C_*, \partial)$ is a homomorphism, but that follows because $f_p : B_p \rightarrow C_p$ is a homomorphism.

Lemma 6.2.5. The association of the abelian group $H_p(C_*, \partial)$ to the chain complex (C_*, ∂) and $H_p(f)$ to a chain map defines a functor $H_p : \mathbf{Ch} \rightarrow \mathbf{Ab}$.

Proof. We must check the axioms of a functor, namely that it sends identities to identities and composition to composition. Both are easy and left to the reader. □

6.3. Singular homology is a functor. We have already seen a class of examples of a chain complex, namely the singular chains $(C_*(X), \partial)$ for a topological space X . In fact, the singular chains form a *functor* $\mathbf{Top} \rightarrow \mathbf{Ch}$.

Definition 6.3.1. For a continuous map $f : X \rightarrow Y$, let $f_* : C_p(X) \rightarrow C_p(Y)$ be the homomorphism defined on generators as $f_*(\sigma) = f \circ \sigma$.

Lemma 6.3.2. In this setting, $f_* : C_*(X) \rightarrow C_*(Y)$ is a chain map. If $g : Y \rightarrow Z$ is another continuous map, then $(g \circ f)_* = g_* \circ f_*$; and $(\text{id}_X)_* : C_*(X) \rightarrow C_*(X)$ is the identity chain map.

Proof. For a generator σ we have

$$f_*(\partial\sigma) = f_*\left(\sum_{i=0}^p (-1)^i \sigma \circ \delta^i\right) = \sum_{i=0}^p (-1)^i f \circ \sigma \circ \delta^i = \partial(f \circ \sigma) = \partial f_*(\sigma).$$

and hence $f_* \circ \partial = \partial \circ f_*$ so f_* is a chain map.

The other claims are similar: check on generators and spell out definitions. □

Hence we have functors

$$\mathbf{Top} \xrightarrow{\text{singular chains}} \mathbf{Ch} \xrightarrow{H_p} \mathbf{Ab}.$$

The composition sends a space X to the singular homology $H_p(X)$ and sends a continuous map $f : X \rightarrow Y$ to the homomorphism

$$f_* : H_p(X) \rightarrow H_p(Y) \\ [\sum c_\sigma \sigma] \mapsto [\sum c_\sigma (f \circ \sigma)],$$

for cycles $c = \sum c_\sigma \sigma$. This functor is also called *singular homology*.

6.4. Exact sequences. The following notion will turn out really useful for us:

Definition 6.4.1. Let A , B , and C be abelian groups, and let $f : A \rightarrow B$ and $g : B \rightarrow C$ be homomorphisms. We say that the sequence

$$A \xrightarrow{f} B \xrightarrow{g} C$$

is *exact* if $\text{Im}(f) = \text{Ker}(g)$.

A sequence of the form

$$0 \rightarrow A \xrightarrow{f} B$$

is exact if and only if f is injective, and a sequence of the form

$$A \xrightarrow{f} B \rightarrow 0$$

is exact if and only if f is surjective.

For a longer string of abelian groups and homomorphisms, for example

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D \xrightarrow{j} E,$$

we say that the sequence is *exact at B* if $\text{Im}(f) = \text{Ker}(g)$, that it is exact at C if $\text{Im}(g) = \text{Ker}(h)$, etc. We say that the sequence is *exact* if it is exact at all spots (in the example, this means exact at B , C , and D).

In this way, we may regard the homology groups $H_p(C_*, \partial)$ of a chain complex as a *measure* of the non-exactness of the sequence $\cdots \rightarrow C_{p+1} \rightarrow C_p \rightarrow C_{p-1} \rightarrow \cdots$: the sequence is exact at C_p if and only if $H_p(C_*, \partial) = 0$.

6.5. Short and long exact sequences. In this section we introduce common terminology for two types of exact sequences appear so often that they have special names: *short exact sequences* and *long exact sequences*. We shall also explain a construction which is often paraphrased as “a short exact sequence of chain complexes induces a long exact sequence of homology groups”.

Definition 6.5.1. A *short exact sequence* is an exact sequence of the form

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0.$$

In other words, a short exact sequence consists of a pair of composable homomorphisms f and g such that f is injective, g is surjective, and $\text{Im}(f) = \text{Ker}(g)$. (This equality of subgroups is usually checked in the two inclusions $\text{Im}(f) \subset \text{Ker}(g)$ and $\text{Im}(f) \supset \text{Ker}(g)$, so to verify that something is short exact there are really *four* things to check.)

Definition 6.5.2. A *long exact sequence* is a sequence of composable homomorphisms

$$\cdots \rightarrow A_{i-1} \xrightarrow{f_{i-1}} A_i \xrightarrow{f_i} A_{i+1} \rightarrow \cdots,$$

which is exact at A_i for all $i \in \mathbb{Z}$.

Definition 6.5.3. A short exact sequence of chain complexes consists of composable chain maps $f : (A_*, \partial) \rightarrow (B_*, \partial)$ and $g : (B_*, \partial) \rightarrow (C_*, \partial)$ such that the sequence

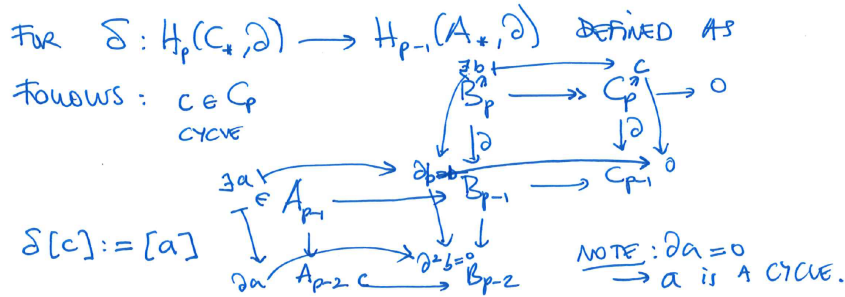
$$0 \rightarrow A_p \xrightarrow{f_p} B_p \xrightarrow{g_p} C_p \rightarrow 0$$

is exact for all $p \in \mathbb{Z}$.

An important construction associates to a short exact sequence of chain complexes as above, a long exact sequence

$$\cdots \rightarrow H_p(B_*, \partial) \xrightarrow{g_*} H_p(C_*, \partial) \xrightarrow{\delta} H_{p-1}(A_*, \partial) \xrightarrow{f_*} H_{p-1}(B_*, \partial) \rightarrow \cdots$$

The homomorphisms f_* and g_* are the ones induced by the chain maps f and g as in Definition 6.2.4. The homomorphism δ is a new construction that we now explain, see also the following figure:



Let $c \in C_p$ be a cycle, i.e. $\partial(c) = 0$. Since $g = g_p : B_p \rightarrow C_p$ is surjective, we may choose $b \in B_p$ with $g(b) = c$. Then, since g is a chain map, $g_{p-1}(\partial b) = \partial c = 0$, so by exactness we have $\partial b \in \text{Ker}(g_{p-1}) = \text{Im}(f_{p-1})$ and hence there exists an $a \in A_{p-1}$ with $f(a) = \partial b$. Since f is a chain map we see that $f(\partial a) = \partial f(a) = \partial^2 b = 0$ and since f is injective (by the exactness of $0 \rightarrow A_{p-1} \rightarrow B_{p-1}$) we see that $\partial a = 0$, i.e. $a \in A_{p-1}$ is a cycle. Hence we are allowed to form the class $[a] \in H_{p-1}(A_*, \partial)$, and we wish to define $\delta[c] = [a]$.

Lemma 6.5.4. In the situation of Definition 6.5.3, there is a well defined homomorphism $\delta : H_p(C_*, \partial) \rightarrow H_{p-1}(A_*, \partial)$, defined by $\delta([c]) = [a]$ when $f(a) = \partial b$ for some $b \in B_p$ with $g(b) = c$.

This homomorphism δ is called the connecting homomorphism.

Before writing out the proof of this Lemma, let us offer the recommendation that the reader reconstruct it on their own rather than read it. It is important to remember the *definitions* (e.g. the construction of δ) and the *statements* of Lemmas, Propositions, and Theorems (e.g. the Lemma above or the Theorem below), but for this kind of result it may be easier to reconstruct the *proofs* on your own rather than reading them from a text.

Proof. We already explained why it is possible to choose b and why the resulting $a \in A_{p-1}$ is a cycle so that we are allowed to write $[a]$. We made two choices: the choice of cycle c representing $[c]$ and the choice of $b \in B_p$ with $g(b) = c$, and we must see that the class $[a] \in H_{p-1}(A_*, \partial)$ is independent of these two choices.

Suppose first that we pick a different lift of the same representative c , i.e. that $c = g(b) = g(\tilde{b})$. Then $g(\tilde{b} - b) = 0$ so by exactness there exists an $x \in A_p$ with $f(x) = \tilde{b} - b$. Then $\partial \tilde{b} = \partial(b + f(x)) = \partial b + f(\partial x) = f(a + \partial x)$. This choice leads to the candidate $[a + \partial x]$ for $\delta([c])$, but $[a + \partial x] = [a]$ by definition of homology class.

Suppose now that we pick different representative $\tilde{c}$ for the same homology class $[c] = [\tilde{c}] \in H_p(C_*, \partial)$. This means that $\tilde{c} = c + \partial x'$ for some $x' \in C_{p+1}$. By surjectivity of g we may write $x' = g(x'')$ for some $x'' \in B_{p+1}$ and hence $\tilde{c} = g(b) + \partial g(x'') = g(b + \partial x'')$, so we may choose the lift $\tilde{b} = b + \partial x''$. But then $\partial \tilde{b} = \partial b + \partial^2 x'' = \partial b = f(a)$, so this also leads to the same candidate $[a]$ for $\delta([c])$.

Finally, we should check that δ is a homomorphism of (abelian) groups, which we leave to the reader. $\square$

With the definition of δ in place, we can now formulate the precise result about “a short exact sequence of chain complexes induces a long exact sequence of homology groups”.

Theorem 6.5.5. *Let $f : (A_*, \partial) \rightarrow (B_*, \partial)$ and $g : (B_*, \partial) \rightarrow (C_*, \partial)$ be two composable chain maps forming a short exact sequence of chain complexes. Let $\delta : H_p(C_*, \partial) \rightarrow H_{p-1}(A_*, \partial)$ be the connecting homomorphism. Then the resulting sequence*

$$\dots \xrightarrow{\delta} H_p(A_*, \partial) \xrightarrow{f_*} H_p(B_*, \partial) \xrightarrow{g_*} H_p(C_*, \partial) \xrightarrow{\delta} H_{p-1}(A_*, \partial) \xrightarrow{f_*} \dots$$

is exact.

The long exact sequence in this theorem is often called the *long exact sequence in homology associated to a short exact sequence of chain complexes*.

Proof. As already mentioned, you should remember the statement of this theorem, as well as the definition of δ . But it is easier and better to (re)construct your own proof of exactness than to read or memorize one that someone else wrote.

In case you insist: see Hatcher, Theorem 2.16 on page 117. $\square$

6.6. Naturality properties. This section could be skipped on a first reading. (In fact I will likely skip it in class as well, but may return to it later, when needed.)

Having explained how to associate a long exact sequence of homology groups to a short exact sequence of chain complexes, we finally explain how to associate a “map of long exact sequences of homology groups” to a “map of short exact sequences of chain complexes”.

Given a commutative diagram of chain complexes of chain maps

$$(6.2) \quad \begin{array}{ccccc} (A_*, \partial) & \xrightarrow{f} & (B_*, \partial) & \xrightarrow{g} & (C_*, \partial) \\ \downarrow h^A & & \downarrow h^B & & \downarrow h^C \\ (A'_*, \partial) & \xrightarrow{f'} & (B'_*, \partial) & \xrightarrow{g'} & (C'_*, \partial) \end{array}$$

such that $0 \rightarrow A_p \xrightarrow{f_p} B_p \xrightarrow{g_p} C_p \rightarrow 0$ and $0 \rightarrow A'_p \xrightarrow{f'_p} B'_p \xrightarrow{g'_p} C'_p \rightarrow 0$ are exact for all p , we obtain a diagram

$$(6.3) \quad \begin{array}{ccccccc} H_p(B_*, \partial) & \xrightarrow{g_*} & H_p(C_*, \partial) & \xrightarrow{\delta} & H_{p-1}(A_*, \partial) & \xrightarrow{f_*} & H_{p-1}(B_*, \partial) \\ \downarrow h_*^B & & \downarrow h_*^C & & \downarrow h_*^A & & \downarrow h_*^B \\ H_p(B'_*, \partial) & \xrightarrow{g'_*} & H_p(C'_*, \partial) & \xrightarrow{\delta'} & H_{p-1}(A'_*, \partial) & \xrightarrow{f'_*} & H_{p-1}(B'_*, \partial) \end{array}$$

continuing infinitely to the left and to the right, whose rows are the long exact sequences associated to the two short exact sequences, and whose vertical maps are induced by h^A , h^B , and h^C .

The two rows of (6.2) are short exact sequences and two vertical chain maps h^A , h^B , and h^C may be regarded as map from the top short exact sequence to the

bottom. Likewise, the rows of (6.3) are long exact sequences of abelian groups, and the horizontal maps may be regarded as a map from the top row to the bottom row. It is natural to expect the diagram (6.3) to be commutative if (6.2) is, but that requires a proof. We have already seen that the squares not involving δ commute: for example $h^B \circ f = f' \circ h^A$, so $(h^B)_* \circ f_* = (f')_* \circ h^A_*$ (since taking homology of a chain complex is a functor) and similarly for the squares involving g . Thus it remains to see that the squares involving δ also commute.

Lemma 6.6.1. *In the above situation, $h^A_* \circ \delta = \delta' \circ (h^C)_* : H_p(C_*, \partial) \rightarrow H_{p-1}(A_*, \partial)$.*

Proof. It may again be easier to trace through definitions on your own, but here is the argument.

Let $c \in C_p$ be a cycle, write $c = g(b)$ and $\partial b = f(a)$: then we have defined $\delta([c]) = [a]$. But then $h^C(c) = h^C(g(b)) = g(h^B(b))$ since $h^C \circ g = g \circ h^B$, and then $\partial(h^B(b)) = h^B(\partial b) = h^B(f(a)) = f'(h^A(a))$ since h^B is a chain map and $h^B \circ f = f' \circ h^A$. But then the definition of δ' gives $\delta'([h^C(c)]) = [h^A(a)]$ or in other words $\delta' \circ (h^C)_*([c]) = h^A_* \circ \delta([c])$. $\square$

(Exercise: define a category whose objects are short exact sequences of chain complexes and whose morphisms are diagrams of the form (6.2), define another category whose objects are long exact sequences of abelian groups, and then explain how to use Lemma 6.6.1 to make the “long exact sequence in homology associated to a short exact sequence of chain complexes” into a *functor*.)

6.7. The 5-lemma. The long exact sequence of homology associated to a short exact sequence of chain complexes, is often combined with the following result, which is known as the *5-lemma*. It is used throughout topology and any other area of mathematics featuring exact sequences.

Theorem 6.7.1. *Let there be given a commutative diagram of abelian groups and homomorphisms*

$$\begin{array}{ccccccccc} A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & D & \longrightarrow & E \\ \downarrow \cong & & \downarrow \cong & & \downarrow & & \downarrow \cong & & \downarrow \cong \\ A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & D' & \longrightarrow & E' \end{array},$$

such that both rows are exact, and such that the vertical arrows labeled “ $\cong$ ” are isomorphisms. Then the middle arrow $C \rightarrow C'$ is also an isomorphism.

Proof. Another exercise in diagram chasing, and you are better off constructing your own proof. If you get stuck, see e.g. Hatcher, page 129 in Section 2.1. $\square$

A typical application is the following “2-out-of-3 property”.

Corollary 6.7.2. *Given a diagram of chain complexes of the form 6.2, if any two of the vertical maps induce isomorphisms on homology in all degrees, then so does the third.*

Proof. Let us assume the maps $B_* \rightarrow B'_*$ and $C_* \rightarrow C'_*$ induce isomorphisms in homology in all degrees and deduce that $A_* \rightarrow A'_*$ does too, the other two cases

are entirely similar. There are induced maps of long exact sequences (6.4)

$$\begin{array}{ccccccccccc} H_p(B_*, \partial) & \xrightarrow{g_*} & H_p(C_*, \partial) & \xrightarrow{\delta} & H_{p-1}(A_*, \partial) & \xrightarrow{f_*} & H_{p-1}(B_*, \partial) & \xrightarrow{g_*} & H_{p-1}(C_*, \partial) \\ \downarrow \cong & & \downarrow \cong & & \downarrow & & \downarrow \cong & & \downarrow \cong \\ H_p(B'_*, \partial) & \xrightarrow{g'_*} & H_p(C'_*, \partial) & \xrightarrow{\delta'} & H_{p-1}(A'_*, \partial) & \xrightarrow{f'_*} & H_{p-1}(B'_*, \partial) & \xrightarrow{g'_*} & H_{p-1}(C'_*, \partial), \end{array}$$

in which the indicated homomorphisms are isomorphisms. By the 5-lemma, the middle one is too. $\square$

6.8. Relative singular chains and homology. The general algebraic results about chain complexes may be applied to obtain long exact sequences of singular homology groups.

Definition 6.8.1. A *pair of spaces* is a pair (X, A) where X is a topological space and $A \subset X$ is a subspace. We shall regard A as a topological space with the subspace topology. A *map of pairs* $(X, A) \rightarrow (Y, B)$ is a continuous map $f : X \rightarrow Y$ with $f(A) \subset B$.

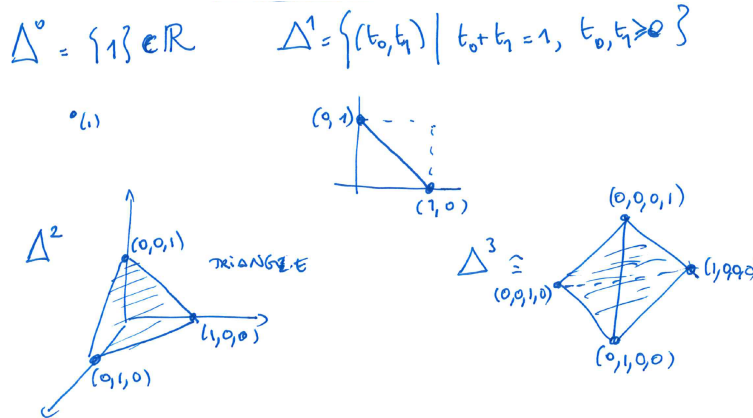
For a pair (X, A) , let $C_p(X, A)$ be the quotient of $C_p(X)$ by the subgroup $C_p(A) \subset C_p(X)$. Let $\partial : C_p(X, A) \rightarrow C_{p-1}(X, A)$ be defined by the commutative diagram

$$\begin{array}{ccccc} C_p(A) & \xrightarrow{i_*} & C_p(X) & \xrightarrow{j_*} & C_p(X, A) \\ \downarrow \partial & & \downarrow \partial & & \downarrow \partial \\ C_{p-1}(A) & \xrightarrow{i_*} & C_{p-1}(X) & \xrightarrow{j_*} & C_{p-1}(X, A), \end{array}$$

where i_* denotes the map induced by the inclusion $i : A \rightarrow X$, and j_* denotes the quotient map.

The *relative homology* of (X, A) is then defined as the homology of the chain complex $C_*(X, A)$, i.e.

$$H_p(X, A) = H_p(C_*(X, A), \partial).$$



Since $C_p(X)$ is free abelian with basis the set of continuous maps $\sigma : \Delta^p \rightarrow X$ and $C_p(A)$ is free abelian on the subset consisting of maps σ with $\sigma(\Delta^p) \subset A$, we see that $C_p(X, A)$ is free abelian with basis the set of (classes modulo $C_p(A)$) of continuous maps $\sigma : \Delta^p \rightarrow X$ with $\sigma(\Delta^p) \cap (X \setminus A) \neq \emptyset$.

Since i_* and j_* are chain maps, and since $0 \rightarrow C_p(A) \rightarrow C_p(X) \rightarrow C_p(X, A) \rightarrow 0$ is exact for all p , we immediately deduce a long exact sequence of homology groups

$$(6.5) \quad \cdots \rightarrow H_p(A) \xrightarrow{i_*} H_p(X) \xrightarrow{j_*} H_p(X, A) \xrightarrow{\delta} H_{p-1}(A) \rightarrow \cdots$$

The map $\delta : H_p(X, A) \rightarrow H_{p-1}(A)$ is the connecting homomorphism of Section 6.5. One can check that it can directly be defined as follows: let $[c] \in H_p(X, A)$ be a relative homology class, represented by $c \in C_p(X)$ with $\partial c \in C_{p-1}(A)$. Then $\delta[c] = [\partial c]$ is the homology class in $H_{p-1}(A)$ defined by its boundary.

Corollary 6.8.2. *Let $f : (X, A) \rightarrow (Y, B)$ be a map such that $f_* : H_p(X) \rightarrow H_p(Y)$ and $(f|_A)_* : H_p(A) \rightarrow H_p(B)$ are isomorphisms for all p . Then $H_p(X, A) \rightarrow H_p(Y, B)$ is also an isomorphism for all p .*

Proof. Apply the 5-lemma to the diagram

$$\begin{array}{ccccccccc} H_p(A) & \longrightarrow & H_p(X) & \longrightarrow & H_p(X, A) & \xrightarrow{\delta} & H_{p-1}(A) & \longrightarrow & H_{p-1}(X) \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ H_p(B) & \longrightarrow & H_p(Y) & \longrightarrow & H_p(Y, B) & \xrightarrow{\delta} & H_{p-1}(B) & \longrightarrow & H_{p-1}(Y), \end{array}$$

whose rows are parts of the long exact sequences of the pairs, and whose vertical maps are the ones induced by f . We have seen that this diagram is commutative (including the square with δ !), have assumed that the four outer vertical homomorphisms are isomorphisms, and may deduce that the middle vertical homomorphism is an isomorphism. $\square$

The long exact sequence of a pair may be slightly generalized as follows.

Lemma 6.8.3. *For $B \subset A \subset X$, the inclusions $i : (A, B) \rightarrow (X, B)$ and $j : (X, B) \rightarrow (X, A)$ induce a short exact sequence of chain complexes*

$$0 \rightarrow C_*(A, B) \xrightarrow{i_*} C_*(X, B) \xrightarrow{j_*} C_*(X, A) \rightarrow 0,$$

and hence a long exact sequence in homology

$$(6.6) \quad \cdots \rightarrow H_p(A, B) \xrightarrow{i_*} H_p(X, B) \xrightarrow{j_*} H_p(X, A) \xrightarrow{\delta} H_{p-1}(A, B) \rightarrow \cdots$$

Before explaining the proof, let us point out that the quotient map $j_* : C_p(X) \rightarrow C_p(X, \emptyset)$ is an isomorphism of chain complexes and hence induces an isomorphism $H_p(X) \cong H_p(X, \emptyset)$. In this way the long exact sequence (6.5) is the special case of (6.6) where $B = \emptyset$.

Proof. This can be seen using the bases of these abelian groups: $C_p(X, B)$ is free on the set

$$\{\sigma : \Delta^p \rightarrow X \mid \sigma(\Delta^p) \cap (X \setminus B) \neq \emptyset\}.$$

This set of maps is the (disjoint) union of those whose image is contained in A and those whose image isn't, i.e. the sets

$$\{\sigma : \Delta^p \rightarrow A \mid \sigma(\Delta^p) \cap (X \setminus B) \neq \emptyset\}, \quad \{\sigma : \Delta^p \rightarrow X \mid \sigma(\Delta^p) \cap (X \setminus A) \neq \emptyset\}.$$

The first of these is the image of a basis for $C_p(A, B)$ and the second maps to a basis for $C_p(X, A)$. This implies that the sequence of chain complexes is short exact. $\square$

6.9. Two important theorems: Homotopy invariance and Excision. With our current knowledge, singular homology seems neither useful nor more calculable: at best know how to calculate H_0 and H_1 ; the only spaces for which we can calculate $H_p(X)$ for all p are more or less the finite sets with discrete topology. In order to proceed, we shall need to following two fundamental results. Their proof is a bit more involved than what we have seen so far.

Theorem 6.9.1. *Let $f : (X, A) \rightarrow (Y, B)$ be a map of pairs such that $f : X \rightarrow Y$ and $f|_A : A \rightarrow B$ are both homotopy equivalences. Then*

$$f_* : H_p(X, A) \rightarrow H_p(Y, B)$$

is an isomorphism for all p .

We shall (time permitting) prove this next week. For now, we shall deduce some consequences.

Remark 6.9.2. It follows from the theorem that if $f : X \rightarrow Y$ is a homotopy equivalence, then $f_* : H_p(X) \rightarrow H_p(Y)$ is an isomorphism for all p . (Proof: since $f|_{\emptyset} : \emptyset \rightarrow \emptyset$ is a homotopy equivalence(!), this is the special case of the theorem where $A = B = \emptyset$.)

Once we've become better at *calculating* singular homology and induced maps, the homotopy invariance theorem will have some obvious applications. For example, we shall soon show that $H_2(S^2) \cong \mathbb{Z}$, from which it then follows that S^2 is not contractible.

Next we state the *excision* theorem. Again we shall postpone the proof and deduce some consequences. This result is essential for most calculations.

Theorem 6.9.3. *Let $A \subset X$ be a subspace, with interior $\text{int}(A)$. Let $Z \subset A$ be a subspace whose closure (in X) is contained in $\text{int}(A)$. Write*

$$i : (X \setminus Z, A \setminus Z) \rightarrow (X, A)$$

for the inclusion. Then the induced map

$$H_p(X \setminus Z, A \setminus Z) \rightarrow H_p(X, A)$$

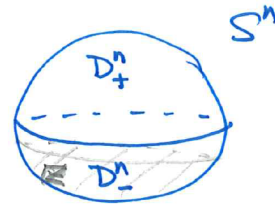
is an isomorphism.

Example 6.9.4. Consider the sphere S^n as the union of the upper and lower hemisphere: $S^n = D_+^n \cup D_-^n$, and let $Z \subset D_-^n$ be a smaller subdisc.

Then excision (Theorem 6.9.3) for $X = S^n$, $A = D_-^n$ and $Z = Z$ implies that the inclusion $(S^n \setminus Z, D_-^n \setminus Z) \hookrightarrow (S^n, D_-^n)$ induces an isomorphism on all homology groups. On the other hand, the inclusion $(D_+^n, \partial D_+^n) \hookrightarrow (S^n \setminus Z, D_-^n \setminus Z)$ is a homotopy equivalence of pairs, and hence also induces an isomorphism on homology groups using now Theorem 6.9.1.

Hence combining both theorems we get an isomorphism $H_p(D_+^n, \partial D_+^n) \xrightarrow{\cong} H_p(S^n, D_-^n)$ for all $p \geq 0$ induced by the inclusion.

We will compute the homology of the sphere S^n in Section 6.11 using the long exact sequence for the pair (S^n, D_-^n) . By the above computation, we will thus need to know the groups $H_p(D_+^n, \partial D_+^n)$, which is the computation we will undertake in the next section.



6.10. Homology of a disk relative to its boundary. Writing $\partial D^n = \{x \in \mathbb{R}^n \mid |x| = 1\} = S^{n-1}$, we may now calculate the relative homology groups $H_p(D^n, \partial D^n)$. The calculation shall use four ingredients: Homotopy invariance, excision, the long exact sequence of a triple, and the explicit calculation of singular homology of a one-point space.

Definition 6.10.1. Define subspaces $\partial_+ D^n$ and $\partial_- D^n$ of ∂D^n by

$$\partial_+ D^n = \{x \in \mathbb{R}^n \mid |x| = 1, x_1 \geq 0\}$$

$$\partial_- D^n = \{x \in \mathbb{R}^n \mid |x| = 1, x_1 \leq 0\}.$$

The subspace $\partial_+ D^n$ shall be called the “upper hemisphere” and $\partial_- D^n$ the “lower hemisphere”.

It is easy to see that $\partial_- D^n$ and $\partial_+ D^n$ are both contractible, provided $n \geq 1$: in fact the projection $(x_1, \dots, x_n) \rightarrow (x_2, \dots, x_n)$ induce homeomorphisms to D^{n-1} , with inverse $(x_2, \dots, x_n) \mapsto (\pm\sqrt{1 - \sum_{i=2}^n x_i^2}, x_2, \dots, x_n)$. After removing a point from the sphere, we also have the following homeomorphism.

Lemma 6.10.2. Let $s = (-1, 0, \dots, 0) \in \partial_- D^n$ be the “south pole”. There exists a homeomorphism $\phi : \partial D^n \setminus \{s\} \rightarrow \mathbb{R}^{n-1}$, restricting to a homeomorphism $\partial_- D^n \setminus \{s\} \rightarrow \mathbb{R}^{n-1} \setminus \text{int}(D^{n-1})$.

Proof sketch. An explicit homeomorphism may be constructed using “stereographic projection”: send $y \in \partial D^n \setminus \{s\}$ to the intersection of the straight line through y and s with the hyperplane $\{0\} \times \mathbb{R}^{n-1}$. In coordinates $\phi(y_1, \dots, y_n) = \frac{1}{1+y_1}(y_2, \dots, y_n)$. $\square$

Suppose now $B \subset X$ and that the inclusion $i : B \rightarrow X$ is a homotopy equivalence. Then $i : H_p(B) \rightarrow H_p(X)$ is an isomorphism, and the long exact sequence of the pair shows that $H_p(X, B) = 0$ for all p . If then $B \subset A \subset X$, the long exact sequence of the triple shows that the connecting homomorphism $H_p(X, A) \rightarrow H_{p-1}(A, B)$ is an isomorphism for all p . In particular this holds when B and X are both contractible, so we may apply it to $\partial_- D^n \rightarrow \partial D^n \rightarrow D^n$, to deduce an isomorphism

$$\delta : H_p(D^n, \partial D^n) \rightarrow H_{p-1}(\partial D^n, \partial_- D^n)$$

for all p and all $n \geq 1$. Excision implies that the inclusion $i : (\partial D^n \setminus \{s\}, \partial_- D^n \setminus \{s\}) \rightarrow (\partial D^n, \partial_- D^n)$ induces an isomorphism

$$H_{p-1}(\partial D^n \setminus \{s\}, \partial_- D^n \setminus \{s\}) \rightarrow H_{p-1}(\partial D^n, \partial_- D^n),$$

and the homeomorphisms ϕ from the lemma above then induces an isomorphism from this group to $H_{p-1}(\mathbb{R}^{n-1}, \mathbb{R}^{n-1} \setminus \text{int}(D^{n-1}))$. Finally, the homotopy equivalence $(D^{n-1}, \partial D^{n-1}) \rightarrow (\mathbb{R}^{n-1}, \mathbb{R}^{n-1} \setminus \text{int}(D^{n-1}))$ induces an isomorphism

$$H_{p-1}(D^{n-1}, \partial D^{n-1}) \rightarrow H_{p-1}(\mathbb{R}^{n-1}, \mathbb{R}^{n-1} \setminus \text{int}(D^{n-1})).$$

Combining these isomorphisms, we get the isomorphism

$$H_p(D^n, \partial D^n) \cong H_{p-1}(D^{n-1}, \partial D^{n-1}),$$

valid for all p and all $n \geq 1$. By induction we get

$$\begin{aligned} H_p(D^n, \partial D^n) &\cong H_{p-n}(D^0, \partial D^0) = H_{p-n}(\{0\}, \emptyset) = H_{p-n}(\{0\}) \\ &\cong \begin{cases} \mathbb{Z} & \text{for } p = n \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

In order for the above calculation to be complete, we must prove the homotopy invariance theorem and the excision theorem. This is the topic of next week. (Next week we shall also explain why $f_* : H_p(X) \rightarrow H_p(Y)$ and $(f|_A)_* : H_p(A) \rightarrow$

$H_p(B)$ being isomorphisms for all p implies that $f_* : H_p(X, A) \rightarrow H_p(Y, B)$ is an isomorphism, but that is much easier than homotopy invariance and excision.)

6.11. Homology of a sphere. Continuing from the calculation of $H_p(D^n, \partial D^n)$, we may calculate $H_p(S^n)$ for all p and n . Let us first deal with $n = 0$, in which case $S^n = \{-1, 1\}$ and hence

$$H_p(S^0) \cong \begin{cases} \mathbb{Z}^2 & \text{for } p = 0 \\ 0 & \text{otherwise.} \end{cases}$$

For $n \geq 1$ we look at the long exact sequence for the pair (D^{n+1}, S^n) . It ends with

$$\dots \rightarrow H_1(D^{n+1}, S^n) \xrightarrow{\delta} H_0(S^n) \rightarrow H_0(D^{n+1}) \rightarrow H_0(D^{n+1}, S^n) \rightarrow 0.$$

Since $n \geq 1$, the space S^n is path connected and hence $H_0(S^n) \cong \mathbb{Z}$ with generator $[y]$ for any $y \in S^n \subset D^{n+1}$. Since also $H_0(D^{n+1}) \cong \mathbb{Z}$ with generator $[y]$, we see that the inclusion induces an isomorphism $H_0(S^n) \rightarrow H_0(D^n)$, and hence the connecting homomorphism $\delta : H_1(D^{n+1}, S^n) \rightarrow H_0(S^n)$ is trivial. Since D^{n+1} is contractible, the two outer terms of the exact sequence

$$H_{p+1}(D^{n+1}) \rightarrow H_{p+1}(D^{n+1}, S^n) \xrightarrow{\delta} H_p(S^n) \rightarrow H_p(D^{n+1}),$$

which is part of the long exact sequence of the pair, must vanish for $p \geq 1$, and hence the connecting homomorphism induces an isomorphism $H_{p+1}(D^{n+1}, S^n) \cong H_p(S^n)$. This then calculates $H_p(S^n)$ for all $p \geq 1$, but since S^n is path connected, we know that $H_0(S^n) \cong \mathbb{Z}$. To summarize,

$$H_p(S^n) \cong \begin{cases} \mathbb{Z}^2 & \text{if } (p, n) = (0, 0) \\ \mathbb{Z} & \text{if } n > 0 \text{ and } n = p \\ \mathbb{Z} & \text{if } n > 0 \text{ and } p = 0 \\ 0 & \text{otherwise.} \end{cases}$$

6.12. A geometric consequence. We now have the tools to prove the following surprising result:

Theorem 6.12.1. *There does not exist a continuous map $r : D^n \rightarrow \partial D^n$ such that $r|_{\partial D^n} = \text{id}_{\partial D^n}$.*

Proof. Let $i : \partial D^n \rightarrow D^n$ be the inclusion and suppose for contradiction that such an r exists, i.e. with $r \circ i = \text{id}_{\partial D^n}$. Then, since H_{n-1} is a functor, we get that the composition

$$r_* \circ i_* : H_{n-1}(\partial D^n) \rightarrow H_{n-1}(D^n) \rightarrow H_{n-1}(\partial D^n)$$

is the identity. For $n > 1$ this composition looks like $\mathbb{Z} \rightarrow 0 \rightarrow \mathbb{Z}$, which clearly cannot be the identity. For $n = 1$ it looks like $\mathbb{Z}^2 \rightarrow \mathbb{Z} \rightarrow \mathbb{Z}^2$ which is also impossible (e.g. the 2×2 matrix must have rank at most 1). $\square$

The case $r : D^2 \rightarrow \partial D^2$ may be argued using fundamental groups instead of H_1 .

7. HOMOTOPY INVARIANCE AND EXCISION (WEEK 6)

We start the week by giving proofs (or sketches thereof in the lectures) of two important theorems we used last week, namely homotopy invariance of homology (Theorem 6.9.1), and excision (Theorem 6.9.3).

7.1. Homotopy invariance. Theorem 6.9.1 states that a map of pairs $f : (X, A) \rightarrow (Y, B)$ such that both $f : X \rightarrow Y$ and $f|_A : A \rightarrow B$ are homotopy equivalences induce an isomorphism in relative homology

$$f_* : H_p(X, A) \xrightarrow{\cong} H_p(Y, B).$$

Now we have a long exact sequence of homology groups for each of the pairs (X, A) and (Y, B) and the map f induces a map of long exact sequences:

$$\begin{array}{ccccccccc} \dots & \longrightarrow & H_p(A) & \longrightarrow & H_p(X) & \longrightarrow & H_p(X, A) & \longrightarrow & H_{p-1}(A) & \longrightarrow & H_{p-1}(X) & \longrightarrow & \dots \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ \dots & \longrightarrow & H_p(B) & \longrightarrow & H_p(Y) & \longrightarrow & H_p(Y, B) & \longrightarrow & H_{p-1}(B) & \longrightarrow & H_{p-1}(Y) & \longrightarrow & \dots \end{array}$$

An application of the 5-lemma to the above map of long exact sequences gives that Theorem 6.9.1 follows from its non-relative version:

Theorem 7.1.1. *Let $f : X \rightarrow Y$ be a homotopy equivalence. Then $f_* : H_p(X) \rightarrow H_p(Y)$ is an isomorphism for all $p \geq 0$.*

So we are left to prove Theorem 7.1.1. For this, we will use the following notion of homotopy for chain maps:

Definition 7.1.2. Let $A = (A_*, \partial^A)$ and $B = (B_*, \partial^B)$ be chain complexes, and let $\phi, \psi : A \rightarrow B$ be two chain maps. Then a *chain homotopy* from ϕ to ψ consists of homomorphisms

$$T_p : A_p \rightarrow B_{p+1}$$

for all $p \in \mathbb{Z}$, such that

$$\partial_{p+1}^B \circ T_p + T_{p-1} \circ \partial_p^A = \psi - \phi.$$

In more brief notation, $\partial \circ T + T \circ \partial = \psi - \phi$.

If a chain homotopy exists, then we say the two chain maps are *chain homotopic*.

Lemma 7.1.3. *If $\phi, \psi : A \rightarrow B$ are chain homotopic, then the induced maps*

$$H_p(\phi), H_p(\psi) : H_p(A_*, \partial^A) \rightarrow H_p(B_*, \partial^B)$$

are equal.

Proof. Let $\alpha \in H_p(A_*, \partial^A)$ and choose a cycle $a \in A_p$ representing it. Then $\partial a = 0$, so

$$\psi(a) - \phi(a) = \partial(Ta) + T(\partial a) = \partial(Ta),$$

i.e. $\psi(a)$ and $\phi(a)$ differ by a boundary, so $[\psi(a)] = [\phi(a)]$. □

The following result explains the name “chain homotopy”.

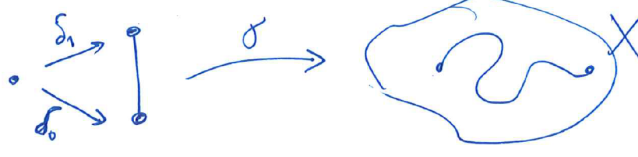


FIGURE 8. A 1-simplex is a path

Theorem 7.1.4. *If $f_0, f_1 : X \rightarrow Y$ are two homotopic continuous maps, then the induced chain maps*

$$(f_0)_*, (f_1)_* : C_*(X) \rightarrow C_*(Y)$$

are chain homotopic.

Sketch proof. We will unfortunately not have time to explain the proof of this important result in sufficient detail. Using a homotopy $h : I \times X \rightarrow Y$, it is possible to write down explicit homomorphisms

$$P : C_p(X) \rightarrow C_{p+1}(Y)$$

called *prism operators*, defined by a geometric procedure which sends a simplex $\sigma : \Delta^p \rightarrow X$ to an alternating sum of compositions of the form

$$\Delta^{p+1} \xrightarrow{\alpha_i} I \times \Delta^p \xrightarrow{\text{id}_I \times \sigma} I \times X \xrightarrow{h} Y.$$

The linear combination is over explicit maps $\alpha_0, \dots, \alpha_p : \Delta^{p+1} \rightarrow I \times \Delta^p$, with suitable signs. See page 112 in Hatcher's book for the precise definition, as well as a proof that these homomorphisms satisfy $P \circ \partial + \partial \circ P = (f_1)_* - (f_0)_*$, i.e. defines a chain homotopy. $\square$

Corollary 7.1.5. *If $f_0, f_1 : X \rightarrow Y$ are two homotopic maps, then the induced maps $(f_0)_*, (f_1)_* : H_p(X) \rightarrow H_p(Y)$ are equal for all p .*

Proof. Combine Lemma 7.1.3 and Theorem 7.1.4. $\square$

Assuming these results, we can now prove Theorem 7.1.1, which itself implies Theorem 6.9.1, as explained above.

Proof of Theorem 7.1.1. Corollary 7.1.5 implies that H_p induces a well defined functor $\mathcal{H} \rightarrow \mathbf{Ab}$, where as before $\mathcal{H}$ denotes the homotopy category, whose morphisms are homotopy classes of maps. We have seen that f gives an isomorphism in $\mathcal{H}$, and that any functor sends isomorphisms to isomorphisms. $\square$

7.2. Small chains. To prove Excision (Theorem 6.9.3), we will use the notion of “small chains”:

Definition 7.2.1. For a topological space X and a set $\mathcal{U}$ of subsets of X , let us say that a singular simplex $\sigma : \Delta^p \rightarrow X$ is $\mathcal{U}$ -small if there exists a $U \in \mathcal{U}$ such that $\sigma(\Delta^p) \subset U$. We then let

$$C_p^{\mathcal{U}}(X) \subset C_p(X)$$

denote the subgroup spanned by the $\mathcal{U}$ -small simplices.

If $\sigma : \Delta^p \rightarrow X$ is $\mathcal{U}$ -small, then $\sigma \circ \delta^i : \Delta^{p-1} \rightarrow X$ is also $\mathcal{U}$ -small and hence $\partial\sigma \in C_{p-1}^{\mathcal{U}}(X)$. Therefore the boundary map on $C_*(X)$ restricts to a homomorphism

$\partial C_p^{\mathcal{U}}(X) \rightarrow C_{p-1}^{\mathcal{U}}(X)$, which still satisfies $\partial \circ \partial = 0$ so we have a chain complex $(C_*^{\mathcal{U}}(X), \partial)$. By definition, the inclusion

$$(7.1) \quad C_*^{\mathcal{U}}(X) \rightarrow C_*(X)$$

is a chain map.

The main technical result about small chains is the following.

Theorem 7.2.2. *Let $\mathcal{U}$ be a collection of subsets of X such that*

$$\bigcup_{U \in \mathcal{U}} \text{int}(U) = X.$$

Then the inclusion (7.1) induces an isomorphism on homology.

The proof of this theorem is a bit lengthy. Before giving it, let us explain why it implies the excision theorem as previously stated. First, let us use the 5-lemma to deduce a relative version.

Definition 7.2.3. Let $\mathcal{U}$ be a set of subsets of X . For $A \subset X$ define

$$\mathcal{U}_A = \{U \cap A \mid U \in \mathcal{U}\}.$$

Then we have defined the chain complex $C_*^{\mathcal{U}_A}(A) \subset C_*(A)$: in degree p it is the set of formal $\mathbb{Z}$ -linear combinations of maps $\sigma : \Delta^p \rightarrow A$ which are $\mathcal{U}$ -small when regarded as simplices of X .

Let $C_p^{\mathcal{U}}(X, A)$ be defined as

$$C_p^{\mathcal{U}}(X, A) = C_p^{\mathcal{U}}(X) / C_p^{\mathcal{U}_A}(A)$$

and define homomorphisms $\partial : C_p^{\mathcal{U}}(X, A) \rightarrow C_{p-1}^{\mathcal{U}}(X, A)$ by applying the boundary operator in $C_*^{\mathcal{U}}(X)$ to representatives.

Write $H_p^{\mathcal{U}}(X) = H_p(C_*^{\mathcal{U}}(X), \partial)$, $H_p^{\mathcal{U}}(A) = H_p(C_*^{\mathcal{U}_A}(A), \partial)$ and $H_p^{\mathcal{U}}(X, A) = H_p(C_*^{\mathcal{U}}(X, A), \partial)$.

Corollary 7.2.4. *Let $\mathcal{U}$ be as in Theorem 7.2.2. Then the inclusion $C_*^{\mathcal{U}}(X, A) \rightarrow C_*(X, A)$ induces an isomorphism $H_p^{\mathcal{U}}(X, A) \rightarrow H_p(X, A)$ for all p .*

Proof. The inclusion homomorphism $C_p(A) \subset C_p(X)$ and the quotient homomorphism $C_p(X) \rightarrow C_p(X, A)$ and the analogous inclusion and quotient homomorphisms for $\mathcal{U}$ -small chains give rise to a commutative diagram of chain maps

$$\begin{array}{ccccccc} 0 & \longrightarrow & C_*^{\mathcal{U}_A}(A) & \longrightarrow & C_*^{\mathcal{U}}(X) & \longrightarrow & C_*^{\mathcal{U}}(X, A) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & C_*(A) & \longrightarrow & C_*(X) & \longrightarrow & C_*(X, A) \longrightarrow 0. \end{array}$$

We have seen that the bottom row of the diagram is a short exact sequence of chain complexes, and the same argument shows that the top row of the diagram is a short exact sequence of chain complexes. By Theorem 7.2.2, the left-most and the middle vertical homomorphisms induce isomorphisms in homology, and hence the right-most one does too, by Corollary 6.7.2 (the “2 out of 3 property”). $\square$

In the setting of the excision theorem, i.e. given subsets $Z \subset A \subset X$ with $\bar{Z} \subset \text{int}(A)$, we may let $\mathcal{U} = \{X \setminus Z, A\}$. Then by assumption we have

$$\text{int}(X \setminus Z) \cup \text{int}(A) = X,$$

so the assumptions of Theorem 7.2.2 apply, and hence the relative homology $H_p(X, A)$ may be calculated as the homology of the chain complex $C_*^{\mathcal{U}}(X, A)$.

Composing a simplex $\sigma : \Delta^p \rightarrow X \setminus Z$ with the inclusion $i : X \setminus Z \rightarrow X$ gives a simplex $i \circ \sigma : \Delta^p \rightarrow X$ which is $\mathcal{U}$ -small, and hence the homomorphism $i_* : C_p(X \setminus Z) \rightarrow C_p(X)$ has image inside $C_p^{\mathcal{U}}(X)$.

Proposition 7.2.5. *The homomorphism $i_* : C_p(X \setminus Z) \rightarrow C_p^{\mathcal{U}}(X)$ induces an isomorphism*

$$C_p(X \setminus Z, A \setminus Z) \rightarrow C_p^{\mathcal{U}}(X, A)$$

for all p .

Proof. An easy set-theoretic consideration shows that the abelian group $C_p(X \setminus Z, A \setminus Z)$ is free abelian with basis the maps $\sigma : \Delta^p \rightarrow X \setminus Z$ with $\sigma(\Delta^p) \cap (A \setminus Z) = \emptyset$. The homomorphism sends these to a basis for the free abelian group $C_p^{\mathcal{U}}(X, A)$. Since the homomorphism takes a basis to a basis, it is an isomorphism. $\square$

Proof of Theorem 6.9.3. The chain map $i_* : C_*(X \setminus Z, A \setminus Z) \rightarrow C_*(X, A)$ is the composition of the isomorphism in Proposition 7.2.5 and the chain map $C_*^{\mathcal{U}}(X, A) \rightarrow C_*(X, A)$ which induces an isomorphism on homology, by theorem 7.2.2. $\square$

However, all of this rests on Theorem 7.2.2, which we have not yet proved.

7.3. Subdivision. The key to proving Theorem 7.2.2 is the definition of a chain map

$$S : C_*(X) \rightarrow C_*(X)$$

homomorphisms

$$T_p : C_p(X) \rightarrow C_{p+1}(X)$$

for all p , satisfying the following criteria.

- (i) S restricts to a chain map $C_*^{\mathcal{U}}(X) \rightarrow C_*^{\mathcal{U}}(X)$ and T_p restricts to a homomorphism $C_p^{\mathcal{U}}(X) \rightarrow C_{p+1}^{\mathcal{U}}(X)$ for all p ,
- (ii) the homomorphisms T_p form a chain homotopy from $S : C_*(X) \rightarrow C_*(X)$ to the identity and from the restriction $S : C_*^{\mathcal{U}}(X) \rightarrow C_*^{\mathcal{U}}(X)$ to the identity.
- (iii) for any simplex $\sigma : \Delta^p \rightarrow X$ there exists a natural number N such that $S^N(\sigma) \in C_p^{\mathcal{U}}(X)$.

Before producing such a chain map, let us explain why that proves theorem 7.2.2.

Lemma 7.3.1. *Let X and $\mathcal{U}$ be as in theorem 7.2.2. If there exist S and T satisfying the conditions above, then the inclusion $i : C_*^{\mathcal{U}}(X) \rightarrow C_*(X)$ induces an isomorphism $H_p^{\mathcal{U}}(X) \rightarrow H_p(X)$ for all p .*

Proof. We first show that the map is surjective. Let $c = \sum n_\sigma \sigma \in C_p(X)$ be a cycle, representing a class $[c] \in H_p(C_*(X), \partial)$. For each of the σ s appearing in c with a non-zero coefficient, there is some natural number $N = N(\sigma)$ such that $S^k(\sigma)$ is $\mathcal{U}$ -small as long as $k \geq N(\sigma)$. Letting N be the maximum of these $N(\sigma)$, all terms in the cycle $c' = S^N(c)$ are $\mathcal{U}$ -small. The chain homotopy T implies that the chain map $S : C_*(X) \rightarrow C_*(X)$ sends induces the identity on homology, so by induction the cycle c' represents the same homology class as c . Therefore $[c] = [c'] \in H_p(C_*(X), \partial)$ is the image of $[c'] \in H_p(C_*^{\mathcal{U}}(X), \partial)$, finishing the proof of surjectivity.

Injectivity is similar. Let $c \in C_p^{\mathcal{U}}(X)$ be a cycle which represents $0 \in H_p(C_*(X), \partial)$, i.e. that $c = \partial d$ for some $d \in C_{p+1}(X)$. Then d need not be $\mathcal{U}$ -small, even though c is, but by the same argument as before the chain $S^N(d)$ will be $\mathcal{U}$ -small for sufficiently large N . Since S is a chain map we have $\partial(S^N(d)) = S^N(\partial d) = S^N(c)$, and hence $[S^N(c)] = 0 \in H_p^{\mathcal{U}}(X)$. Since S is chain homotopic to the identity also when restricted to a chain map on $\mathcal{U}$ -small chains, we have $[c] = [S^N(c)] \in H_p^{\mathcal{U}}(X)$. $\square$

It remains to construct the chain map S and the chain homotopy T . In each degree, the homomorphisms $S_p : C_p(X) \rightarrow C_p(X)$ and $T_p : C_p(X) \rightarrow C_{p+1}(X)$ are uniquely given by their values on simplices, and the idea of the construction is to send a simplex $\sigma : \Delta^p \rightarrow X$ to a suitable linear combination of compositions of the form

$$\Delta^p \xrightarrow{\delta} \Delta^p \xrightarrow{\sigma} X,$$

where δ runs over certain continuous maps from the p -simplex to itself, i.e. we wish to define

$$(7.2) \quad S(\sigma) = \sum_{\delta} n_{\delta}(\sigma \circ \delta)$$

for suitable numbers $n_{\sigma} \in \mathbb{Z}$, only finitely many of which are non-zero. The idea is then to choose the maps δ carefully, so that the compositions $\sigma \circ \delta : \Delta^p \rightarrow X$ have “smaller” image than σ itself does, such that after finitely many iterations they actually become $\mathcal{U}$ -small.

The numbers n_{δ} should be non-zero for only finitely many maps $\delta : \Delta^p \rightarrow \Delta^p$ and we decide to choose them independent of σ . Then the homomorphism S defined by (7.2) may be encoded via the chain

$$c = \sum_{\delta} n_{\delta} \sigma \in C_p(\Delta^p)$$

and then the definition (7.2) may be concisely written as $S(\sigma) = \sigma_*(c)$, where $\sigma_* : C_p(\Delta^p) \rightarrow C_p(X)$ is the map induced by σ regarded as a continuous map $\Delta^p \rightarrow X$. Let us also decide that we want the self-maps $\delta : \Delta^p \rightarrow \Delta^p$ to be *affine*, as in the following definition.

Definition 7.3.2. For a convex subset $X \subset \mathbb{R}^n$, a simplex $\sigma : \Delta^p \rightarrow X$ is *affine* if it satisfies

$$\sigma(t_0, \dots, t_p) = \sum t_i \sigma(e_i).$$

In particular, affine simplices are uniquely determined by where they send the vertices $e_0, \dots, e_p \in \Delta^p$.

Let $L_p(X) \subset C_p(X)$ denote the span of all the affine simplices.

(Sometimes affine maps are called “linear”, but that seems misleading since Δ^p is not a vector space.)

The subspace $\Delta^p \subset \mathbb{R}^{p+1}$ is convex and the face maps $\delta^i : \Delta^{p-1} \rightarrow \Delta^p$ are affine. Moreover, the composition of two affine maps is again affine. Hence $\partial\sigma = \sum_{i=0}^p (-1)^i \sigma \circ \delta^i$ is affine if σ is affine, so ∂ restricts to a homomorphism $L_p(X) \rightarrow L_{p-1}(X)$ when $X \subset \mathbb{R}^n$ is affine. This restriction still satisfies $\partial \circ \partial = 0$ so we have a chain complex $(L_*(X), \partial)$.

In order to define $S : C_p(X) \rightarrow C_p(X)$ and $T : C_p(X) \rightarrow C_{p+1}(X)$ in general, the strategy is as follows (i) define S and T on affine chains of convex subsets of euclidean space, and in particular $S : L_p(\Delta^p) \rightarrow L_p(\Delta^p)$ and $T : L_p(\Delta^p) \rightarrow L_{p+1}(\Delta^p)$, (ii) let $c = S(\iota_p) = \sum n_{\delta} \delta \in L_p(\Delta^p)$, (iii) for general chains in topological spaces define $S(\sigma) = \sum n_{\delta}(\sigma \circ \delta)$, and (iv) define T on linear chains and extend to general chains in a similar way.

Definition 7.3.3. Let $X \subset \mathbb{R}^n$ be convex and let $x \in X$. Define the *cone operator*

$$C_x : L_p(X) \rightarrow L_{p+1}(X)$$

by sending a generator $\sigma : \Delta^p \rightarrow X$ to the affine simplex $C_x(\sigma) : \Delta^{p+1}(X)$ which sends $e_0 \mapsto x$ and $e_{i+1} \mapsto \sigma(e_i)$.

Lemma 7.3.4. *Let $\sigma : \Delta^p \rightarrow X$ be affine and $x \in X$. Then*

$$\partial C_x(\sigma) = \begin{cases} \sigma - C_x(\partial\sigma) & \text{for } p \geq 1 \\ \sigma - x & \text{for } p = 0 \end{cases}$$

Proof. The affine map $\delta^j : \Delta^{p-1} \rightarrow \Delta^p$ is characterized by

$$\delta^j(e_i) = \begin{cases} e_i & \text{for } i < j \\ e_{i+1} & \text{for } i \geq j. \end{cases}$$

From this we verify $C_x(\sigma \circ \delta^j) = (C_x\sigma) \circ \delta^{j+1}$ for $j = 0, \dots, p$ and any affine $\sigma : \Delta^p \rightarrow X$. Indeed, both have value x on e_0 , value $\sigma(e_{i-1})$ for $i = 1, \dots, j-1$, and value $\sigma(e_i)$ for $i = j, \dots, p-1$. Since they are affine simplices agreeing on vertices of Δ^{p-1} they are equal. By a similar argument we see $(C_x\sigma) \circ \delta^0 = \sigma$. Plugging these expressions for $(C_x\sigma) \circ \delta^j$ into $\partial(C_x\sigma) = \sum_{j=0}^{p+1} (-1)^j (C_x\sigma) \circ \delta^j$ gives the desired formula. $\square$

Definition 7.3.5. Let $X \subset \mathbb{R}^n$ be convex and $\sigma : \Delta^p \rightarrow X$ an affine simplex. The *barycenter* is the point $b(\sigma) = \sigma(1/(p+1), \dots, 1/(p+1)) \in X$. Define a homomorphism $S : L_p(X) \rightarrow L_p(X)$ by induction on p as follows. For $p = 0$ we let S_0 be the identity. Assume S_{p-1} has been defined, let $\sigma \in L_p(X)$ be an affine simplex and define

$$S_p(\sigma) = C_{b(\sigma)}(S_{p-1}(\partial\sigma)).$$

Lemma 7.3.6. *The homomorphisms $S_p : L_p(X) \rightarrow L_p(X)$ form a chain map.*

Proof. We calculate

$$\partial S_p(\sigma) = \partial C_{b(\sigma)}(S_{p-1}(\partial\sigma)) = S_{p-1}(\partial\sigma) - C_{b(\sigma)}(\partial S_{p-1}(\partial\sigma)).$$

Now by induction on p we may assume that we already proved $\partial S_{p-1} = S_{p-2}\partial$, so the above formula gives

$$\partial S_p(\sigma) = S_{p-1}(\partial\sigma) - C_{b(\sigma)}(S_{p-2}(\partial\partial\sigma)) = S_{p-1}(\partial\sigma),$$

i.e. $\partial S_p = S_{p-1}\partial$. By induction we have proved that S is a chain map. $\square$

Lemma 7.3.7. *Let $X \subset \mathbb{R}^n$ and $Y \subset \mathbb{R}^m$ be convex and let $f : X \rightarrow Y$ be an affine map, i.e. $f(t_0x_0 + \dots + t_px_p) = t_0f(x_0) + \dots + t_pf(x_p)$ for all $(t_0, \dots, t_p) \in \Delta^p$ and all $x_0, \dots, x_p \in X$. Then $f_* : L_p(X) \rightarrow L_p(Y)$ is a well defined chain map, and $f_* \circ S = S \circ f_* : L_p(X) \rightarrow L_p(Y)$.*

Proof sketch. We first claim that $\phi_*(\sigma)$ is an affine chain if σ is an affine simplex. This just follows from $\delta^j : \Delta^{p-1} \rightarrow \Delta^p$ being affine, and from composition of affine maps being affine.

To see $f_* \circ S = S \circ f_*$, the main observation is that affine maps preserve barycenters. The claim then follows inductively from the defining equation $S_p(\sigma) = C_{b(\sigma)}(S_{p-1}(\partial\sigma))$. $\square$

We are now ready to define $S : C_p(X) \rightarrow C_p(X)$ in general, and prove that it is a chain map.

Definition 7.3.8. Let $\iota_p \in L_p(\Delta^p)$ be the simplex given by the identity map of Δ^p and let $c = \sum n_\delta \delta \in L_p(\Delta^p) \subset C_p(\Delta^p)$ be the chain defined by $c = S(\iota_p)$. Then define $S_p : C_p(X) \rightarrow C_p(X)$ on generators by $S(\sigma) = \sigma_*(S(\iota_p)) = \sum n_\delta (\sigma \circ \delta)$.

Lemma 7.3.9. *This defines a chain map $S : C_*(X) \rightarrow C_*(X)$.*

Proof. Let us first note that the i th face map $\delta^i : \Delta^{p-1} \rightarrow \Delta^p$ may be regarded as an affine chain $\delta^i \in L_{p-1}(\Delta^p)$ and that $\delta^i = (\delta^i)_*(\iota_{p-1})$. Hence $S(\delta^i) = S(\delta^i_*(\iota_{p-1})) = \delta^i_*(S(\iota_{p-1}))$.

Then we may calculate

$$\begin{aligned} \partial(S(\sigma)) &= \partial(\sigma_*(S(\iota_p))) = \sigma_*(S(\partial\iota_p)) = \sum_{i=0}^p (-1)^i \sigma_*(S(\delta^i)) \\ &= \sum_{i=0}^p (-1)^i \sigma_* \delta^i_*(S(\iota_{p-1})) = \sum_{i=0}^p (-1)^i (\sigma \circ \delta^i)_*(S(\iota_{p-1})) = S(\partial\sigma). \quad \square \end{aligned}$$

It is easy to see that this chain map S satisfies at least some of the criteria we wanted. For example, it is clear that if we're given a collection $\mathcal{U}$ of subsets of X , then $S(\sigma)$ is $\mathcal{U}$ -small if σ is $\mathcal{U}$ -small.

Lemma 7.3.10. *Both $S : C_*(X) \rightarrow C_*(X)$ and its restriction to $\mathcal{U}$ -small chains are chain homotopic to the identity.*

Proof sketch. The construction of a chain homotopy $C_p(X) \rightarrow C_{p+1}(X)$ proceeds in a very similar manner to how we defined S , namely one first defines $T_p : L_p(X) \rightarrow L_{p+1}(X)$ for convex $X \subset \mathbb{R}^n$ and then uses $T_p(\iota_p) \in L_{p+1}(\Delta^p) \subset C_{p+1}(\Delta^p)$ to define $T_p : C_p(X) \rightarrow C_{p+1}(X)$ in general. See e.g. Hatcher's text (page 122) for details. $\square$

It remains to see one crucial property, which unfortunately we didn't have time for in class, namely that if X is covered by $\text{int}(U)$ for $U \in \mathcal{U}$, then $S^N(\sigma)$ is $\mathcal{U}$ -small for sufficiently large N .

Lemma 7.3.11. *For an affine map $\sigma : \Delta^p \rightarrow \mathbb{R}^n$, let $d(\sigma)$ denote the diameter of σ , i.e. the maximal distance between any of the vertices $\sigma(e_i)$. Then $S(\sigma)$ is a linear combinations of (affine) simplices with diameter $\leq (p/(p+1))d(\sigma)$.*

Proof sketch. For now, see page 120 in Hatcher's book. $\square$

By iterating we get the following consequence.

Corollary 7.3.12. *Let $\sigma : \Delta^p \rightarrow \mathbb{R}^n$ be an affine map. Then $S^N(\sigma)$ is a linear combination of (affine) simplices of diameter $\leq (p/(p+1))^N d(\sigma)$.* $\square$

Corollary 7.3.13. *Let $\mathcal{U}$ be a collection of open subsets of Δ^p , such that*

$$\bigcup_{U \in \mathcal{U}} \text{int}(U) = \Delta^p.$$

Then there exists an N such that $S^N(\iota_p)$ is $\mathcal{U}$ -small.

Proof. We may pick a Lebesgue number, i.e. an $\varepsilon > 0$ such that any ε -ball in the metric space Δ^p is contained in $\text{int}(U)$ for some $U \in \mathcal{U}$. Choose $N \in \mathbb{N}$ large enough that $d(\iota_p)(p/(p+1))^N < \varepsilon$. Then $S^N(\iota_p)$ is a linear combination of simplices of diameter $< \varepsilon/2$ and hence are contained in a ball of radius ε and hence in $\text{int}(U)$ for some $U \in \mathcal{U}$. $\square$

Corollary 7.3.14. *For any $\sigma : \Delta^p \rightarrow X$ and any set $\mathcal{U}$ of subsets of X with*

$$\bigcup_{U \in \mathcal{U}} \text{int}(U) = X,$$

there exists an $N \in \mathbb{N}$ with $S^N(\sigma) \in C_p^{\mathcal{U}}(X)$.

Proof. Apply Corollary 7.3.13 to the collection $\mathcal{U}' = \{\sigma^{-1}(U) \subset \Delta^p \mid U \in \mathcal{U}\}$ to get an N with $S^N(\iota_p)$ a linear combination of $\mathcal{U}'$ -small chains in Δ^p . But then $S^N(\sigma) = \sigma_*(S^N \iota_p)$ is $\mathcal{U}$ -small. $\square$

As we have explained, this finishes the proof of excision.

7.4. Excision revisited. Let X be a space and $A \subset X$ a subspace. Let us write X_+ for the disjoint union of X and a one-point space $\{+\}$, with the topology that makes $X \hookrightarrow X_+$ a homeomorphism onto an open and closed subset. Let $X/A = (X_+)/\sim$ be the quotient by the equivalence relation generated by $a \sim +$ for all $a \in A$. In words, we say that X/A is obtained from X by “collapsing $A \subset X$ to a point”. For example, $I/\partial I$ is homeomorphic to S^1 and more generally $D^n/\partial D^n$ is homeomorphic to S^n for any $n \geq 0$.

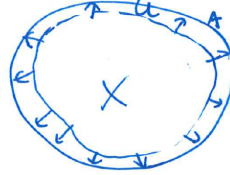
The quotient map $X_+ \rightarrow X/A$ gives rise to a map of pairs $q : (X, A) \rightarrow (X/A, A/A)$, where $A/A \subset X/A$ is the one-point subspace consisting of the equivalence class consisting of the point $+$ and all the $a \in A$. It is “usually” the case that the homomorphism induced by the quotient map

$$q_* : H_p(X, A) \rightarrow H_p(X/A, A/A)$$

is actually an isomorphism. Here, “usually” means that $A \subset X$ shouldn’t be too pathological a subspace. More precisely we have the following.

Definition 7.4.1. Let $A \subset X$ be a closed subspace. We say that A is a *deformation retract of a neighborhood* in X , or simply that (X, A) is a *good pair*, if there exists a set $U \subset X$ with $A \subset \text{int}(U)$ and a homotopy $h : I \times U \rightarrow U$ relative to A , from the identity to a map $r : U \rightarrow A \subset U$.

Example 7.4.2. The pair $(X, A) = (D^2, S^1)$, with $S^1 \subset D^2$ the boundary of the unit disc, is a good pair, where we can take U to be a small annulus containing S^1 .



Theorem 7.4.3. If a closed subspace $A \subset X$ is a deformation retract of an neighborhood, then the quotient map induces an isomorphism $H_p(X, A) \rightarrow H_p(X/A, A/A)$.

Example 7.4.4. We have that $D^2/S^1 \cong S^2$, showing that $H_p(D^2, S^1) \cong H_p(S^2, \{p\})$.



The proof is based on excision, and the fact that $X \setminus A$ is homeomorphic to $(X/A) \setminus (A/A)$. In order to apply excision, we first enlarge the space A in much the same way as when we proved that for a cell attachment the map $(D^n, \partial D^n) \rightarrow (A \cup_e D^n, A)$ induces an isomorphism.

Proof. Let $A \subset X$ be a deformation retract of a neighborhood U . This implies that the inclusion $A \hookrightarrow U$ is a homotopy equivalence, with homotopy inverse $r : U \rightarrow A$. This homotopy equivalence descends to a homotopy equivalence $A/A \hookrightarrow U/A$. We

then have the diagram

$$\begin{array}{ccccc}
 H_p(X, A) & \xrightarrow[\cong]{\text{htpy invariance}} & H_p(X, U) & \xleftarrow[\cong]{\text{excision}} & H_p(X \setminus A, U \setminus A) \\
 \downarrow & & \downarrow & & \downarrow \\
 H_p(X/A, A/A) & \xrightarrow[\cong]{\text{htpy invariance}} & H_p(X/A, U/A) & \xleftarrow[\cong]{\text{excision}} & H_p((X/A) \setminus (A/A), (U/A) \setminus (A/A))
 \end{array}$$

The horizontal homomorphisms are isomorphisms by the homotopy and excision theorems, and the rightmost homomorphism is an isomorphism because the underlying map of spaces are homomorphisms. $\square$

7.5. Cell attachments. Let us explain why it is useful to know $H_p(D^n, \partial D^n)$: excision will allow us to calculate $H_p(X, A)$ for any pair (X, A) obtained by a so-called “cell attachment”. If we know the homology of A , we can then use the long exact sequence of the pair to calculate the homology of X .

Recall that two sets X and Y have a *disjoint union* $X \sqcup Y$ obtained by taking the union of “disjoint copies” of X and Y . This set comes with “inclusion” maps $i_X : X \rightarrow X \sqcup Y$ and $i_Y : Y \rightarrow X \sqcup Y$. (More formally, let $X \sqcup Y = (\{0\} \times X) \cup (\{1\} \times Y)$ and $i_X(x) = (0, x)$ and $i_Y(y) = (1, y)$.) If X and Y are topological spaces, we let $X \sqcup Y$ have the topology in which $U \subset X \sqcup Y$ is open if and only if $i_X^{-1}(U) \subset X$ and $i_Y^{-1}(U)$ are both open. This is called the *disjoint union topology* and has the property that a function $f : X \sqcup Y \rightarrow Z$ is continuous if and only if $f \circ i_X$ and $f \circ i_Y$ are both continuous.

Definition 7.5.1. Let A be a space and let $e : \partial D^n \rightarrow A$. Define a new space

$$X = A \cup_e D^n$$

as the quotient space of $A \sqcup D^n$ under the equivalence relation generated by $i_{D^n}(v) \sim i_A(e(v))$ for all $v \in \partial D^n$.

In words, $A \cup_e D^n$ is obtained from A by *gluing* D^n along $e : \partial D^n \rightarrow A$. We shall also often say that this space is obtained from A by *attaching an n -cell*, and that $e : \partial D^n \rightarrow A$ is the *attaching map*.

In this situation, there are canonical maps

$$\begin{aligned}
 i : A &\rightarrow A \cup_e D^n \\
 c : D^n &\rightarrow A \cup_e D^n,
 \end{aligned}$$

obtained by composing the inclusions $i_A : A \rightarrow A \sqcup D^n$ and $i_{D^n} : D^n \rightarrow A \sqcup D^n$ with the quotient map to $A \cup_e D^n$.

Lemma 7.5.2. *The image $i(A) \subset A \cup_e D^n$ is closed, and i is a homeomorphism onto its image.*

Proof. Left to the reader. $\square$

By a slight “abuse of notation”, justified by this lemma, we shall identify A with its image $i(A) \subset A \cup_e D^n$.

The map $c : D^n \rightarrow X = A \cup_e D^n$ is in general not a homeomorphism onto its image, for example it is not injective unless $e : \partial D^n \rightarrow A$ is injective. We may use excision to deduce the following result.

Proposition 7.5.3. *Let $e : \partial D^n \rightarrow A$ be a continuous map, and let $X = A \cup_e D^n$ be obtained from A by attaching a cell. The map of pairs $c : (D^n, \partial D^n) \rightarrow (X, A)$ induces an isomorphism*

$$c_* : H_p(D^n, \partial D^n) \rightarrow H_p(X, A)$$

for all p .

Proof. From the definition of the quotient topology on X , we see that $A \subset X$ is closed and that the set $U = X \setminus \{c(0)\}$ is open. The map $c : D^n \rightarrow X$ has $c^{-1}(A) = \partial D^n$ and $c^{-1}(U) = D^n \setminus \{0\}$ and hence there is a commutative diagram

$$\begin{array}{ccccc} H_p(D^n, \partial D^n) & \xrightarrow[\cong]{\text{htpy eq}} & H_p(D^n, D^n \setminus \{0\}) & \xleftarrow[\cong]{\text{excision}} & H_p(D^n \setminus \partial D^n, D^n \setminus (\{0\} \cup \partial D^n)) \\ \downarrow c_* & & \downarrow c_* & & \downarrow c_* \\ H_p(X, A) & \xrightarrow[\cong]{\text{htpy eq}} & H_p(X, U) & \xleftarrow[\cong]{\text{excision}} & H_p(X \setminus A, U \setminus A), \end{array}$$

whose rows are induced by the obvious inclusion and whose vertical maps are induced by restrictions of the map c . The horizontal maps are isomorphisms because of the homotopy invariance and excision theorems, and the righthand vertical map is an isomorphism because the underlying maps of spaces are homeomorphisms.

It follows that the lefthand vertical map is also a homeomorphism. $\square$

Corollary 7.5.4. *The homomorphism*

$$i_* : H_p(A) \rightarrow H_p(X)$$

is an isomorphism if $p \notin \{n-1, n\}$. For $p = n-1$ it is surjective, and the kernel of i_ agrees with the image of the composition*

$$\mathbb{Z} \cong H_n(D^n, \partial D^n) \rightarrow H_{n-1}(\partial D^n) \xrightarrow{e_*} H_{n-1}(A).$$

If this composition is denoted $\phi : \mathbb{Z} \rightarrow H_{n-1}(A)$, then there is a short exact sequence

$$0 \rightarrow H_n(A) \rightarrow H_n(X) \rightarrow \text{Ker}(\phi).$$

Proof. Since $H_p(X, A) = 0$ for $p \neq n$, the long exact sequence of the pair gives isomorphisms $i_* : H_p(A) \rightarrow H_p(X)$ for $p \neq n, n-1$. The part of the long exact sequence containing $H_n(X, A)$ looks as follows.

$$0 \rightarrow H_n(A) \xrightarrow{i_*} H_n(X) \rightarrow \mathbb{Z} \xrightarrow{\phi} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(X) \rightarrow 0,$$

where the term $H_n(X, A)$ has been replaced by the isomorphic group $\mathbb{Z}$. To see identify the connecting homomorphism with the composition ϕ above we compare the long exact sequences for $(D^n, \partial D^n)$ and (X, A) , which includes the commutative square

$$\begin{array}{ccc} H_n(D^n, \partial D^n) & \xrightarrow{\delta} & H_{n-1}(\partial D^n) \\ \downarrow & & \downarrow e_* \\ H_n(X, A) & \xrightarrow{\delta} & H_{n-1}(A). \end{array}$$

$\square$

8. MORE TOOLS FOR COMPUTATIONS (WEEK 6-7)

Today we shall first discuss the Mayer-Vietoris exact sequence, a very useful tool to compute homology groups. After that, we will introduce Δ -complexes and *simplicial homology*, which is in Hatcher section 2.1. Simplicial homology is a different approach to homology using a much smaller chain complex.

8.1. The Mayer-Vietoris sequence. Let U_0 and U_1 be subsets of a topological space X and write $i_0 : U_0 \rightarrow X$ and $i_1 : U_1 \rightarrow X$ for the two inclusions. Let $\mathcal{U} = \{U_0, U_1\}$ and define the chain complex $C_*^{\mathcal{U}}(X)$ of $\mathcal{U}$ -small chains as above.

The abelian group $C_p^{\mathcal{U}}(X)$ is free on those continuous maps $\sigma : \Delta^p \rightarrow X$ whose image is contained in either U_0 or U_1 , and is therefore equal to the image of the homomorphism

$$(8.1) \quad \begin{aligned} C_p(U_0) \oplus C_p(U_1) &\rightarrow C_p(X) \\ (\alpha, \beta) &\mapsto (i_0)_*(\alpha) + (i_1)_*(\beta) \end{aligned}$$

The kernel of (8.1) consists of the pairs of formal linear combinations $(\sum_{\sigma} n_{\sigma} \sigma, \sum_{\sigma} n'_{\sigma} \sigma)$ of simplices in U_0 and U_1 , such that $n_{\sigma} + n'_{\sigma} = 0$ when $\sigma(\Delta^p) \subset U_0 \cap U_1$, and such that the coefficients of any simplex not contained in $U_0 \cap U_1$ is zero. This kernel is equal to the image of the map

$$(8.2) \quad \begin{aligned} C_p(U_{01}) &\rightarrow C_p(U_0) \oplus C_p(U_1) \\ c &\mapsto ((j_0)_*(c), -(j_1)_*(c)), \end{aligned}$$

where we write $U_{01} = U_0 \cap U_1$ and let $j_0 : U_{01} \rightarrow U_0$ and $j_1 : U_{01} \rightarrow U_1$ be the two inclusions. The homomorphism (8.2) is injective (in fact each coordinate $(j_0)_*$ and $-(j_1)_*$ is injective), so we have a short exact sequence

$$(8.3) \quad 0 \rightarrow C_p(U_{01}) \rightarrow C_p(U_0) \oplus C_p(U_1) \rightarrow C_p^{\mathcal{U}}(X) \rightarrow 0.$$

This may be viewed as a short exact sequence of chain complexes, where the middle term is a chain complex with boundary map $(\alpha, \beta) \mapsto (\partial\alpha, \partial\beta)$, and hence induces a long exact sequence on homology.

Assuming in addition that $\text{int}(U_0) \cup \text{int}(U_1) = X$, the homology of $C_*^{\mathcal{U}}(X)$ agrees with the homology of X , so the long exact sequence may be written

$$\cdots \rightarrow H_p(U_{01}) \rightarrow H_p(U_0) \oplus H_p(U_1) \rightarrow H_p(X) \xrightarrow{\delta} H_{p-1}(U_{01}) \rightarrow \cdots,$$

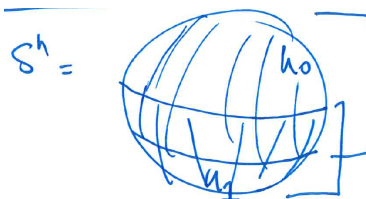
where the two maps are

$$(i_0)_* + (i_1)_* : H_p(U_0) \oplus H_p(U_1) \rightarrow H_p(X)$$

and

$$((j_0)_*, -(j_1)_*) : H_p(U_{01}) \rightarrow H_p(U_0) \oplus H_p(U_1)$$

Example 8.1.1. One can inductively compute the homology of the spheres S^n starting from that of S^0 using the Mayer-Vietoris sequence for $S^n = U_0 \cup U_1$ for U_0 a neighborhood of the North hemisphere and U_1 a neighborhood of the South hemisphere, noting that $U_0 \cap U_1 \simeq S^{n-1}$.



8.2. Relative homology of a simplex. We start by producing an isomorphism $\mathbb{Z} \rightarrow H_n(\Delta^n, \partial\Delta^n)$. It is not too hard to construct a homeomorphism $(D^n, \partial D^n) \approx (\Delta^n, \partial\Delta^n)$, from which it follows that $H_n(\Delta^n, \partial\Delta^n)$ is *isomorphic* to $\mathbb{Z}$, but in this section it is important to have an explicit isomorphism.

Let $\iota_p \in C_p(\Delta^p)$ be the chain given by the identity map of Δ^p and $\delta^i \in C_{p-1}(\Delta^p)$ be the chain given by the i th face map, for $i = 0, \dots, p$. Then we have

$$\partial \iota_p = \sum_{i=0}^p (-1)^i \delta^i \in C_{p-1}(\Delta^p).$$

In particular ι_p is not a cycle (unless $p = 0$), but its image in relative chains $C_*(\Delta^p, \partial\Delta^p)$ is a cycle, where as usual we write $\partial\Delta^p = \cup_{i=0}^p \delta^i(\Delta^{p-1})$.

Lemma 8.2.1. *The chain $\iota_p \in C_p(\Delta^p, \partial\Delta^p)$ is a cycle, representing a generator of the group $H_p(\Delta^p, \partial\Delta^p) \cong \mathbb{Z}$.*

Proof. We already explained why ι_p is a cycle, so it represents a class $[\iota_p] \in H_p(\Delta^p, \partial\Delta^p)$. I will skip this in class for reasons of time, but here are the details.

Let $\Lambda \subset \partial\Delta^p$ be the subspace $\Lambda = \cup_{i=1}^p \delta^i(\Delta^{p-1})$. Notice that we left out the face $\delta^0(\Delta^{p-1})$. It is easy to see that Λ is contractible for $p \geq 1$: in fact the identity map of Λ is homotopic to the constant map e_0 by the straight-line homotopy.

Now, (exercise!) the face map $\delta^0 : \Delta^{p-1} \rightarrow \partial\Delta^p$ induces a homeomorphism $\Delta^{p-1}/\partial\Delta^{p-1} \rightarrow \partial\Delta^p/\Lambda$ and hence by excision we get an isomorphism

$$(\delta^0)_* : H_{p-1}(\Delta^{p-1}, \partial\Delta^{p-1}) \rightarrow H_{p-1}(\partial\Delta^p, \Lambda).$$

By induction, we may assume that $[\iota_{p-1}] \in H_{p-1}(\Delta^{p-1}, \partial\Delta^{p-1}) \cong \mathbb{Z}$ is a generator, so we deduce that $(\delta^0)_*([\iota_{p-1}]) = [\delta^0] \in H_{p-1}(\partial\Delta^p, \Lambda)$ is also a generator.

For the induction step, first note that the inclusion map $\Lambda \rightarrow \Delta^p$ is a homotopy equivalence since both spaces are contractible, and that hence $H_i(\Delta^p, \Lambda) = 0$ for all i . By the long exact sequence for the triple $(\Delta^p, \partial\Delta^p, \Lambda)$, the connecting homomorphism

$$(8.4) \quad H_p(\Delta^p, \partial\Delta^p) \xrightarrow{\delta} H_{p-1}(\partial\Delta^p, \Lambda)$$

is an isomorphism. Using the *definition* of the connecting homomorphism in the long exact sequence, we see that in this case it will send the class $[\iota_p] \in H_p(\Delta^p, \partial\Delta^p)$ to the class of the cycle $\sum_{i=0}^p (-1)^i \delta^i \in C_{p-1}(\Delta^p, \Lambda)$. But clearly $\delta^i \in C_{p-1}(\Lambda) \subset C_{p-1}(\partial\Delta^p)$ for $i = 1, \dots, p$ so the only remaining term is $\delta^0 \in C_{p-1}(\partial\Delta^p, \Lambda)$. We have proved that the isomorphism (7.1) sends $[\iota_p] \mapsto [\delta^0] \in C_{p-1}(\partial\Delta^p, \Lambda)$. Since $[\delta_0] \in H_{p-1}(\partial\Delta^p, \Lambda_n)$ is a generator, it follows that $[\iota_p] \in H_p(\Delta^p, \partial\Delta^p)$ is too, as desired. $\square$

As a corollary, we may obtain a generator for $H_p(D^p, \partial D^p)$ as $[\sigma]$ for any choice of homeomorphism $\sigma : (\Delta^p, \partial\Delta^p) \rightarrow (D^p, \partial D^p)$: indeed, $\sigma_*([\iota_p]) = [\sigma]$.

8.3. Δ -complexes. Recall that if X is a topological space, we write $\text{Sin}_p(X)$ for the set of continuous maps $\Delta^p \rightarrow X$, and that we have defined

$$d_i : \text{Sin}_p(X) \rightarrow \text{Sin}_{p-1}(X)$$

and that these satisfy $d_i \circ d_j = d_{j-1} \circ d_i$ when $i < j$.

Definition 8.3.1. A Δ -complex structure on a topological space X is a tuple $\Sigma = (\Sigma_p)_{p \in \mathbb{N}}$ consisting of subsets $\Sigma_p \subset \text{Sin}_p(X)$ for each $p \geq 0$, satisfying

- (i) $d_i(\Sigma_p) \subset \Sigma_{p-1}$ for all $i = 0, \dots, p$ and all $p \in \mathbb{N}$;
- (ii)

$$\bigcup_{p \in \mathbb{N}} \bigcup_{\sigma \in \Sigma_p} \sigma(\Delta^p \setminus \partial \Delta^p) = X,$$

the subsets $\sigma(\Delta^p \setminus \partial \Delta^p) \subset X$ in this union are all disjoint, and all restrictions $\sigma|_{\Delta^p \setminus \partial \Delta^p}$ are injective;

- (iii) a subset $A \subset X$ is open if and only if $\sigma^{-1}(A) \subset \Delta^p$ is open for all $\sigma \in \Sigma_p$ and all $p \in \mathbb{N}$.

A Δ -complex is then a pair (X, Σ) consisting of a topological space X and a Δ -complex structure Σ on X . In that case, the elements of Σ_p are called the p -simplices of the Δ -complex, and the subsets $\sigma(\Delta^p \setminus \partial \Delta^p) \subset X$ are called the *open simplices*.

It is customary to omit the Σ from the notation, for example by saying “let X be a Δ -complex” and then using the same notation X for the underlying topological space. (It is also customary to use the same letter for the sets of simplices, i.e. write X_p for the set of p -simplices of X , but we shall refrain from that here.)

Remark 8.3.2. Let us explain a more conceptual point of view on the axioms, according to which they express that the underlying topological space X can be completely reconstructed from the sets Σ_p and all the maps $d_i : \Sigma_p \rightarrow \Sigma_{p-1}$. This is best described in terms of the *disjoint union*

$$(8.5) \quad \coprod_{p \in \mathbb{N}} \Sigma_p \times \Delta^p.$$

The set $\Sigma_p \times \Delta^p$ consists of one “copy” of Δ^p for each element of Σ_p , and the symbol “ $\coprod$ ” means that we take their union in a way that makes these disjoint also as $p \in \mathbb{N}$ varies. (More formally, this could be defined as e.g. the actual union of the sets $\{p\} \times \Sigma_p \times \Delta^p$.)

By axiom (i) we have well defined maps $d_i : \Sigma_p \rightarrow \Sigma_{p-1}$ for $i = 0, \dots, p$, and we let $\sim$ be the equivalence relation on the set (8.5) generated by $(\sigma, \delta^i(t)) \sim (d_i \sigma, t)$, over all $p \in \mathbb{N}$, $\sigma \in \Sigma_p$, $t \in \Delta^{p-1}$, and all $i = 0, \dots, p$. For each p there is a canonical “evaluation” map $\Sigma_p \times \Delta^p \rightarrow X$ given by $(\sigma, t) \mapsto \sigma(t)$, and there is a well defined induced map

$$(8.6) \quad \left(\coprod_{p \in \mathbb{N}} \Sigma_p \times \Delta^p \right) / \sim \xrightarrow{\text{ev}} X$$

$$(\sigma, t) \quad \mapsto \sigma(t),$$

which is continuous when the domain is given the quotient topology, i.e. the finest topology making the canonical maps from $\Sigma_p \times \Delta^p$ continuous, for all $p \in \mathbb{N}$.

As long as axiom (i) holds, we get a well defined and continuous map (8.6). Now axiom (ii) expresses that (8.6) is a bijection of sets, and axiom (iii) expresses that X has the quotient topology from (8.6).

Therefore the two axioms (ii) and (iii) may be replaced by one: that the map (8.6) be a homeomorphism. This formulation of the axioms takes a bit longer to explain, but is possibly more conceptual: it expresses that the topological space X may be reconstructed, up to homeomorphism, from the abstract sets Σ_p and the maps d_i between them.

As a final remark that we shall not spend time on in class, one may conversely consider tuples Σ consisting of *abstract* sets Σ_p and maps $d_i : \Sigma_p \rightarrow \Sigma_{p-1}$ between

them for $i = 0, \dots, p$, required to satisfy $d_i \circ d_j = d_{j-1} \circ d_i$ for $i < j$. Such data is called a *semi-simplicial set* and we may *define* a topological space by the gluing formula in (8.6). This space is called the geometric realization of Σ , is denoted $|\Sigma|$, and comes with a canonical Δ -complex structure.

Verifying the third axiom can often be handled by the following compactness argument.

Lemma 8.3.3. *Let X be a Hausdorff space and let $\Sigma = (\Sigma_p)_{p \in \mathbb{N}}$ satisfy axiom (i) and (ii) of Definition 8.3.1, and assume in addition that the sets Σ_p are finite, and that all but finitely many of them are empty. Then axiom (iii) is also satisfied.*

Proof. Let $B \subset X$ be a subset satisfying that $\sigma^{-1}(B) \subset \Delta^p$ is closed for all $p \in \mathbb{N}$ and all $\sigma \in \Sigma_p$. Then each such $\sigma^{-1}(B)$ is compact and since $\sigma : \Delta^p \rightarrow X$ is a continuous function to a Hausdorff space, $\sigma(\sigma^{-1}(B)) \subset X$ is compact and therefore closed. Now $\sigma(\sigma^{-1}(B)) = B \cap \sigma(\Delta^p)$, and axiom (ii) implies that the subsets $\sigma(\Delta^p) \subset X$ cover all of X . Therefore

$$B = \bigcup_{p \in \mathbb{N}} \bigcup_{\sigma \in \Sigma_p} \sigma(\sigma^{-1}(B))$$

is a finite union of closed subsets of X , and therefore closed. If $A \subset X$ is a subset such that all $\sigma^{-1}(A)$ are open, then all $\sigma^{-1}(X \setminus A)$ are closed, so $X \setminus A$ is closed and hence A is open. This verifies the “if” part of axiom (iii), and “only if” is trivial. $\square$

Example 8.3.4. We shall construct a Δ -complex structure on the torus, defined as the quotient space $T = \mathbb{R}^2 / \mathbb{Z}^2$ where $\mathbb{Z}^2$ acts by translation. We set $\Sigma_2 = \{\rho, \rho'\}$, where the two maps $\rho, \rho' : \Delta^2 \rightarrow T$, given by the formulas in the proof of Lemma 6.1.2, i.e.

$$\begin{aligned} \rho(t_0, t_1, t_2) &= t_0(0, 0) + t_1(1, 0) + t_2(1, 1) \\ \rho'(t_0, t_1, t_2) &= t_0(0, 0) + t_1(0, 1) + t_2(1, 1). \end{aligned}$$

Setting $\Sigma_1 = \{a, b, c\}$ where

$$\begin{aligned} a(t_0, t_1) &= t_0(0, 0) + t_1(1, 0) \\ b(t_0, t_1) &= t_0(0, 0) + t_1(0, 1) \\ c(t_0, t_1) &= t_0(0, 0) + t_1(1, 1), \end{aligned}$$

we see, using translation by $(-1, 0)$ and $(0, -1)$, that $d_0\rho = b$, $d_1\rho = c$, $d_2\rho = a$, $d_0\rho' = a$, $d_1\rho' = c$, and $d_2\rho' = b$. Then $d_i(\Sigma_2) \subset \Sigma_1$ for $i = 0, 1, 2$. Finally, set $\Sigma_0 = \{x_0\}$, where $x_0 : \Delta^0 \rightarrow T$ is the constant map to $(0, 0)$ and check that $d_i(\Sigma_1) \subset \Sigma_0$ for $i = 0, 1$. This proves the first axiom in the definition.

Axiom (ii) is verified by a case-by-case argument: any element $[(x, y)] \in T$ has a unique representative with $(x, y) \in [0, 1)^2$, and for this representative we have the following mutually exclusive cases to consider.

- $(x, y) = (0, 0)$: this is the zero-simplex x_0 .
- $x > 0, y = 0$: then $(x, y) = a(t)$ with $t = (1 - x, x) \in \Delta^1 \setminus \partial\Delta^1$, and this is the unique way to write it as the image of an interior point in a simplex.
- $x = 0, y > 0$: then $(x, y) = b(t)$ with $t = (1 - y, y) \in \Delta^1 \setminus \partial\Delta^1$, and this is the unique way to write it as the image of an interior point in a simplex.
- $0 < x = y$: then $(x, y) = c(t)$ with $t = (1 - x, x) \in \Delta^1 \setminus \partial\Delta^1$, and this is the unique way to write it as the image of an interior point in a simplex.
- $0 < y < x$: then $(x, y) = \rho(t)$ with $t = (1 - x, x - y, y) \in \Delta^2 \setminus \partial\Delta^2$, and this is the unique way to write it as the image of an interior point in a simplex.

- $0 < x < y$: then $(x, y) = \rho'(t)$ with $t = (1 - y, y - x, x) \in \Delta^2 \setminus \partial\Delta^2$, and this is the unique way to write it as the image of an interior point in a simplex.

This case-by-case analysis verifies axiom (ii), and axiom (iii) follows by the lemma above.

We shall mostly be interested in *finite* Δ -complexes, those where each Σ_p is a finite set, and all but finitely many Σ_p are empty. It is an important point that a finite Δ -complex (X, Σ) may be specified by finitely much data, namely the finitely many finite set Σ_p and for each simplex $\sigma \in \Sigma_p$ the faces $d_i(\sigma) \in \Sigma_{p-1}$ that it is glued to.

Finite Δ -complexes give a way of representing some topological spaces on a computer or in a programming language. The special case of *oriented graphs* is ubiquitous: it is the special case where Σ_0 is the set of vertices, Σ_1 is the set of edges, and $\Sigma_p = \emptyset$ for $p > 1$.

8.4. Simplicial chains and simplicial homology. Simplicial homology is defined using a chain complex of simplicial chains, as we define now:

Definition 8.4.1. Let (X, Σ) be a Δ -complex. The *simplicial p -chains* of X is the subgroup

$$\Delta_p(X) \subset C_p(X)$$

spanned by the maps $\sigma : \Delta^p \rightarrow X$ which are in Σ_p .

Then $\sigma \circ \delta^i \in \Delta_{p-1}(X)$ for all $i = 0, \dots, p$ and we define $\partial\sigma = \sum_{i=0}^p (-1)^i \sigma \circ \delta^i$ as we did for singular chains. This again satisfies $\partial \circ \partial = 0$, and we define

$$H_p^\Delta(X) = H_p(\Delta_*(X), \partial).$$

Theorem 8.4.2. Let X be a Δ -complex. The inclusion $\Delta_*(X) \rightarrow C_*(X)$ of simplicial chains into singular chains is a chain map and induces an isomorphism

$$(8.7) \quad H_p^\Delta(X) \rightarrow H_p(X)$$

for all p .

Example 8.4.3. One reason this theorem is valuable is that the left hand side is usually quite computable. For example, for the torus the simplicial chain complex is of the form

$$\dots \rightarrow 0 \rightarrow \mathbb{Z}^2 \xrightarrow{\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \end{bmatrix}} \mathbb{Z}^3 \xrightarrow{0} \mathbb{Z},$$

from which it is easy to read off the homology. Indeed, by changing basis to $(1, 0), (1, -1) \in \mathbb{Z}^2$ and $(1, 1, -1), (1, 0, 0), (0, 1, 0) \in \mathbb{Z}^3$, we see that the chain complex is isomorphic to the chain complex

$$\dots \rightarrow 0 \rightarrow \mathbb{Z}^2 \xrightarrow{\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}} \mathbb{Z}^3 \xrightarrow{0} \mathbb{Z},$$

from which it is immediate that $H_0 \cong \mathbb{Z}$ generated by $[x]$, $H_1 \cong \mathbb{Z}^2$ generated by $[a]$ and $[b]$, and $H_2 \cong \mathbb{Z}$ generated by $\rho - \rho'$.

The calculation of $H_p^\Delta(X)$ from the data of the sets Σ_p and the maps $d_i : \Sigma_p \rightarrow \Sigma_{p-1}$ may be programmed into a computer, at least when X is a finite Δ -complex.

For example, the mathematical software SAGE has implemented Δ -complexes as a data type, and an algorithm for calculating homology and other invariants.

Besides calculations, Theorem 8.4.2 also has “theoretical” consequences.

Corollary 8.4.4. *Let X be a topological space admitting the structure of a finite Δ -complex. Then $H_p(X)$ is a finitely generated abelian group for all p and is the trivial group if p is larger than the dimension of any of the simplices in X .*

Proof. The kernel of $\partial : \Delta_p(X) \rightarrow \Delta_{p-1}(X)$ is a finitely generated abelian group, since it is a subgroup of the finitely generated abelian group $\Delta_p(X)$. Hence the quotient $H_p(X) \cong H_p^\Delta(X)$ is too. $\square$

In order to prove theorem 8.4.2, it is convenient to formulate a version for relative homology.

Definition 8.4.5. Let (X, Σ^X) be a Δ -complex. A subspace $A \subset X$ is a *subcomplex* if it is the union of some of the simplices of X . In other words, if

$$A = \bigcup_{p \in \mathbb{N}} \bigcup_{\sigma \in \Sigma'_p} \sigma(\Delta^p)$$

for some subsets $\Sigma'_p \subset \Sigma_p^X$.

We leave it as an exercise to verify that in this situation we may define a Δ -complex structure on A by setting $\Sigma_p^A = \{\sigma \in \Sigma_p^X \mid \sigma(\Delta^p) \subset A\}$. (This set could be larger than Σ'_p .)

Example 8.4.6. For any Δ -complex X , the n -skeleton is the subcomplex $X^n \subset X$ given as

$$X^n = \bigcup_{p \leq n} \bigcup_{\sigma \in \Sigma_p^X} \sigma(\Delta^p)$$

It has p -simplices $\Sigma_p^{X^n} = \Sigma_p^X$ for $p \leq n$ and $\Sigma_p^{X^n} = \emptyset$ for $p > n$.

Definition 8.4.7. Let X be a Δ -complex and A a subcomplex. Then $\Delta_p(A) \subset \Delta_p(X)$ and we define $\Delta_p(X, A) = \Delta_p(X)/\Delta_p(A)$. This inherits a homomorphism $\partial : \Delta_p(X, A) \rightarrow \Delta_{p-1}(X, A)$ satisfying $\partial \circ \partial = 0$, and hence we may define $H_p^\Delta(X, A) = H_p(\Delta_*(X, A), \partial)$.

In fact we have a diagram

$$(8.8) \quad \begin{array}{ccccccc} 0 & \longrightarrow & \Delta_p(A) & \longrightarrow & \Delta_p(X) & \longrightarrow & \Delta_p(X, A) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & C_p(A) & \longrightarrow & C_p(X) & \longrightarrow & C_p(X, A) \longrightarrow 0, \end{array}$$

where both rows define short exact sequences of chain complexes. We will prove the following generalization of Theorem 8.4.2 to relative homology.

Theorem 8.4.8. *Let X be a Δ -complex and $A \subset X$ a subcomplex. Then the homomorphism*

$$H_p^\Delta(X, A) \rightarrow H_p(X, A)$$

induced by including simplicial chains into singular chains, i.e. the rightmost vertical map $\Delta_(X, A) \rightarrow C_*(X, A)$ in (8.8), is an isomorphism for all p .*

In fact Theorem 8.4.8 is easily deduced from Theorem 8.4.2 by a 5-lemma argument, but the relative statement is more convenient to prove. The following proposition contains the key step: to prove the relative statement in the case where all relative cells have the same dimension.

Proposition 8.4.9. *Let X be a Δ -complex and $n \in \mathbb{N}$. Then Theorem 8.4.8 holds for the pair (X^n, X^{n-1}) , i.e. the natural map*

$$H_p^\Delta(X^n, X^{n-1}) \rightarrow H_p(X^n, X^{n-1})$$

is an isomorphism for all p .

Proof. This is done by calculating both sides. In the domain we just use the chain complex, which looks like

$$\cdots \rightarrow 0 \rightarrow \mathbb{Z}\Sigma_n \rightarrow 0 \rightarrow \cdots,$$

from which we get that $H_p^\Delta(X^n, X^{n-1}) = 0$ for $p \neq n$ and that $H_n^\Delta(X^n, X^{n-1})$ is the free abelian group with basis $[\sigma]$, ranging over $\sigma \in \Sigma_n$.

Hence it remains to prove that the same is true for singular homology, i.e. that $H_p(X^n, X^{n-1}) = 0$ for $p \neq n$ and that

$$\begin{aligned} \mathbb{Z}\Sigma_n &\rightarrow H_n(X^n, X^{n-1}) \\ \sigma &\mapsto [\sigma] \end{aligned}$$

is an isomorphism. We will obtain that by combining excision with Lemma 8.2.1.

We have a canonical “evaluation” map

$$\begin{aligned} \Sigma_n \times \Delta^n &\xrightarrow{\text{ev}} X^n \\ (\sigma, t) &\mapsto \sigma(t) \end{aligned}$$

which is continuous when Σ_n is given the discrete topology and $\Sigma_n \times \Delta^n$ the product topology. Since $\partial\Delta^n = \cup_{i=0}^n \delta^i(\Delta^{n-1})$ we see that $\text{ev}(\sigma, \delta^i(t)) = \sigma(\delta^i t) = (d_i \sigma)(t) \in X^{n-1}$, so we have a map of pairs $(\Sigma_n \times \Delta^n, \Sigma_n \times \partial\Delta^n) \rightarrow (X^n, X^{n-1})$. It is easily verified that both are *good pairs* (we shall leave this as an exercise, with the hint to remove the barycenters of each Δ^n and use a straight-line homotopy to retract each simplex to its boundary) and that the induced map of quotient spaces

$$\frac{\Sigma_n \times \Delta^n}{\Sigma_n \times \partial\Delta^n} \xrightarrow{\text{ev}} \frac{X^n}{X^{n-1}}$$

is a homeomorphism. By excision we then get the isomorphism

$$\text{ev}_* : H_p(\Sigma_n \times \Delta^n, \Sigma_n \times \partial\Delta^n) \rightarrow H_p(X^n, X^{n-1}).$$

Finally, connectivity of Δ^p and discreteness of Σ_n gives a chain level isomorphism

$$\bigoplus_{\sigma \in \Sigma_n} C_p(\Delta^n, \partial\Delta^n) \rightarrow C_p(\Sigma_n \times \Delta^n, \Sigma_n \times \partial\Delta^n)$$

inducing an isomorphism

$$\bigoplus_{\sigma \in \Sigma_n} H_p(\Delta^n, \partial\Delta^n) \rightarrow H_p(\Sigma_n \times \Delta^n, \Sigma_n \times \partial\Delta^n),$$

which combined with Lemma 8.2.1 gives the desired formula for $H_p(X^n, X^{n-1})$. $\square$

Proof of Theorems 8.4.2 and 8.4.8. We shall only give full details in the case X is *finite dimensional*, i.e. when $X = X^n$ for some n .

Given a Δ -complex X , let us prove Theorem 8.4.2 for X^n , i.e. that $H_p^\Delta(X^n) \rightarrow H_p(X^n)$ is an isomorphism for all p , by induction on n . We may begin the induction at $n = -1$ where $X^n = \emptyset$. (Or begin at $n = 0$ where X^0 is the set Σ_0 with the discrete topology, for which we may calculate singular homology by hand.)

Now the diagram (8.8) for the pair (X^n, X^{n-1}) induces long exact sequences in simplicial and singular homology, and a ladder diagram between the two long exact sequences. We have proved that $H_p^\Delta(X^n, X^{n-1}) \rightarrow H_p(X^n, X^{n-1})$ is an isomorphism for all p , and by induction we may assume $H_p^\Delta(X^{n-1}) \rightarrow H_p(X^{n-1})$ is an isomorphism for all p . The induction step then follows by the 5-lemma.

When X is assumed finite dimensional Δ -complex we have $X = X^n$ for sufficiently large n , so in this case we have now proved Theorem 8.4.2. The relative case in Theorem 8.4.8 follows by another 5-lemma. $\square$

Remark 8.4.10. We shall not discuss this in class, but in fact the theorem also holds for infinite dimensional Δ -complexes. The proof has one extra point-set topological ingredient, namely that any compact subset of a Δ -complex X must be contained in some finite skeleton (proof: if $K \subset X$ is not contained in any X^n , pick a sequence of points $x_n \in K \setminus X^n$ and use Axiom (iii) to see that the countably infinite set $\{x_0, x_1, \dots\} \subset K$ is closed; therefore K is not sequentially compact and hence not compact). In particular any $\sigma \in \text{Sin}_p(X)$ comes from $\text{Sin}_p(X^n)$ for some n .

Then one sees that $H_p(X, X^n) = 0$ for $n \geq p$: indeed, by the compactness observation any cycle in $C_p(X, X^n)$ must come from a cycle in $C_p(X^m, X^n)$ for some finite m , but by the finite-dimensional case we have $H_p(X^m, X^n) \cong H_p^\Delta(X^m, X^n) = 0$, so the cycle is a boundary in $C_*(X^m, X^n)$ and hence in $C_*(X, X^n)$. By the long exact sequence, the inclusion induces an isomorphism $H_p(X^n) \cong H_p(X)$ for $n > p$. For fixed p , the general case is then reduced to the finite dimensional case, which we have proved.

9. APPLICATIONS OF HOMOLOGY (WEEK 7)

Finally, we shall discuss some applications: the euler characteristic of a finite Δ -complex and the degree of a map $f : S^n \rightarrow S^n$. Both are treated in Hatcher's section 2.2. We also discuss the Jordan Curve theorem and its higher-dimensional generalization (see section 2.B in Hatcher's book).

9.1. Euler characteristic. To define the Euler characteristic, we will use the following notion of rank:

Definition 9.1.1. Let A be a finitely generated abelian group. By a theorem from algebra, we may find an isomorphism

$$A \cong \mathbb{Z}^d \oplus T,$$

where T is a finite abelian group. Then the *rank* of A is the number d .

It is not hard to see that the rank is well defined, i.e. independent of choice of isomorphism: indeed, if we let $pA = \{pa \mid a \in A\}$ for a prime number p , the quotient A/pA is a vector space over the finite field $\mathbb{F}_p$ with p elements; if we choose p so that no element of A has order p , then $d = \dim_{\mathbb{F}_p}(A/pA)$.

Definition 9.1.2. Let X be a topological space such that $H_p(X)$ is finitely generated for all p and such that there exists an N with $H_p(X) = 0$ for all $p > N$. For such X we define

$$\chi(X) = \sum_{i=0}^{\infty} (-1)^i \text{rank}(H_i(X)).$$

This number $\chi(X) \in \mathbb{Z}$ is called the *euler characteristic* of X .

Some spaces do not have finitely generated homology, and for such spaces $\chi(X)$ is not defined. By Corollary 8.4.4, the euler characteristic is defined for any space homotopy equivalent to a finite Δ -complex.

As we shall see next, it turns out to be *much* easier to calculate the euler characteristic than to calculate the whole homology: it only depends on the number of simplices in each dimension, not on how they are glued together!

Theorem 9.1.3. Let X be a finite Δ -complex and let $n_i \in \mathbb{N}$ be the number of i -simplices in the Δ -complex structure. Then

$$\chi(X) = \sum_{i=0}^{\infty} (-1)^i n_i.$$

There are usually many different Δ -complex structures on the same space. The theorem implies that the alternating sum of the number of simplices is independent of choice of Δ -complex structure. For example one easily calculates $\chi(S^2) = 2$, so given any Δ -complex structure on S^2 with V zero-simplices, E one-simplices, and F two-simplices we have $E - V + F = 2$, independently of Δ -complex structure. This special case is often called *Euler's polyhedron theorem*.

Notice that $n_i = \text{rank}(\Delta_n(X))$. Hence the theorem follows from the following completely algebraic theorem.

Theorem 9.1.4. *Let (A_*, ∂) be a chain complex such that A_i is finitely generated for all i and non-trivial for only finitely many i . Write $H_i = H_i(A_*, \partial)$. Then*

$$\sum_i (-1)^i \text{rank}(A_i) = \sum_i (-1)^i \text{rank}(H_i).$$

Proof. Let us write $Z_i = \text{Ker}(\partial : A_i \rightarrow A_{i-1})$ and $B_i = \text{Im}(\partial : A_{i+1} \rightarrow A_i)$. Then we have short exact sequences

$$\begin{aligned} 0 \rightarrow B_i &\xrightarrow{\text{incl}} Z_i \rightarrow H_i \rightarrow 0 \\ 0 \rightarrow Z_i &\xrightarrow{\text{incl}} A_i \xrightarrow{\partial} B_{i-1} \rightarrow 0. \end{aligned}$$

By Lemma 9.1.5 below, this implies that $\text{rank}(Z_i) = \text{rank}(B_i) + \text{rank}(H_i)$ and $\text{rank}(A_i) = \text{rank}(Z_i) + \text{rank}(B_{i-1})$. These two equations imply $(-1)^i \text{rank}(H_i) = (-1)^i \text{rank}(A_i) + (-1)^i \text{rank}(B_i) - (-1)^{i-1} \text{rank}(B_{i-1})$, and adding over all i gives the desired result. $\square$

Lemma 9.1.5. *Let A , B , and C be finitely generated abelian groups and let*

$$0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$$

be a short exact sequence. Then $\text{rank}(B) = \text{rank}(A) + \text{rank}(C)$.

Proof. Without loss of generality we may assume $C = \mathbb{Z}^r \oplus T$ for a finite abelian group T . Let $N = |T|$. Then we may write $g(b) = (g_1(b), \dots, g_r(b), t(b))$ for homomorphisms $g_i : B \rightarrow \mathbb{Z}$ and $t : B \rightarrow T$. By surjectivity of g we may choose elements $c_j \in C$ such that $g_i(c_i) = 1$ and $g_i(c_j) = 0$ for $i \neq j$ and $t(c_j) = 0$. Define a homomorphism $s : C \rightarrow B$ by $s(n_1, \dots, n_r, \alpha) = N(n_1 c_1 + \dots + n_r c_r)$. Then an easy calculation shows that $g \circ s(c) = Nc$ for all $c \in C$, i.e. $g \circ s = N1_C$.

It follows that $g \circ (N1_B - s \circ g) = Ng - g \circ s \circ g = 0$, so that the homomorphism $N1_B - s \circ g : B \rightarrow B$ takes values in the kernel of g . Therefore we may define a homomorphism $r : B \rightarrow A$ as $r = f^{-1} \circ (N1_B - s \circ g)$.

Now define homomorphisms $\phi : A \oplus C \rightarrow B$ and $\psi : B \rightarrow A \oplus C$ by $\phi(a, c) = f(a) + s(c)$ and $\psi(b) = (r(b), g(b))$. These are not quite inverse, but satisfy $\phi \circ \psi = N1_B$ and $\psi \circ \phi = N1_{A \oplus C}$.

For any prime p , we may form the quotients $\bar{\phi} : A/pA \oplus C/pC \rightarrow B/pB$ and $\bar{\psi} : B/pB \rightarrow A/pA \oplus C/pC$ induced by ϕ and ψ , and deduce that $\bar{\phi} \circ \bar{\psi}$ and $\bar{\psi} \circ \bar{\phi}$ are both multiplication by $N \pmod{p}$. If p is a prime not dividing N , we may choose $M \in \mathbb{Z}$ with $NM \equiv 1 \pmod{p}$ and deduce that $\bar{\phi}$ is an isomorphism with inverse $M\bar{\psi}$. Hence $\dim_{\mathbb{F}_p}(B/pB) = \dim_{\mathbb{F}_p}(A/pA) + \dim_{\mathbb{F}_p}(C/pC)$, which proves the claim. $\square$

9.2. Degree. Recall that we calculated $H_n(S^n) = \mathbb{Z}$ for $n > 0$. For any continuous $f : S^n \rightarrow S^n$ we therefore get an induced map $f_* : H_n(S^n) \rightarrow H_n(S^n)$, which must be multiplication by some integer, since $H_n(S^n)$ is an infinite cyclic group.

Definition 9.2.1. For $n > 0$ and $f : S^n \rightarrow S^n$ continuous, let $\deg(f) \in \mathbb{Z}$ be the unique integer such that $f_* : H_n(S^n) \rightarrow H_n(S^n)$ is $\deg(f)1_{H_n(S^n)}$. This number is called the *degree* of f .

Lemma 9.2.2. *The degree of 1_{S^n} is 1. If $f, g : S^n \rightarrow S^n$, then $\deg f \circ g = \deg(f) \deg(g)$. If $f \simeq g$ then $\deg(f) = \deg(g)$.*

Proof. Easy consequence of H_n being a homotopy invariant functor. $\square$

Theorem 9.2.3. *Let $f : S^n \rightarrow S^n$ be reflection in a hyperplane, i.e. $f(x) = x - 2\langle x, v \rangle v$ for some $v \in S^n \subset \mathbb{R}^{n+1}$. Then $\deg(f) = -1$.*

Proof sketch. First let $g : S^n \rightarrow S^n$ be given by multiplication by an orthogonal matrix with v as first column, so that $g(e_1) = v$. Then the lemma gives $\deg(g^{-1} \circ f \circ g) = \deg(f)$, so by replacing f by $g^{-1} \circ f \circ g$ we may assume $v = e_1$.

The key point is now the existence of a Δ -complex structure on S^n with no simplices of dimension greater than n , and precisely two n -simplices $\Sigma_n = \{\sigma_+, \sigma_-\}$ satisfying $f \circ \sigma_+ = \sigma_-$ and $\sigma_+|_{\partial \Delta^n} = \sigma_-|_{\partial \Delta^n}$. Given that, $\sigma_+ - \sigma_- \in \Delta_n(S^n)$ will be a cycle and it must represent a non-trivial element in $H_n(S^n) = H_n^\Delta(S^n) \cong \mathbb{Z}$, since $\Delta_{n+1}(S^n) = 0$. Now $f_*(\sigma_+ - \sigma_-) = -(\sigma_+ - \sigma_-)$ so we deduce that $\deg(f) = -1$.

To construct such a Δ -complex structure the idea is to let σ_+ parametrize the entire upper hemisphere and σ_- the lower hemisphere. In slightly more detail, we may pick a homeomorphism $h : \Delta^n \rightarrow D^n$, define two maps

$$\begin{aligned} \sigma_+, \sigma_- : \Delta^n &\rightarrow S^n \\ t &\mapsto (\pm \sqrt{1 - |h(t)|^2}, h(t)), \end{aligned}$$

and give S^n the Δ -complex structure in which $\Sigma_n = \{\sigma_-, \sigma_+\}$ and whose lower-dimensional simplices are all possible composition of face maps $\Delta^i \rightarrow \partial \Delta^n$ composed with $\sigma_\pm|_{\partial \Delta^n}$. $\square$

Corollary 9.2.4. *Let $f : S^n \rightarrow S^n$ be the antipodal map, i.e. $f(x) = -x$. Then $\deg(f) = (-1)^{n+1}$. In particular it is not homotopic to the identity when n is even.*

Proof. The antipodal map is the composition of $n + 1$ reflections. $\square$

The following theorem is sometimes called the “hairy ball theorem”, because of an interpretation in the case $n = 2$.

Theorem 9.2.5. *For even n and any continuous map $v : S^n \rightarrow \mathbb{R}^{n+1}$ such that $\langle x, v(x) \rangle = 0$ for all $x \in S^n \subset \mathbb{R}^{n+1}$, there must exist an $x \in S^n$ with $v(x) = 0$.*

Proof. Assume for contradiction that no such x exists. After replacing v by $x \mapsto v(x)/|v(x)|$ we may assume $|v(x)| = 1$, so that $\{x, v(x)\}$ forms an orthonormal basis for the 2-dimensional vector space spanned by x and $v(x)$. Then the formula

$$h(t, x) = \cos(\pi t)x + \sin(\pi t)v(x)$$

defines a map $h : I \times S^n \rightarrow S^n$ which is a homotopy from the identity to the antipodal map, a contradiction. $\square$

9.3. Reduced homology. A reduced version of homology can be defined as follows:

Definition 9.3.1. For a topological space X , the reduced (or “augmented”) singular chain complex is defined as

$$\tilde{C}_p(X) = \begin{cases} C_p(X) & \text{for } p \geq 0 \\ \mathbb{Z} & \text{for } p = -1 \\ 0 & \text{for } p < -1. \end{cases}$$

The boundary map $\tilde{C}_p(X) \rightarrow \tilde{C}_{p-1}(X)$ is defined as the singular boundary when $p > 0$, while the boundary map

$$C_0(X) = \tilde{C}_0(X) \rightarrow \tilde{C}_{-1}(X) = \mathbb{Z}$$

is defined by sending all zero-simplices $\sigma : \Delta^0 \rightarrow X$ to $1 \in \mathbb{Z}$.

Reduced singular homology is then defined as

$$\tilde{H}_i(X) = H_i(\tilde{C}_*(X), \partial).$$

It is easily verified that $(\tilde{C}_*(X), \partial)$ is indeed a chain complex: it suffices to observe that if $\sigma : \Delta^1 \rightarrow X$ then $\partial\sigma = d_0\sigma - d_1\sigma$, and $\partial(d_0\sigma - d_1\sigma) = 1 - 1 = 0$.

Obviously $\tilde{H}_i(X) = H_i(X)$ when $i > 0$ and when $i < -1$. In degree 0 and -1 there is an exact sequence

$$0 \rightarrow \tilde{H}_0(X) \rightarrow H_0(X) \rightarrow \mathbb{Z} \rightarrow \tilde{H}_{-1}(X) \rightarrow 0$$

$$[\sum_{x \in X} n_x \cdot x] \mapsto \sum_{x \in X} n_x.$$

If $X = \emptyset$, this exact sequence shows $\tilde{H}_0(\emptyset) = 0$ and $\tilde{H}_{-1}(\emptyset) \cong \mathbb{Z}$. If X is non-empty the middle map is surjective so $\tilde{H}_{-1}(X) = 0$ in that case, and we have a short exact sequence $0 \rightarrow \tilde{H}_0(X) \rightarrow H_0(X) \rightarrow \mathbb{Z} \rightarrow 0$. Choosing a point $x \in X$ gives a splitting $\mathbb{Z} \rightarrow H_0(X)$ inducing an isomorphism $H_0(X) \cong \mathbb{Z} \oplus \tilde{H}_0(X)$. (The isomorphism depends on the choice of x but the short exact sequence is independent of choices.)

Reduced singular homology is a functor from topological spaces to abelian groups in the same way as unreduced. It inherits most properties that unreduced singular homology has, in particular a homotopy equivalence $X \rightarrow Y$ induced an isomorphism of reduced singular homology groups.

Lemma 9.3.2. *If X is contractible then $\tilde{H}_i(X) = 0$ for all i . If the number of path components of X is $r \in \mathbb{Z}_{>0}$, then $\tilde{H}_0(X) \cong \mathbb{Z}^{r-1}$.*

Proof. The short exact sequence $0 \rightarrow \tilde{H}_0(X) \rightarrow H_0(X) \rightarrow \mathbb{Z} \rightarrow 0$ shows that the abelian group $\tilde{H}_0(X) \subset H_0(X)$ has no torsion (since $H_0(X) \cong \mathbb{Z}[\pi_0(X)]$ doesn't) and that it is finitely generated of rank $r - 1$. Hence it must be isomorphic to $\mathbb{Z}^{r-1}$. In particular $\tilde{H}_0(X) = 0$ for path connected spaces.

If X is in addition contractible then $\tilde{H}_i(X) \cong H_i(X) = 0$ for $i > 0$. □

There is also a Mayer–Vietoris sequence for reduced homology. Recall that the (unreduced) Mayer–Vietoris sequence arose from a short exact sequence of chain complexes $0 \rightarrow C_*(U_{01}) \rightarrow C_*(U_0) \oplus C_*(U_1) \rightarrow C_*^{\mathcal{U}}(X) \rightarrow 0$ when $X = U_0 \cup U_1$ for two open sets $U_0, U_1 \subset X$ with $U_0 \cup U_1 = X$ and $U_{01} = U_0 \cap U_1$. The last term is the $\mathcal{U}$ -small chains with $\mathcal{U} = \{U_0, U_1\}$. Using that all 0-chains are small, this short

exact sequence may be extended to the reduced chains ending in

$$\begin{array}{ccccccc}
 & & \downarrow \partial & & \downarrow \partial & & \downarrow \partial \\
 0 & \longrightarrow & C_0(U_{01}) & \longrightarrow & C_0(U_0) \oplus C_0(U_1) & \longrightarrow & C_0(X) \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \mathbb{Z} & \longrightarrow & \mathbb{Z} \oplus \mathbb{Z} & \longrightarrow & \mathbb{Z} \longrightarrow 0,
 \end{array}$$

where the two bottom maps are $n \mapsto (n, -n)$ and $(n, m) \mapsto n + m$. The result is a short exact sequence of chain complexes, which by Lemma 6.5.4 induces a long exact sequence. This long exact sequence looks like the Mayer–Vietoris sequence but with reduced homology. (It of course agrees with the actual Mayer–Vietoris in positive degrees.)

9.4. The Jordan curve theorem. We write $I^k = I \times \cdots \times I$ for the k -fold product of $I = [0, 1]$, often called the k -cube. It is not hard to construct a homeomorphism $I^k \approx D^k$.

Theorem 9.4.1. *For any $k \geq 0$ and any injective map $j : I^k \rightarrow S^n$ we have $\tilde{H}_i(S^n \setminus j(I^k)) = 0$ for all i .*

In particular (but perhaps unsurprisingly), for any such embedding j the complement $S^n \setminus j(I^k)$ will be path connected: by Lemma 9.3.2 the number of path components is one more than the rank of the group $\tilde{H}_0(S^n \setminus j(I^k)) \cong 0$.

Proof. For $k = 0$ the subspace $j(I^0) \subset S^n$ is a single point. Stereographic projection shows that the complement is homeomorphic to $\mathbb{R}^n$. In particular it is contractible and therefore $\tilde{H}_i(S^n \setminus j(I^0)) = 0$ for all i . We proceed by induction on k .

For the induction step, we may assume $S^n \setminus j(\{t\} \times I^{k-1})$ has vanishing reduced homology, for each $t \in I$. Assume for contradiction that $S^n \setminus j(I^k)$ does not have vanishing reduced homology, and choose a cycle $c \in \tilde{C}_p(S^n \setminus j(I^k))$ representing a non-zero element $[c] \in \tilde{H}_p(S^n \setminus j(I^k))$. Cover $S^n \setminus j(\{\frac{1}{2}\} \times I^{k-1})$ by the two open sets

$$\begin{aligned}
 U &= S^n \setminus j([0, \tfrac{1}{2}] \times I^{k-1}) \\
 V &= S^n \setminus j([\tfrac{1}{2}, 1] \times I^{k-1}).
 \end{aligned}$$

They satisfy $U \cup V = S^n \setminus j(\{\frac{1}{2}\} \times I^{k-1})$ so by induction $\tilde{H}_i(U \cup V) = 0$ for all i and hence the Mayer–Vietoris sequence for reduced homology degenerates to isomorphisms

$$\tilde{H}_i(U \cap V) \rightarrow \tilde{H}_i(U) \oplus \tilde{H}_i(V).$$

In particular the non-zero class $[c] \in \tilde{H}_i(U \cap V)$ must remain non-zero in either $\tilde{H}_i(U)$ or $\tilde{H}_i(V)$, possibly both. Hence we must have

$$\begin{aligned}
 \tilde{H}_p(S^n \setminus j(I \times I^{k-1})) &\rightarrow \tilde{H}_p(S^n \setminus j(I_1 \times I^{k-1})) \\
 [c] &\mapsto \text{incl}_*([c]) \neq 0
 \end{aligned}$$

for either $I_1 = [0, \frac{1}{2}]$ or $I_1 = [\frac{1}{2}, 1]$. By the same argument we may partition the interval I_1 into two subintervals of length $\frac{1}{4}$ and choose one of them $I_2 \subset I_1$ with the property that

$$\begin{aligned}
 \tilde{H}_p(S^n \setminus j(I_1 \times I^{k-1})) &\rightarrow \tilde{H}_p(S^n \setminus j(I_2 \times I^{k-1})) \\
 [c] &\mapsto \text{incl}_*([c]) \neq 0.
 \end{aligned}$$

Repeating this argument infinitely many times we get a descending sequence of intervals $I \supset I_1 \supset I_2 \supset \dots$ where I_s has length 2^{-s} , with the property that the cycle c is not a boundary in any of the sets $S^n \setminus j(I_s \times I^{k-1})$.

The intersection of the intervals I_s consists of a single point $t \in I$, and hence the union of the open subsets $S^n \setminus j(I_s \times I^{k-1})$ is $S^n \setminus j(\{t\} \times I^{k-1})$. Since by induction we have $\tilde{H}_p(\{t\} \times I^{k-1})$, the cycle $c \in \tilde{C}_p(S^n \setminus j(I^k)) \subset \tilde{C}_p(S^n \setminus j(\{t\} \times I^{k-1}))$ must be a boundary of some cycle $d \in \tilde{C}_{p+1}(S^n \setminus j(\{t\} \times I^{k-1}))$. But the finitely many $(p+1)$ -simplices appearing in d with non-zero coefficient must all have positive distance from the compact set $j(\{t\} \times I^{k-1})$ and hence in fact be disjoint from $j(I_s \times I^{k-1})$ for sufficiently large s . This contradicts the property that c represents a non-zero class in $\tilde{H}_p(S^n \setminus j(I_s \times I^{k-1}))$. $\square$

Theorem 9.4.2. *For any $k \geq 0$, any $n \geq 0$, and any injective map $j : S^k \rightarrow S^n$ we have*

$$\tilde{H}_i(S^n \setminus j(S^k)) \cong \begin{cases} \mathbb{Z} & \text{for } i = n - k - 1 \\ 0 & \text{otherwise.} \end{cases}$$

In particular, for any embedding $j : S^{n-1} \rightarrow S^n$ the complement $S^n \setminus S^{n-1}$ has precisely two path components. Similarly, any injective map $j : S^n \rightarrow S^n$ must be surjective and hence a homeomorphism, because $\tilde{H}_{-1}(S^n \setminus j(S^n)) \cong \mathbb{Z}$ but only the empty set has non-trivial $\tilde{H}_{-1}$.

Proof. For $k = 0$ we are embedding the two-point space $S^0 = \{-1, +1\}$. By stereographic projection the complement $S^n \setminus j(+1)$ is homeomorphic to $\mathbb{R}^n$ and hence $S^n \setminus j(S^0)$ is homeomorphic to $\mathbb{R}^n$ with a point removed, which is homeomorphic to $S^{n-1} \times (0, \infty)$:

$$\begin{aligned} \mathbb{R}^n \setminus \{x_0\} &\rightarrow S^{n-1} \times (0, \infty) \\ x &\mapsto \left(\frac{x}{|x|}, |x| \right). \end{aligned}$$

Hence it is homotopy equivalent to S^{n-1} whose reduced homology is as stated.

Now proceed by induction on k , using that $S^k = \partial D^{k+1} = \partial_- D^{k+1} \cup \partial_+ D^{k+1}$, where both $\partial_- D^{k+1}$ and $\partial_+ D^{k+1}$ are homeomorphic to D^k and therefore to I^k . Therefore the open sets $U = S^n \setminus j(\partial_- D^{k+1})$ and $V = S^n \setminus j(\partial_+ D^{k+1})$ both have trivial reduced homology in all degrees, by the previous theorem, so the Mayer–Vietoris sequence for reduced homology degenerates to isomorphisms

$$\tilde{H}_{i+1}(U \cup V) \xrightarrow{\cong} \tilde{H}_i(U \cap V)$$

for all i . But $U \cap V = S^n \setminus j(S^k)$ and $U \cup V = S^n \setminus j(\partial_- D^{k+1} \cap \partial_+ D^{k+1})$. Since $\partial_- D^{k+1} \cap \partial_+ D^{k+1}$ is homeomorphic to S^{k-1} (it is the “equator” in S^k), the induction hypothesis implies that $\tilde{H}_{i-1}(U \cup V) \cong \mathbb{Z}$ when $i+1 = n - (k-1) - 1$ and that it vanishes otherwise. This finishes the induction step. $\square$

Corollary 9.4.3. *Let $k > 0$ and let $j : S^k \rightarrow \mathbb{R}^n$ be an injective map. Then*

$$\tilde{H}_i(\mathbb{R}^n \setminus S^k) \cong \begin{cases} \mathbb{Z} & \text{if } i = n - 1 \text{ or } i = n - k - 1 \\ 0 & \text{otherwise.} \end{cases}$$

Proof sketch. Use stereographic projection to identify $\mathbb{R}^n$ with $S^n \setminus \{N\}$, where $N = (1, 0, \dots, 0)$, and cover S^n by the open sets $U = S^n \setminus \{N\}$ and $V = S^n \setminus j(S^k)$. We are then interested in the reduced homology of $U \cap V = (S^n \setminus \{N\}) \setminus j(S^k)$, which is easily deduced from the Mayer–Vietoris sequence for the open cover of S^n by U and V . $\square$

The classical case $n = 2$ and $k = 1$ is the Jordan Curve theorem: it asserts that if a subset $C \subset \mathbb{R}^n$ is homeomorphic to a S^1 (a “curve”), then $\mathbb{R}^2 \setminus C$ has precisely two path components.

10. ENGELSK–DANSK ORDBOG

Some of the concepts introduced in this class have established translations to Danish. In order for these Danish terms not to die out, I hope those of you who are Danish will learn and adopt the Danish terms as well. Here is an incomplete dictionary.

English	Dansk
algebraic topology	algebraisk topologi
homotopy theory	homotopiteori
fundamental group	fundamentalgruppe
base point	basispunkt
covering map	overlejring / overlejningsafbildning
open covering	åben overdækning
simply connected	enkeltsammenhængende
path	sti
loop	løkke
concatenation	konkatenering
homotopy	homotopi
homotopy equivalence	homotopiækvivalens
contractible	kontraktibel
path connected	stisammenhængende
category	kategori
functor	funktor
chain complex	kædekompleks
chain map, chain homotopy	kædeafbildning, kædehomotopi
boundary	rand
cycle	cykel
exact sequence	eksakt følge
homology	homologi
singular homology	singulær homologi
simplicial homology	simpliciel homologi
simplex	simpleks
barycentric subdivision	barycentrisk underdeling
euler characteristic	eulerkarakteristik